NetApp MetroCluster over IP

April 2023 | LD00045 Version 1.2.0

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1 Introduction

There is an increased need to provide both near-zero recovery time (RTO) and zero recovery point (RPO) for business critical applications and data. This need demands high availability at both the compute and storage layers. The combination of VMware Infrastructure and NetApp MetroCluster provides this kind of availability from both the server and the storage perspectives.

VMware vSphere Metro Storage Cluster (vMSC) provides a robust server consolidation solution with high application availability. MetroCluster provides continuously available storage between racks in a data center, across a campus, or in a metropolitan area.

MetroCluster is a scalable solution that provides high availability (HA) at the local cluster level and cross-site synchronous disaster recovery (DR). At the local level, redundant nodes are members of a cluster HA pair. Cross-site disaster recovery is provided through an HA pair in a cluster in one location, and a partner HA pair in a cluster at another location. The two clusters are peered, and they provide bi-directional DR where each cluster can be the source and backup for the other cluster.

The purpose of this lab is to acquaint you with how the VMware vSphere Infrastructure and NetApp MetroCluster work together to provide business continuity, and to understand the basic steps necessary to perform a switchover of operations from one site to another. You will simulate a site failure, perform a switchover, and prepare and perform a site switchback. You will also perform a planned switchover used in cases of necessary maintenance or DR testing. At the end of this lab you should have a basic understanding of the specific actions necessary to perform those activities.

This lab introduces an IP-based configuration of MetroCluster (MCC-IP) that provides an alternative to the MetroCluster with dedicated back-end FC fabrics resulting in the following differences:

- MCC-IP uses Ethernet ports rather than FC host adapter ports.
- Routable IP addresses are used to access the remote site, which allows for ISLs that are shared with other network traffic.
- Remote disks are accessed using iSCSI protocol with the remote DR node acting as the iSCSI target; thus, supporting platforms with integrated storage.
- Uses IP switches instead of Fibre Channel switches.
- Uses internal or direct attached storage instead of using SAS/FC bridges.
- Aggregate healing (resynchronization) prior to switchback is performed automatically.
- MCC-IP uses the ONTAP Mediator.

The following diagram depicts a MetroCluster environment during normal operation.

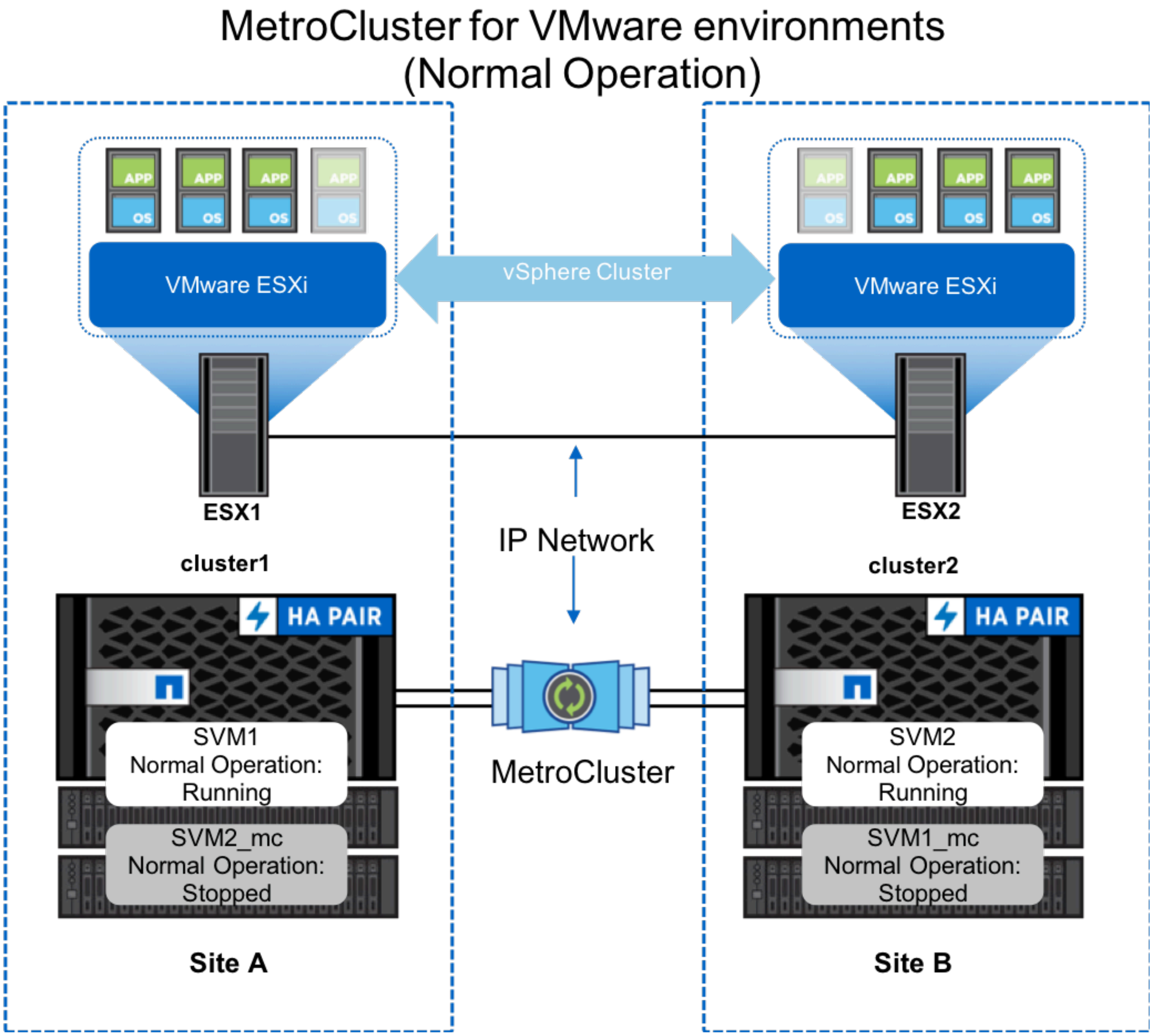

Figure 1-1:

The resulting operation after a site failure and switchover is depicted below.

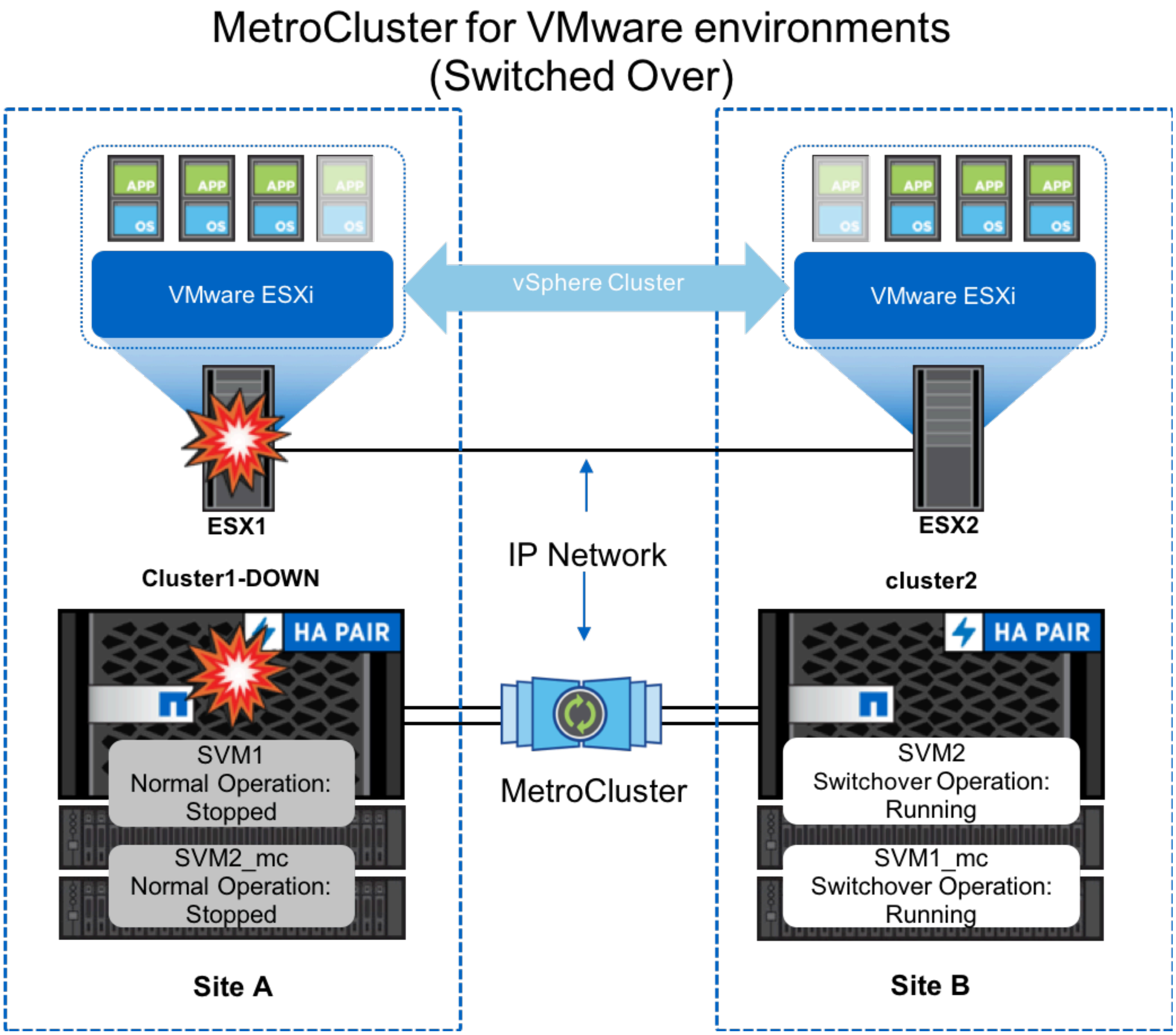

Figure 1-2:

1.1 Lab Objectives

This lab demonstrates various failure scenarios (host, network, and storage), and identifies NetApp best practices where applicable.

When you complete this lab you should have a basic understanding of the following subjects:

- The differences between the IP-based and Fibre Channel-based versions of NetApp MetroCluster.
- How to verify proper configuration and status of VMware Infrastructure and MetroCluster.
- The proper operation of this environment during both planned and unplanned site switchover scenarios.
- How a return to normal operation is performed after a site switchover (switchback).

2 Lab Activities

Key NetApp capabilities will be highlighted in the following activities:

- Tour the Environment
    - Tour the Environment Using the Web UI
    - MetroCluster Status using the CLI
- Unplanned Switchover: High Availability for Business Continuity
- MetroCluster Switchback
- Planned Switchover: High Availability for Non-Disruptive Operations
    - Planned Switchover
    - Optional: Switchback from Planned Maintenance

2.1 Tour the Environment

In this activity, you review the configuration of MetroCluster and VMware vSphere. Best practices for configuraing the solution include using VM/Host groups and VM/Host Rules to organize resources so that they align with the MetroCluster site where the data stores are located. You also review how to use the MetroCluster Health Check feature in ONTAP to validate MetroCluster. The tasks you will perform in this activity are listed below:

Tasks
Use the VMware vSphere Web Client to power on the VM and verify proper settings.
Use System Manager to verify the status for MetroCluster.
Using the CLI to check MetroCluster configuration and status.

2.1.1 Tour the Environment Using the Web UI

In this activity, you review the configuration of MetroCluster and VMware vSphere using the web browser management interfaces.

1. Open **Firefox** from the task bar.

Figure 2-1:

2. Maximize the browser window.

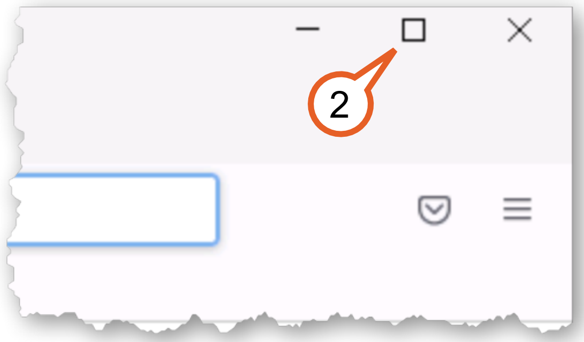

Figure 2-2:

3. Click the **vCenter** bookmark.

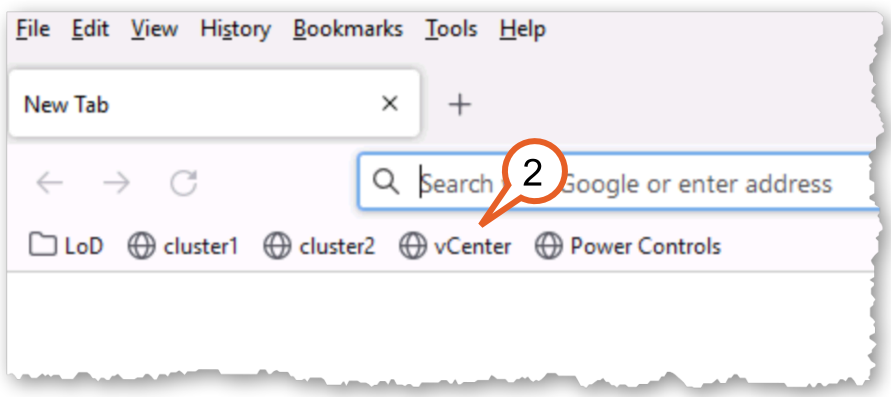

Figure 2-3:

4. Log in with the following credentials:
   - User name:`administrator@demo.netapp.com`
   - Password: `Netapp1!`

5. Click **Login**.

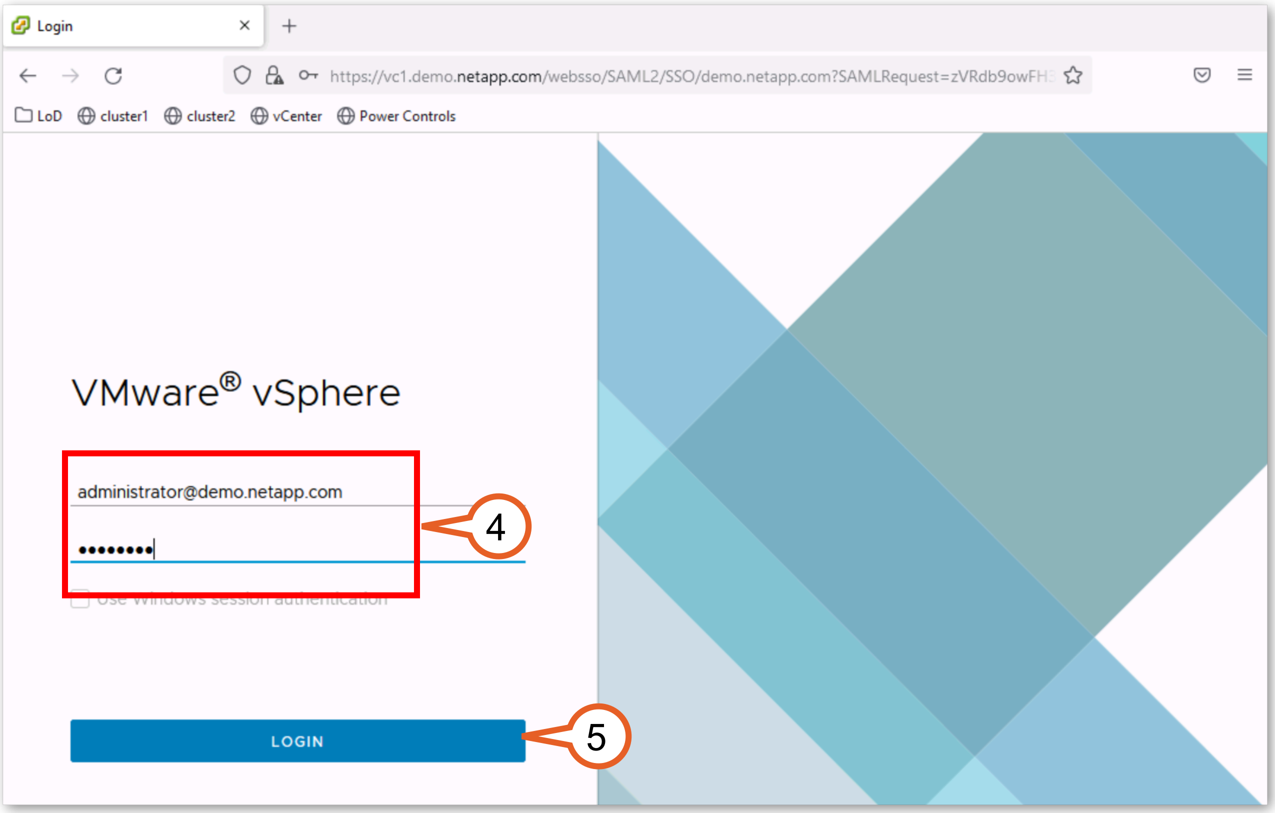

Figure 2-4:

6. If the menus are collapsed, expand **vc1.demo.netapp.com** > **Datacenter** > **ESX Cluster** to locate "vm1".

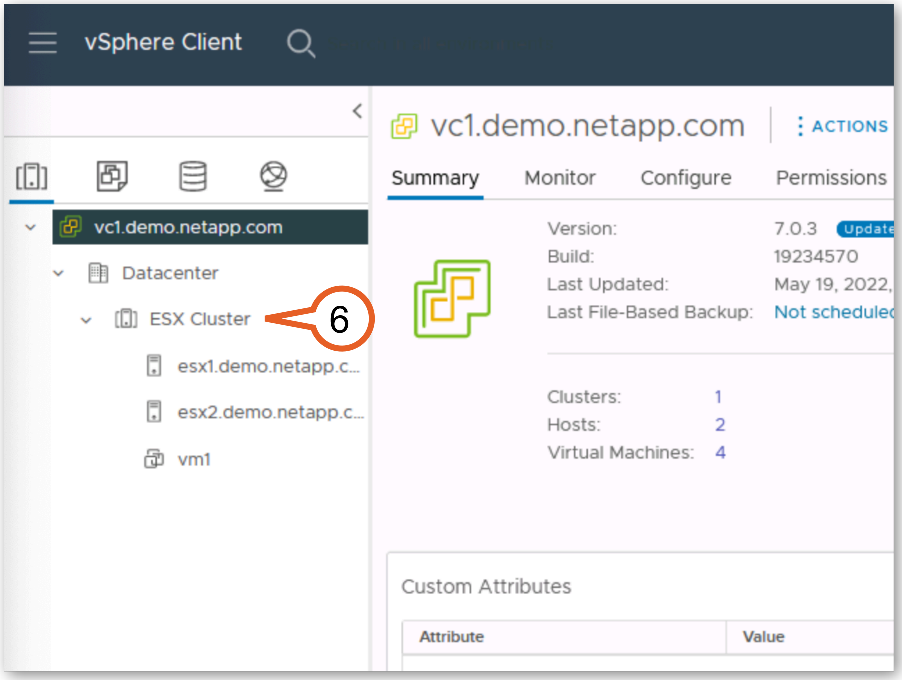

Figure 2-5:

7. Right-click **vm1**.

8. Select **Power** > **Power On**.

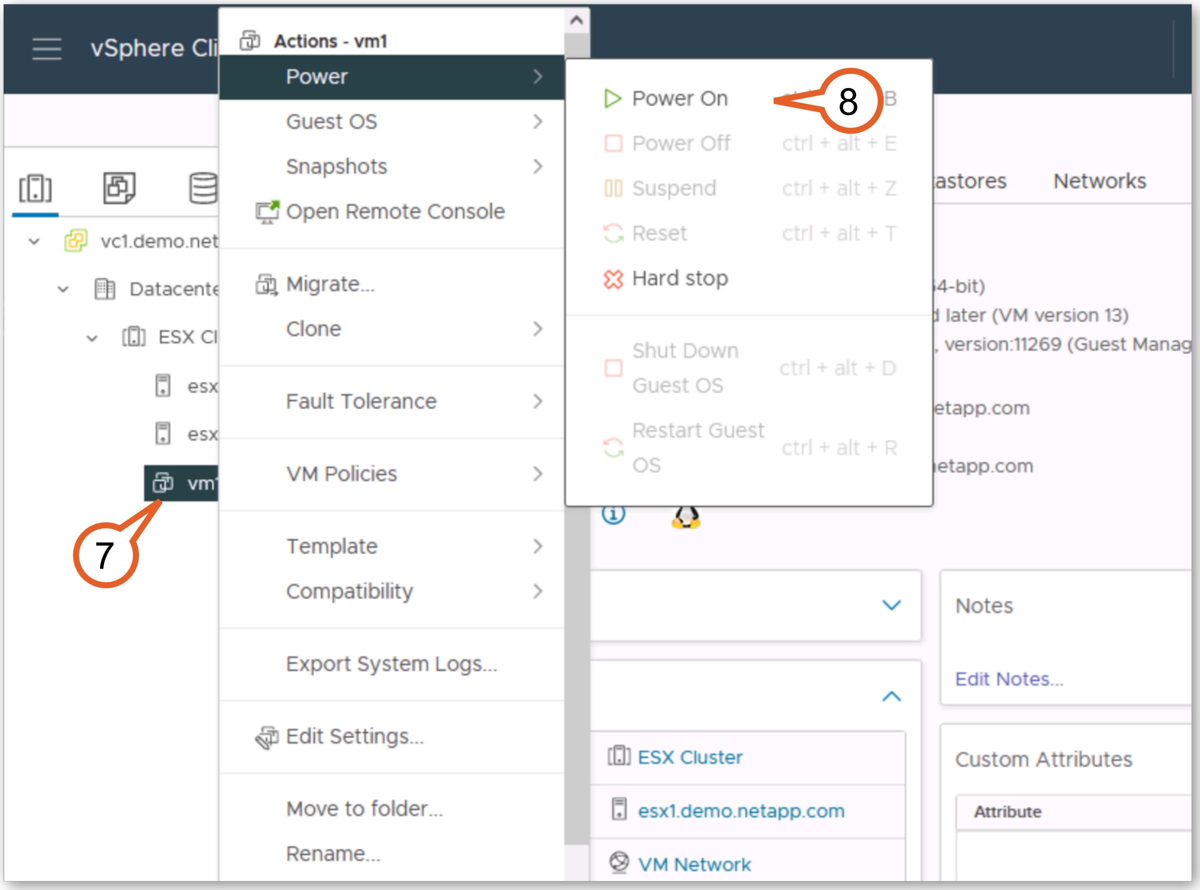

Figure 2-6:

While the VM starts, note that there are two ESXi hosts for this lab demonstration: esx1 provides the compute layer at Site A, while esx2 is at Site B. Both hosts belong to the same VMware High Availability (HA) Cluster. This is defined as a stretched VMware cluster.

9. Right-click **ESX Cluster**.

10. Select **Settings**.

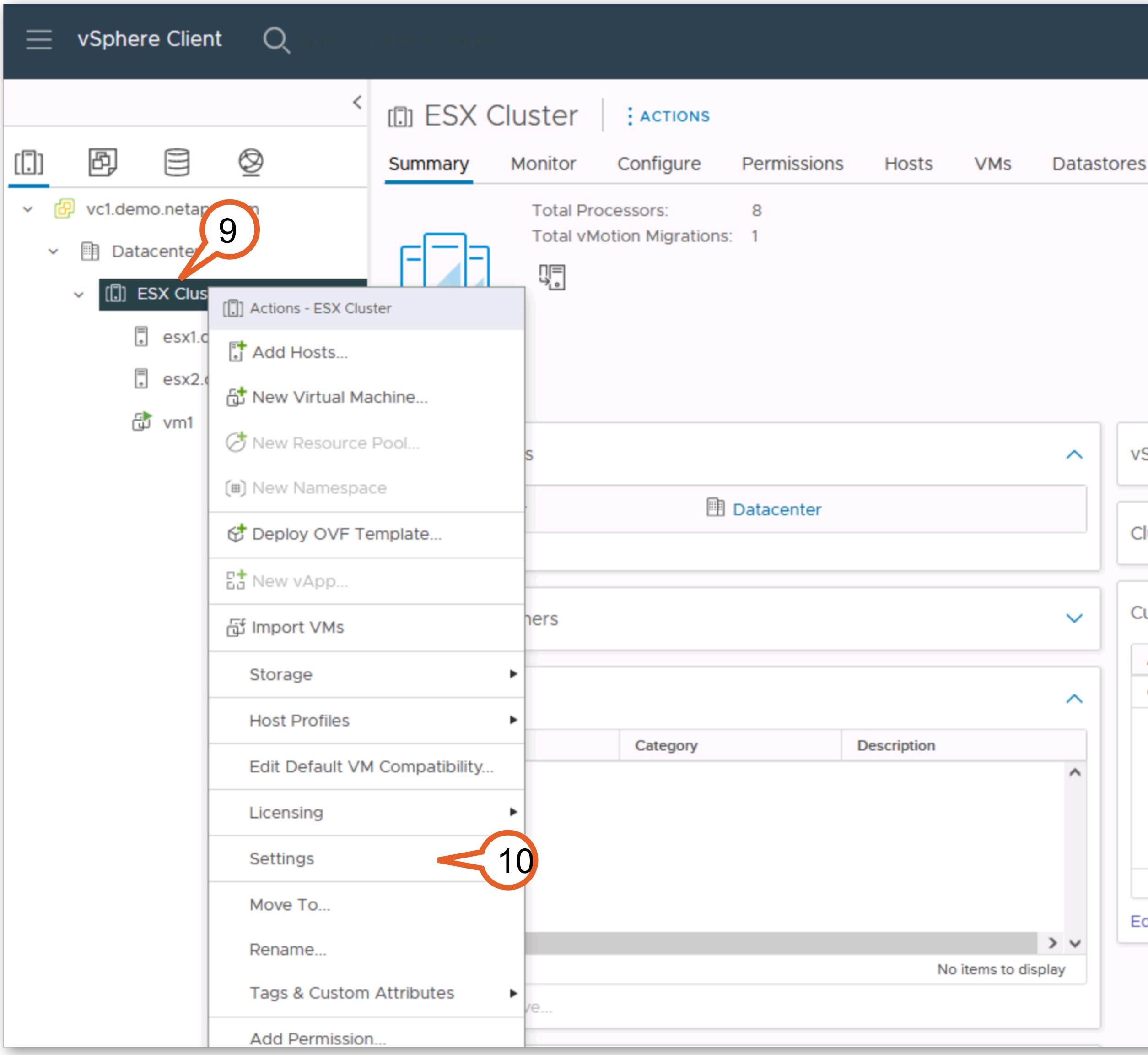

Figure 2-7:

11. Select **VM/Host Groups**.

Note: If the settings list is not visible, expand the browser window if possible.

.

12. Note that there are three groups defined, “sitea_hosts”, “siteb_hosts”, and a VM Group for the “sitea_vms”.

13. If you click on each of these groups you can see that “esx1” is in the “sitea_hosts” group, “esx2” is in the “siteb_hosts” group, and “vm1” is in the “sitea_vms” group.

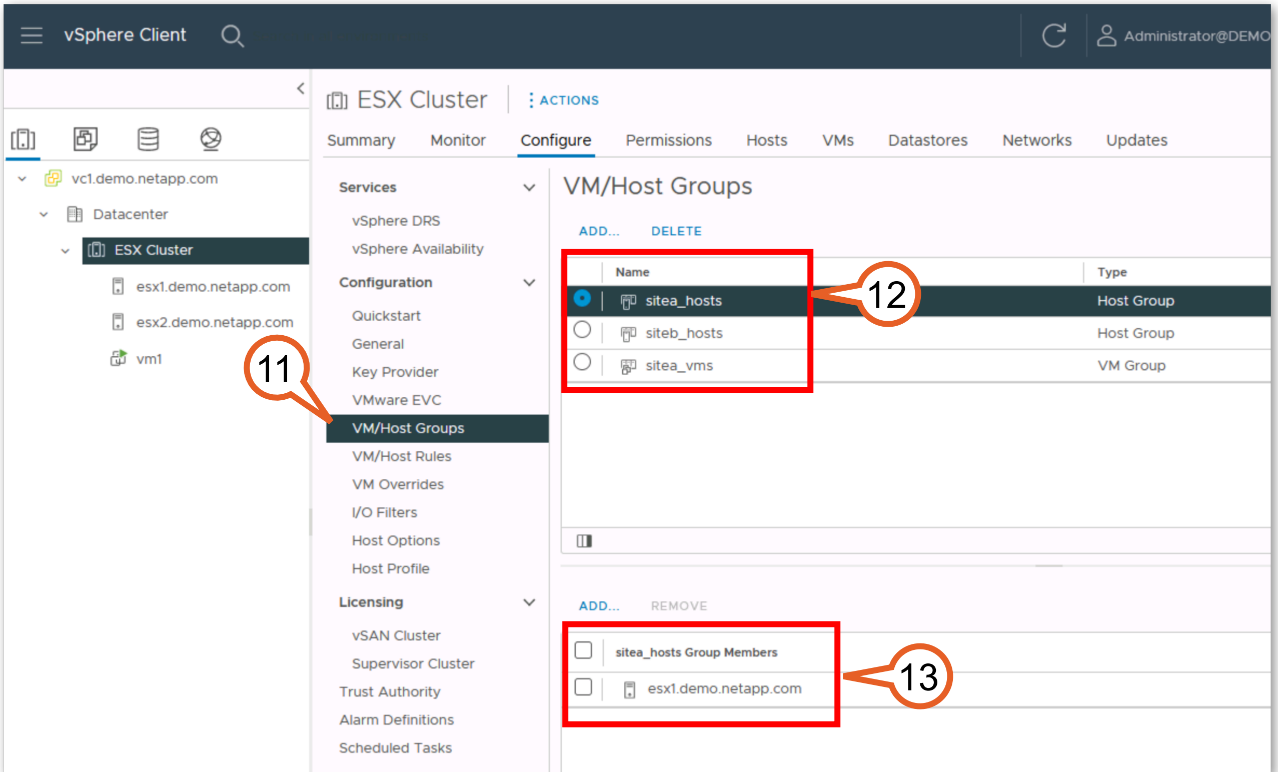

Figure 2-8:

14. Click **VM/Host Rules**.
15. Select **sitea_affinity**.

16. Click **Edit**.

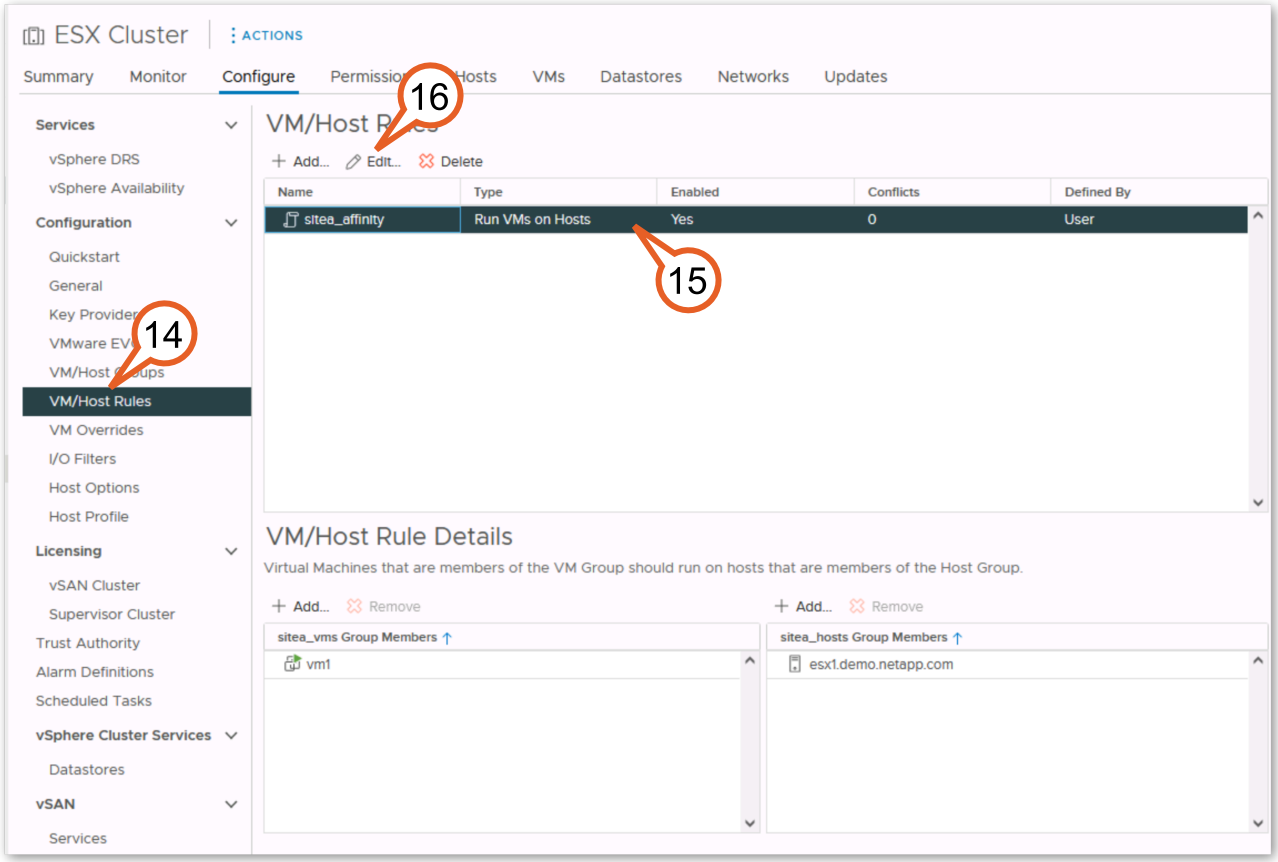

Figure 2-9:

17. Examine the rule. This illustrates that the VMs in the sitea_vms group (vm1) should normally run on the hosts in the sitea_hosts group (esx1). Click **Cancel** to close.

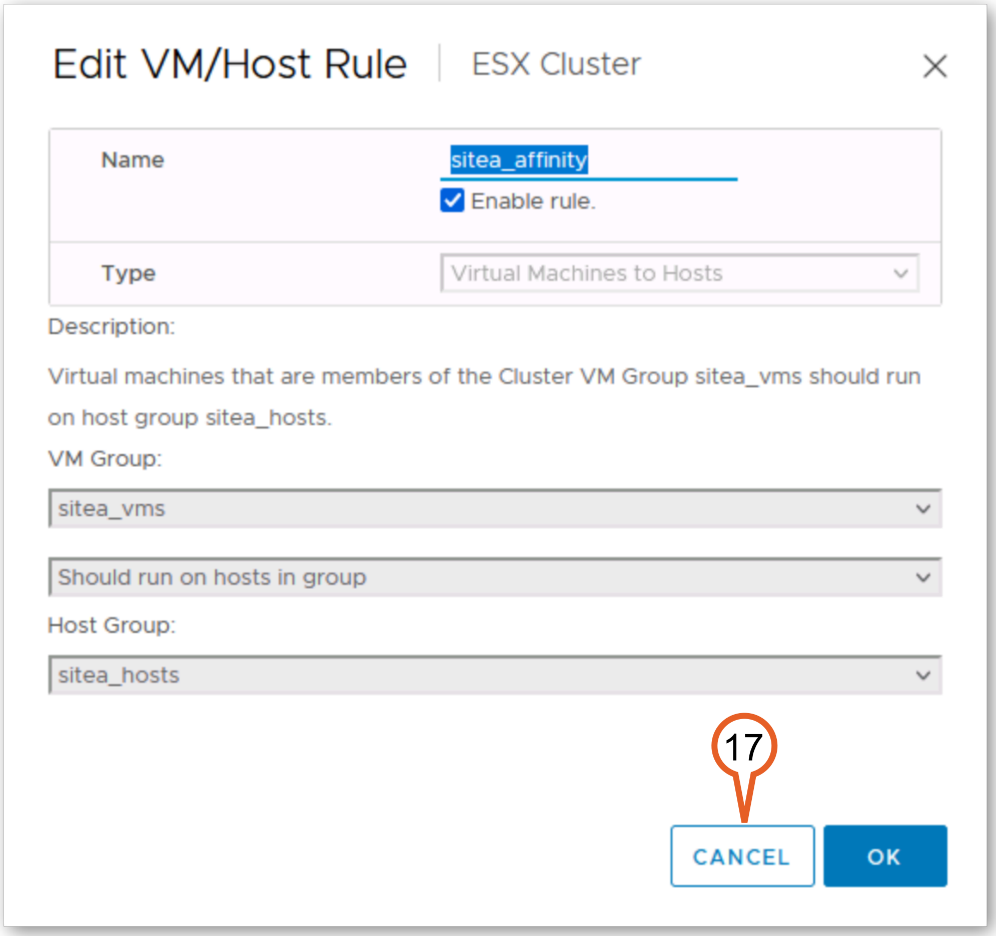

Figure 2-10:

Now that you have had a brief look at the VMware environment through the vCenter interface, verify the proper status of the SVMs on each cluster.

18. Open a new browser tab and click the bookmark for cluster1's ONTAP System Manager (**cluster1**).

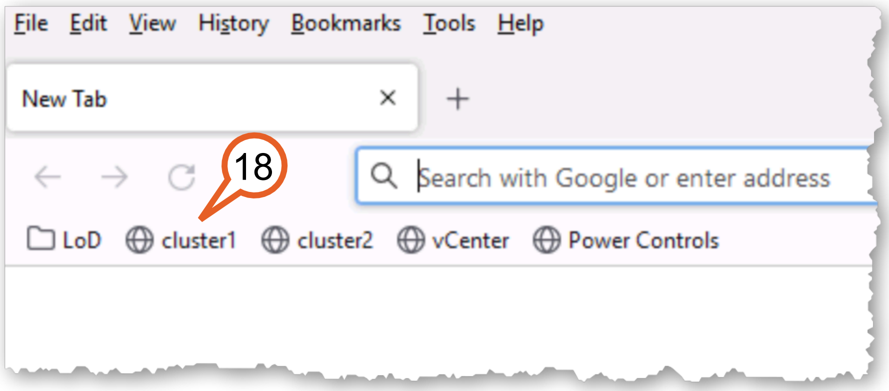

Figure 2-11:

19. Enter the following credentials:
    - User name: **`admin`**
    - Password: **`Netapp1!`**
20. Click **Sign in**.

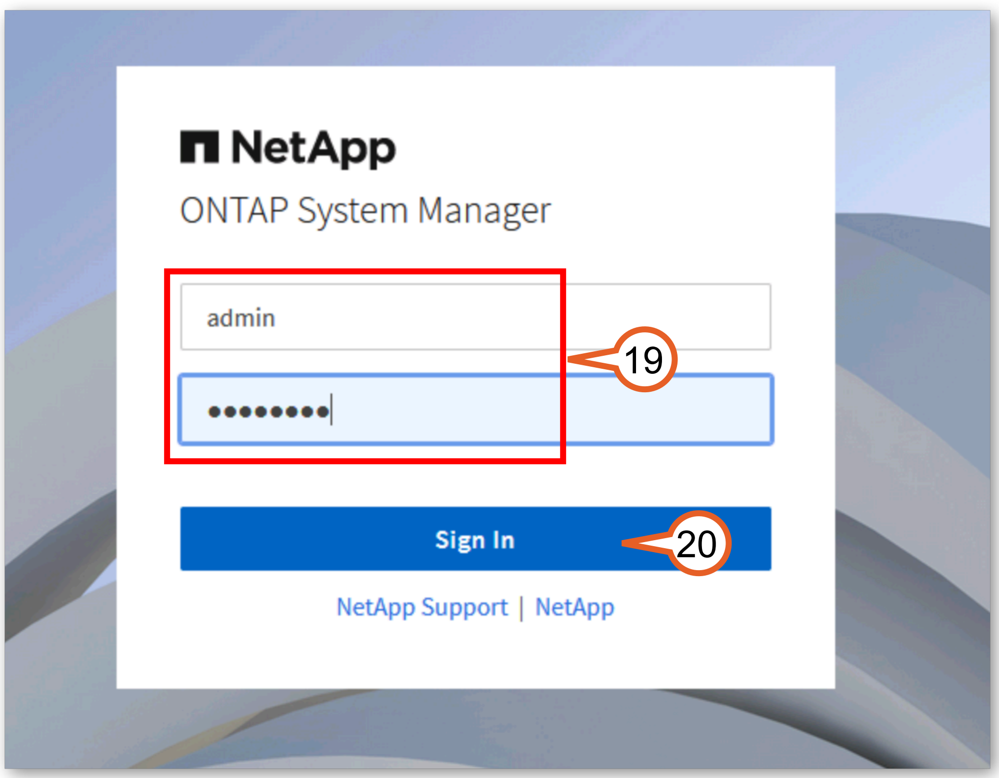

Figure 2-12:

21. From the Dashboard you can get an overview of the cluster. On the bottom left of the Dashboard is a MetroCluster panel that shows the overall status. Click the **menu icon** in the “cluster1” box.

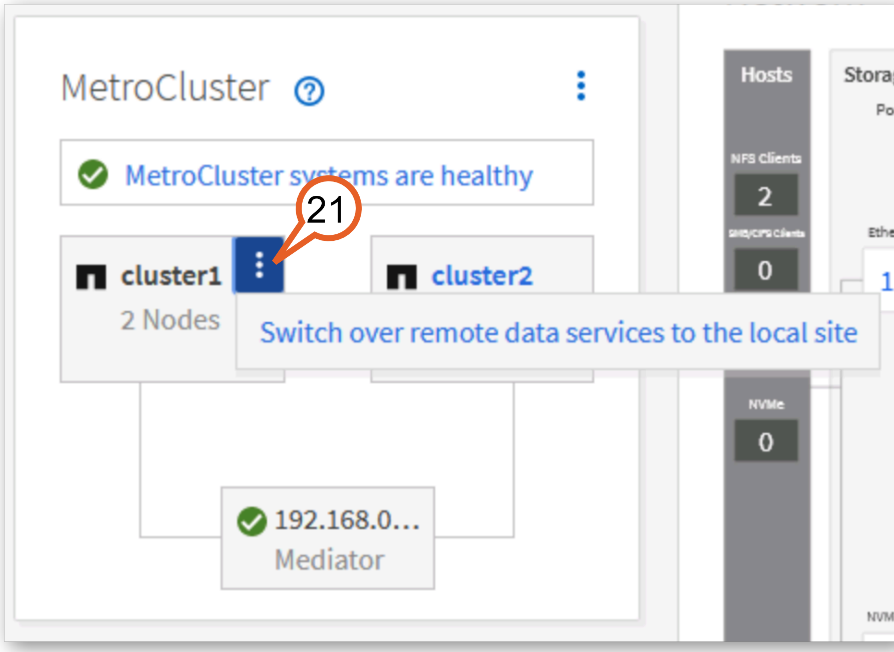

Figure 2-13:

This menu allows you to switchover the services from the remote site (cluster2) to the local site (cluster1).

22. Click outside of the menu, or press the **Escape** key, to close the menu.

23. Hover the mouse over the green check box between the cluster1 and cluster2 boxes. This shows that MetroCluster is active. This changes to show the current MetroCluster state.

24. Click **MetroCluster systems are healthy**.

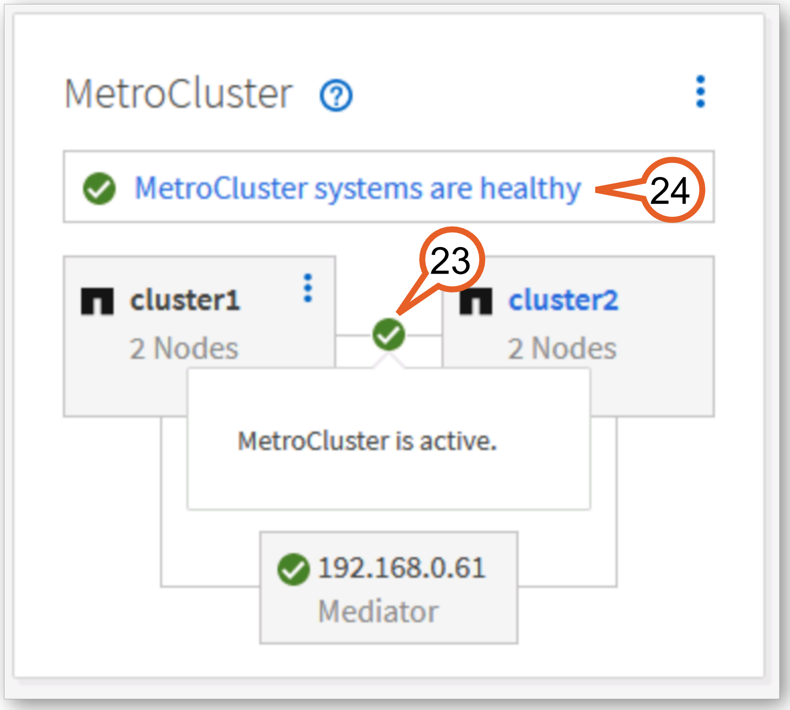

Figure 2-14:

You are taken to the “MetroCluster Health” page. This shows the status from the most recent health check.

25. Click **Check MetroCluster Health**.

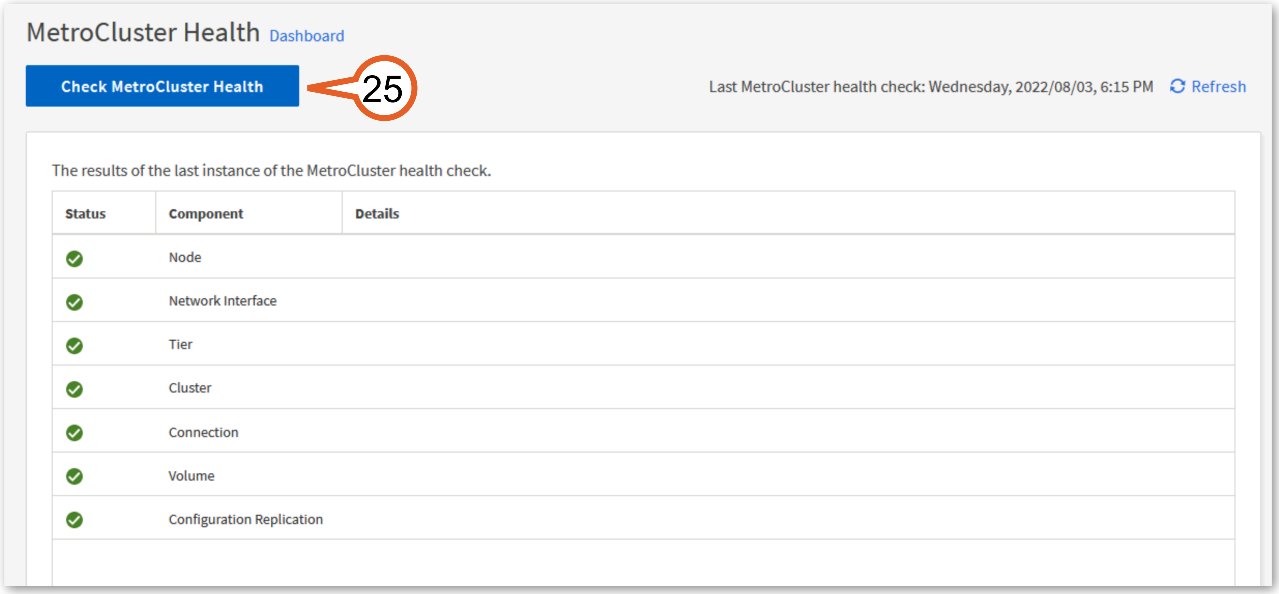

Figure 2-15:

This will take a minute or two to complete. While it is running the button shows the status.

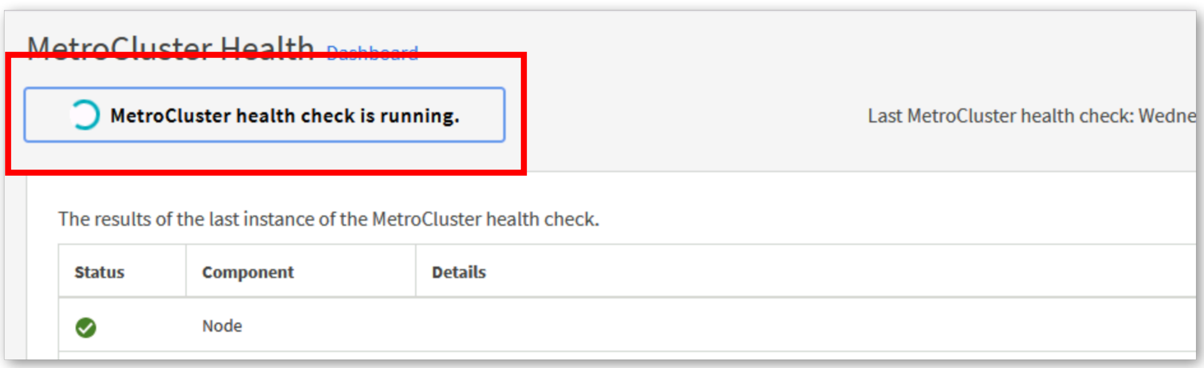

Figure 2-16:

Once completed a banner appears and the page is updated with the results.

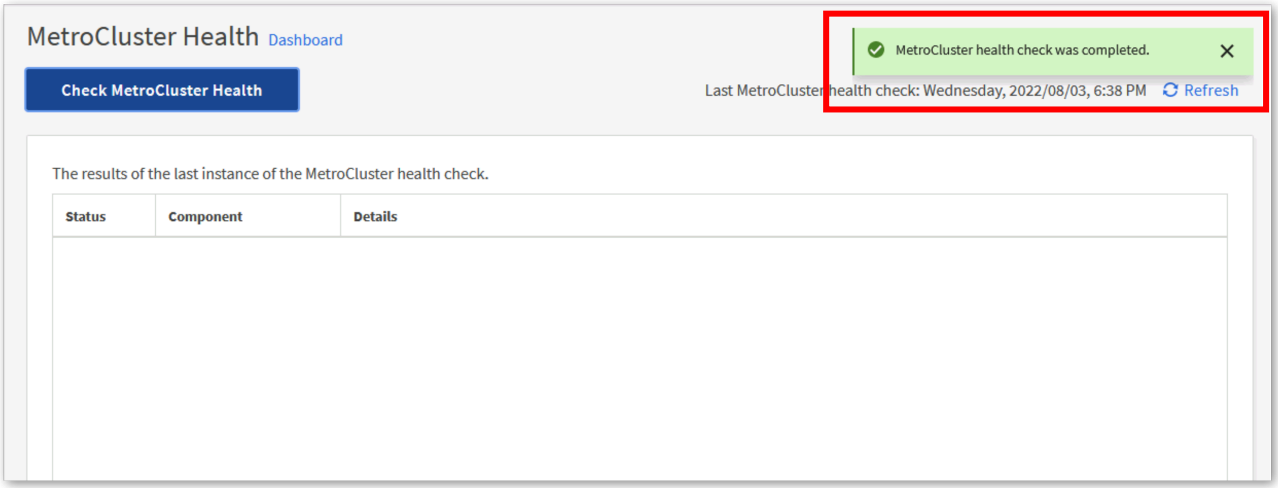

Figure 2-17:

26. The status shows all checks are OK. Click **Dashboard** to return to the Dashboard.

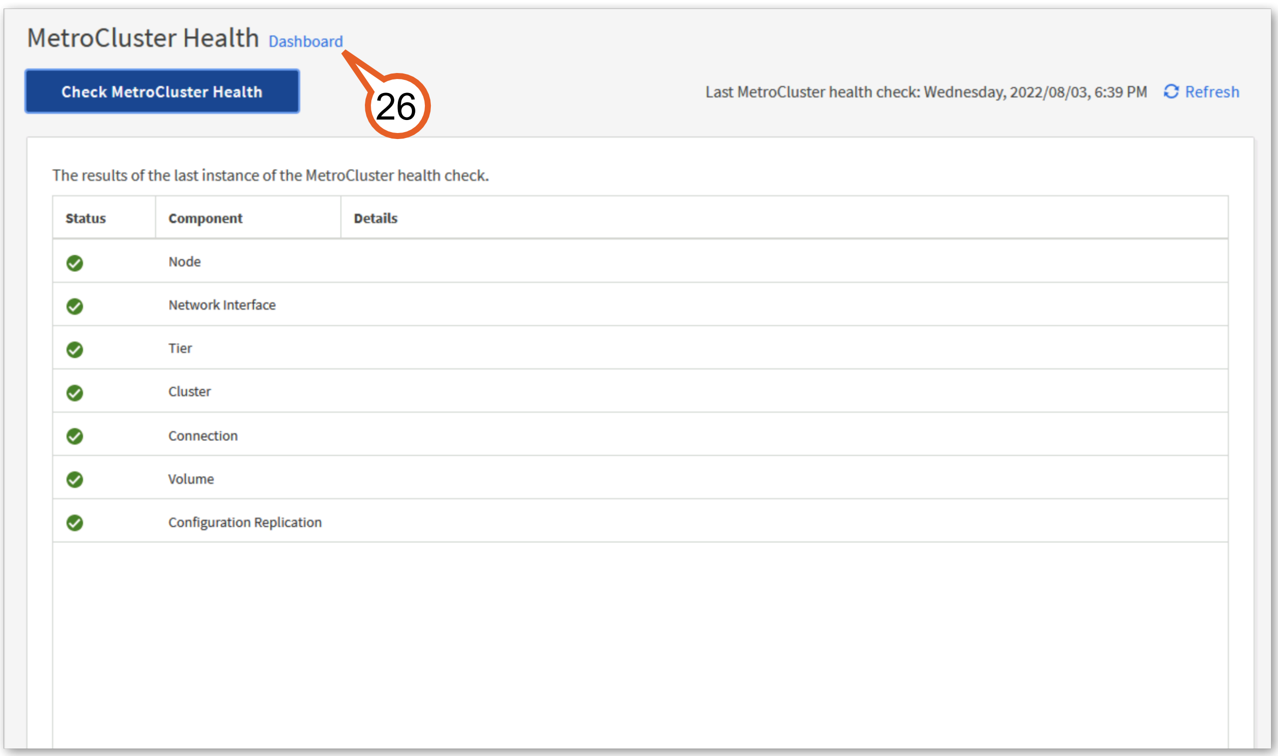

Figure 2-18:

27. The MetroCluster panel on the Dashboard also shows the connection to the ONTAP Mediator. Hover the mouse over the green check box to see the status is “Connected”.

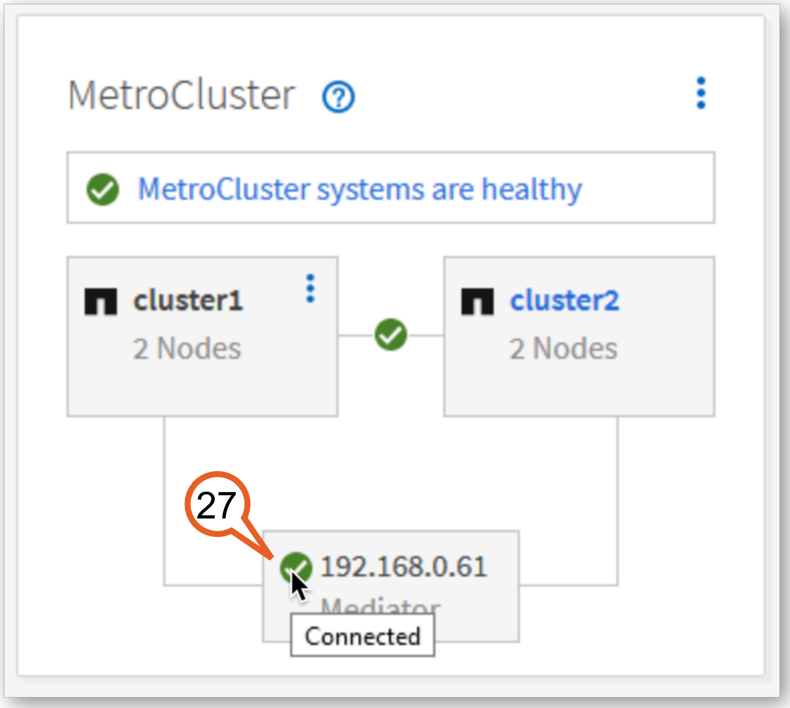

Figure 2-19:

This activity provided a review of the vSphere and MetroCluster environments from various perspectives that allow you to verify the proper configuration and status.

2.1.2 MetroCluster Status using the CLI

This activity familiarizes you with the command line interface (CLI) for MetroCluster. The CLI provides more details on the MetroCluster Health Check that you used in the previous activity.

1. Click the **PuTTY** icon on the taskbar of the Jumphost.

Figure 2-20:

2. Select the saved session for **cluster2**.

3. Click **Open**.

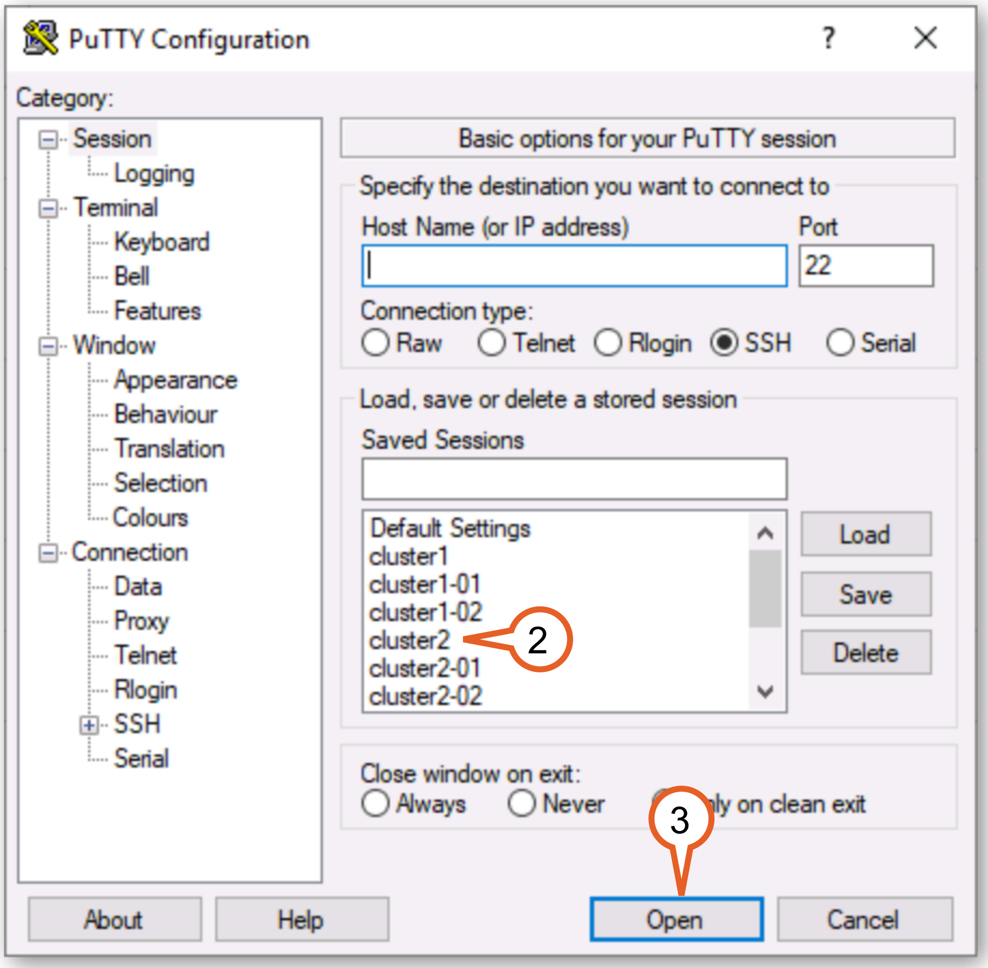

Figure 2-21:

4. Login as **`admin`**, with password **`Netapp1!`**.
5. Issue the `metrocluster show` command to verify the overall status is "Configured" and the state is "Normal" for both clusters.

```
cluster2::> metrocluster show

Configuration: IP-fabric

Cluster                        Entry Name             State
------------------------------ ---------------------- ---------------------
Local: cluster2
                                 Configuration State  configured
                                                Mode  normal
                                 AUSO Failure Domain  auso-on-cluster-disaster
Remote: cluster1
                                 Configuration State  configured
                                                Mode  normal
                                 AUSO Failure Domain  auso-on-cluster-disaster
```

Note: You will see the "AUSO Failure Domain" for both clusters is set to "auso-on-cluster-disaster", signifying that communications have been established with the NetApp Mediator typically located at a third site. These communications allow MetroCluster to determine whether complete site switchover over is necessary.

MetroCluster IP uses the ONTAP Mediator to provide heartbeat status and to determine if automatic switchover is possible. The ONTAP Mediator does not monitor or initiate a switchover the way TieBreaker does for MetroCluster FC. The ONTAP Mediator provides iSCSI mailbox disks that allow the nodes in the clusters the ability to communicate through a third site that allows ONTAP to determine if it should automatically switchover in the event of a disaster. When you add the ONTAP Mediator to MetroCluster IP, “auso-on-cluster-disaster” is automatically enabled on both clusters.

Tip: Automatic switchover can be disabled by issuing the `metrocluster modify -auto-switchover-failure-domain auso-disabled` command.

6. Verify all four nodes are “enabled”.

```
cluster2::> metrocluster node show
DR                              Configuration  DR
Group Cluster Node              State          Mirroring Mode
----- ------- ----------------- -------------- --------- --------------------
1     cluster2
              cluster2-01       configured     enabled   normal
              cluster2-02       configured     enabled   normal
      cluster1
              cluster1-01       configured     enabled   normal
              cluster1-02       configured     enabled   normal
4 entries were displayed.
```

The figure above indicates all nodes of the MetroCluster are up and operational.

7. Verify “cluster2” is peered to “cluster1”.

```
cluster2::> cluster peer show
Peer Cluster Name         Cluster Serial Number Availability   Authentication
------------------------- --------------------- -------------- --------------
cluster1                  1-80-000011           Available      ok
```

This tells you there is an operational peer relationship between clusters. This is a requirement for MetroCluster because the configuration is replicated using the intercluster peering network.

8. Issue the `metrocluster check run` command.

```
cluster2::> metrocluster check run

The operation has been started and is running in the background. Wait for
it to complete and run "metrocluster check show" to view the results. To
check the status of the running metrocluster check operation, use the command,
"metrocluster operation history show -job-id 160"
```

:

9. Check the status of the operation using the job-id shown in the results of the previous output using the`metrocluster operation history show -job-id NNN` command (where NNN is the job ID from the previous output):

```
cluster2::> metrocluster operation history show -job-id 160
Operation                     State          Start Time       End Time
----------------------------- -------------- ---------------- ----------------
check                         in-progress    8/4/2022 01:18:11
```

Note: The job ID may be different from the one in this guide. Use the ID from the output of the `metrocluster check run` command you executed in the lab.

The `metrocluster check run` command performs analysis on the MetroCluster configuration and reports configuration errors if any. This normally takes a few minutes to complete.

10. Issue the `metrocluster check show` command to check the results of the “metrocluster check” operation:

```
cluster2::> metrocluster check show

Last Checked On: 8/4/2022 01:18:11

Component           Result
```

```
------------------- ---------
nodes               ok
lifs                ok
config-replication  ok
aggregates          ok
clusters            ok
connections         ok
volumes             ok
7 entries were displayed.
```

The preceding steps indicate a properly configured and operational MetroCluster.

Tip: You can see more details for each of the components of the check operation. For example, the `metrocluster check node show` command will provide more details on the "nodes" component.

11. It is also possible to determine the status of the ONTAP Mediator from the CLI. You can view that the mailbox disks are connected with `storage disk show -container-type mediator`.

```
cluster2::> storage disk show -container-type mediator
                     Usable           Disk    Container   Container
Disk                   Size Shelf Bay Type    Type        Name      Owner
---------------- ---------- ----- --- ------- ----------- --------- --------
NET-1.1                   -     -   0 VMDISK  mediator    -         cluster2-02
NET-1.2                   -     -   1 VMDISK  mediator    -         cluster1-02
NET-1.3                   -     -   2 VMDISK  mediator    -         cluster2-01
NET-1.4                   -     -   3 VMDISK  mediator    -         cluster1-01
4 entries were displayed.

cluster2::>
```

.

12. You can check the status of the ONTAP Mediator connection using the advanced mode of the CLI.

```
cluster2::> set advanced

Warning: These advanced commands are potentially dangerous; use them only when
         directed to do so by NetApp personnel.
Do you want to continue? {y|n}: y

cluster2::*> storage iscsi-initiator show -label mediator
                                                                       Status
Node Type Label    Target Portal      Target Name                      Admin/Op
---- ---- -------- ------------------ -------------------------------- --------
cluster2-01
     mailbox
          mediator 192.168.0.61       iqn.2012-05.local:mailbox.target.2f5b8d35-
9bc1-11ea-8e35-005056b03a56:4611eeb6-9bc1-11ea-88e6-005056b01aaf:1
                                                                         up/up
cluster2-02
     mailbox
          mediator 192.168.0.61       iqn.2012-05.local:mailbox.target.2f5b8d35-
9bc1-11ea-8e35-005056b03a56:4611eeb6-9bc1-11ea-88e6-005056b01aaf:1
                                                                         up/up
2 entries were displayed.

cluster2::*>
```

This activity provided a more detailed review of MetroCluster configuration and status using the CLI. In the next activity you will simulate a site failure and recover from the disaster.

2.2 Unplanned Switchover: High Availability for Business Continuity

In this activity you act as the storage and virtualization administrator. The MetroCluster and VMware solution provide business continuity between two data centers on the same campus. You simulate a "site failure" on Site A (cluster1), perform a switchover, and monitor the switchover of Site A (cluster1) operations to Site B (cluster2). With MetroCluster IP it's possible to do either a manual or automatic switchover in the event of a failure that affects a site's ability to serve data. In this lab, there are limitations in creating a site failure. This activity does a shutdown of one site and uses a manual switchover after cluster1 is shutdown gracefully.

This activity includes the following tasks:

Tasks
Shut down the devices at Site A to create a “Site Failure”.
Initiate a switchover event to move Site A storage services to Site B.
Verify successful switchover of operations.

1. To create the site failure, open a new browser tab and click the **Power Controls** bookmark.

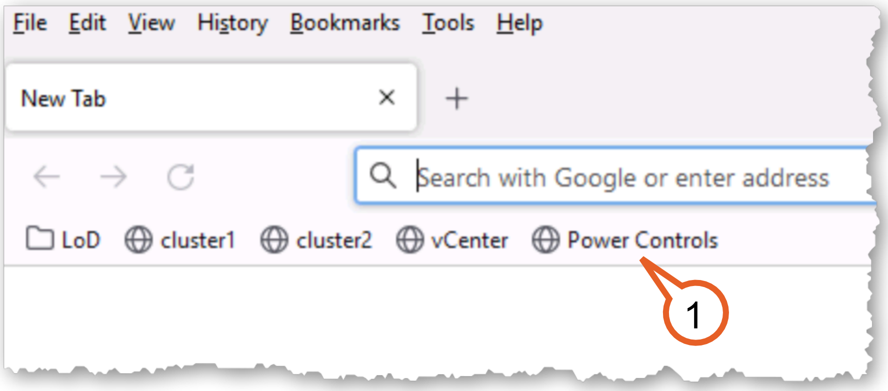

Figure 2-22:

2. You will see a list of the systems in this lab grouped by Site. You will be inducing a site failure for “Site A”. Click the yellow **Shutdown Guest** button for the Full Site in the row in “Site A”.

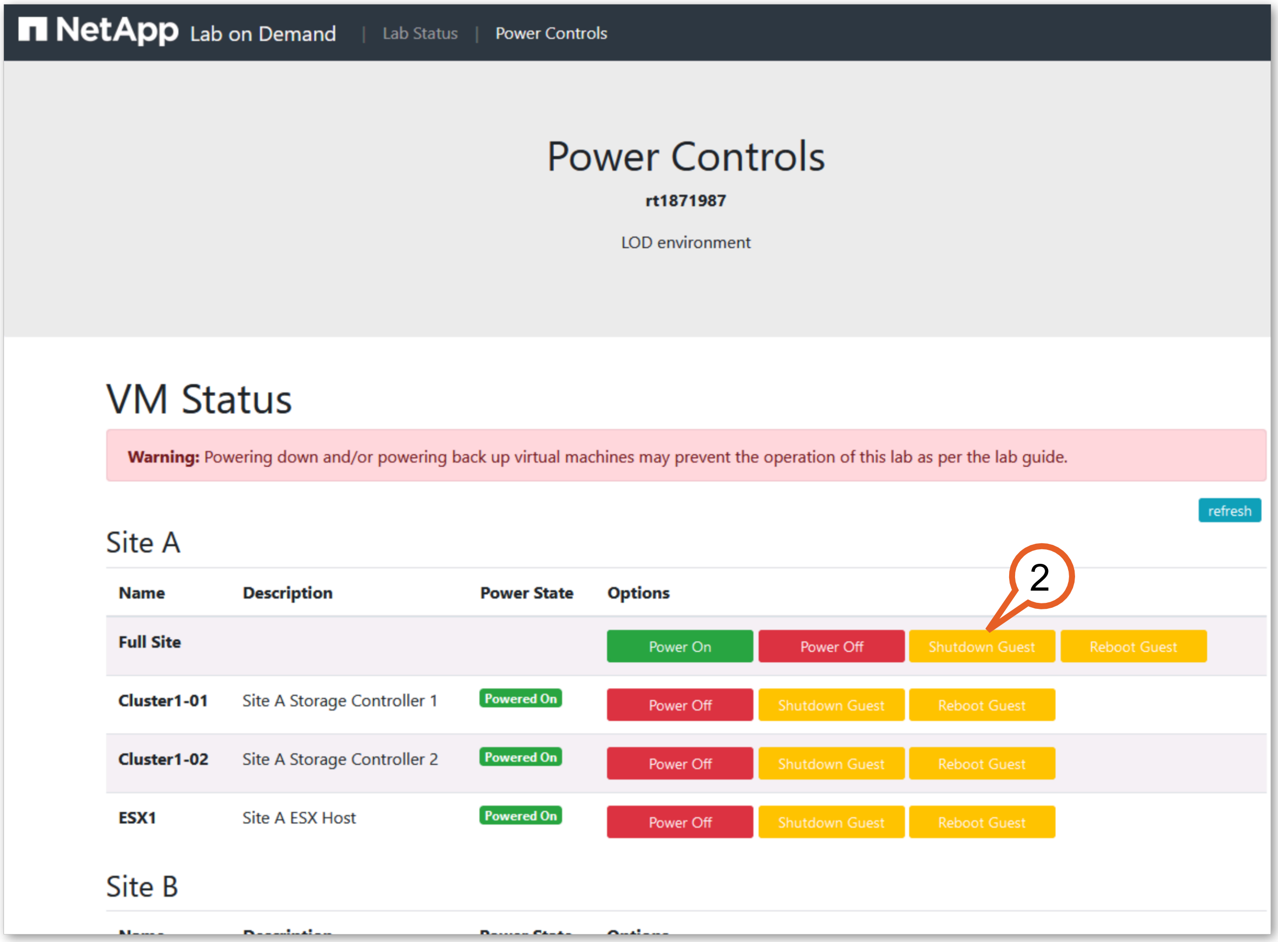

Figure 2-23:

Leave the “Power Controls” browser tab open. You will return to it later.

3. If the “cluster2” System Manager is not already open in the browser, click **cluster2** bookmark and log in with the user name **`admin`** and the password **`Netapp1!`**.

4. View the Dashboard to confirm the MetroCluster status panel changes to show errors.

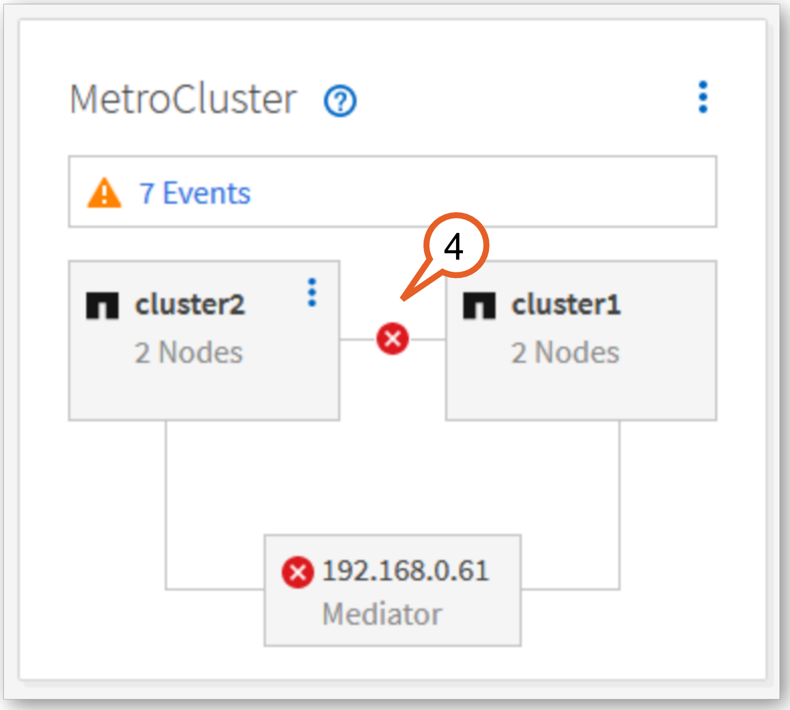

Figure 2-24:

5. Start the switchover process by clicking the menu on “cluster2”.

6. Select **Switch over remote data services to the local site**.

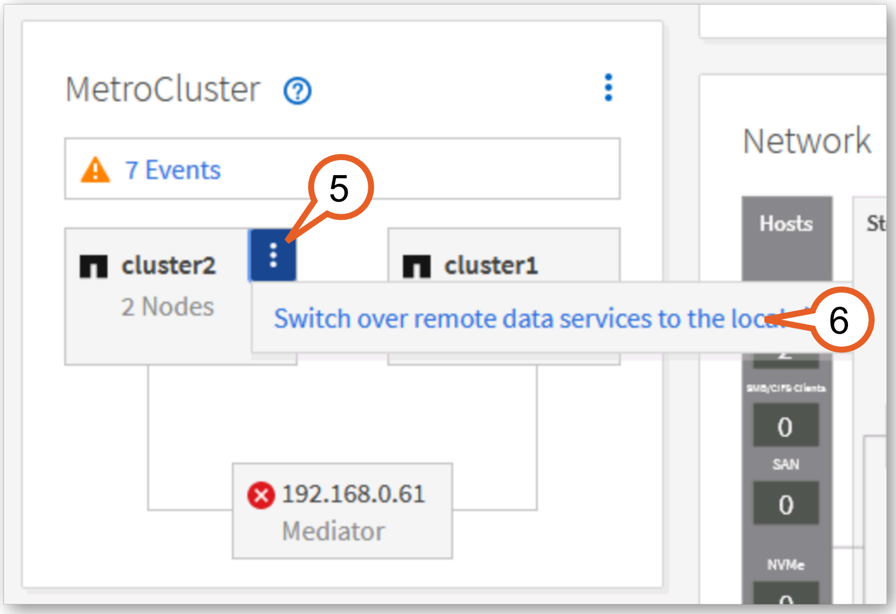

Figure 2-25:

7. A dialog confirms the switchover. Click **OK**.

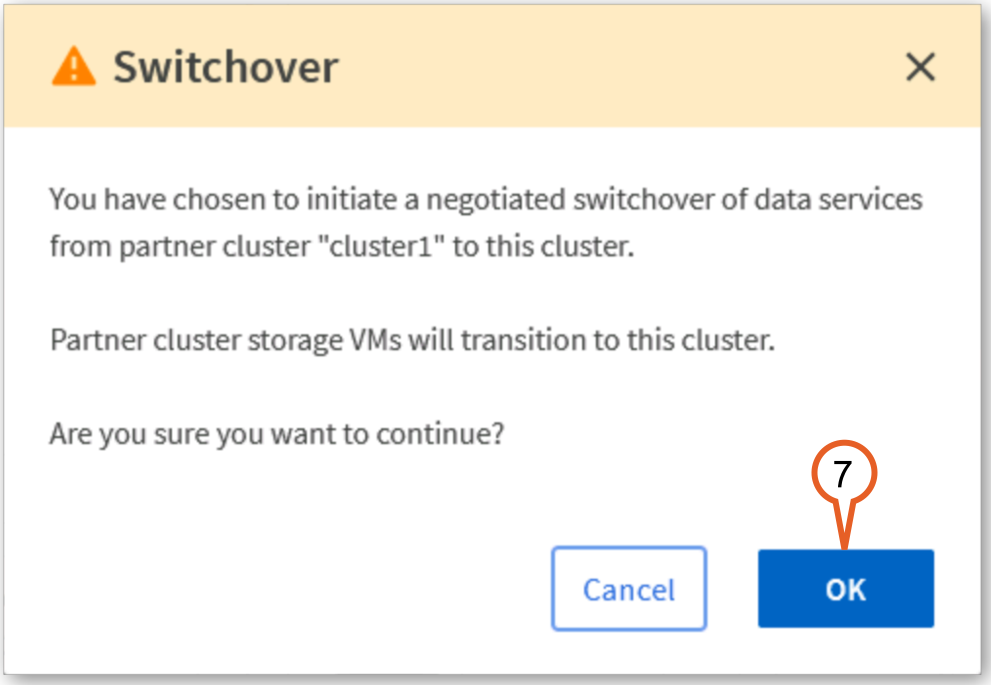

Figure 2-26:

ONTAP checks to see if it is possible to perform a negotiated switchover, then informs you it is not possible.

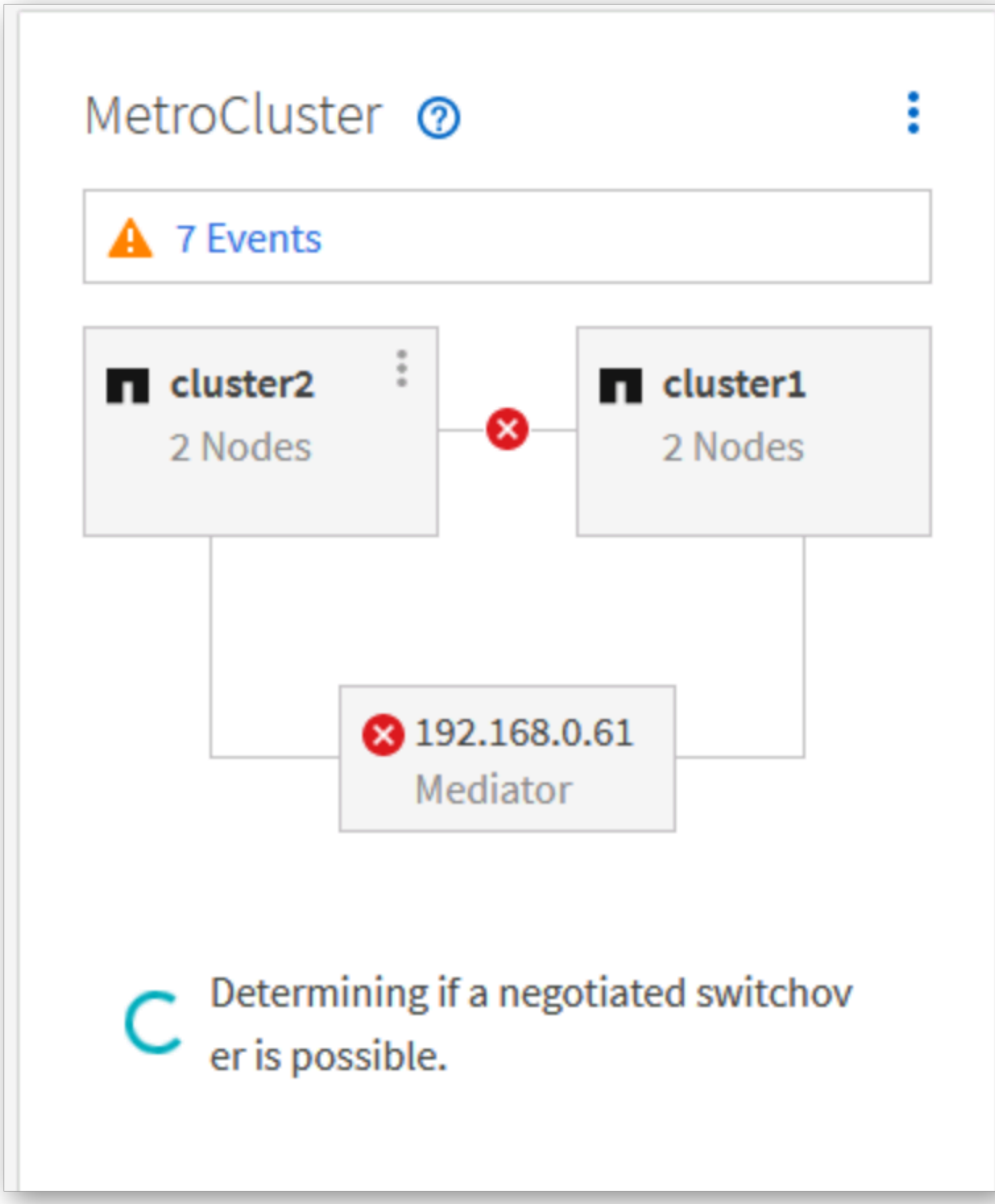

Figure 2-27:

Tip: MetroCluster uses the terms planned and unplanned to differentiate between doing a switchover of a system that is operating normally compared to a system that has a failure. This lab uses the manual switchover process. There is also an automatic unplanned switchover that happens when there is a site failure, such as if both nodes in a cluster go down.

8. The MetroCluster Status panel changes to inform you that a negotiated switchover was not initiated. Click **View details**.

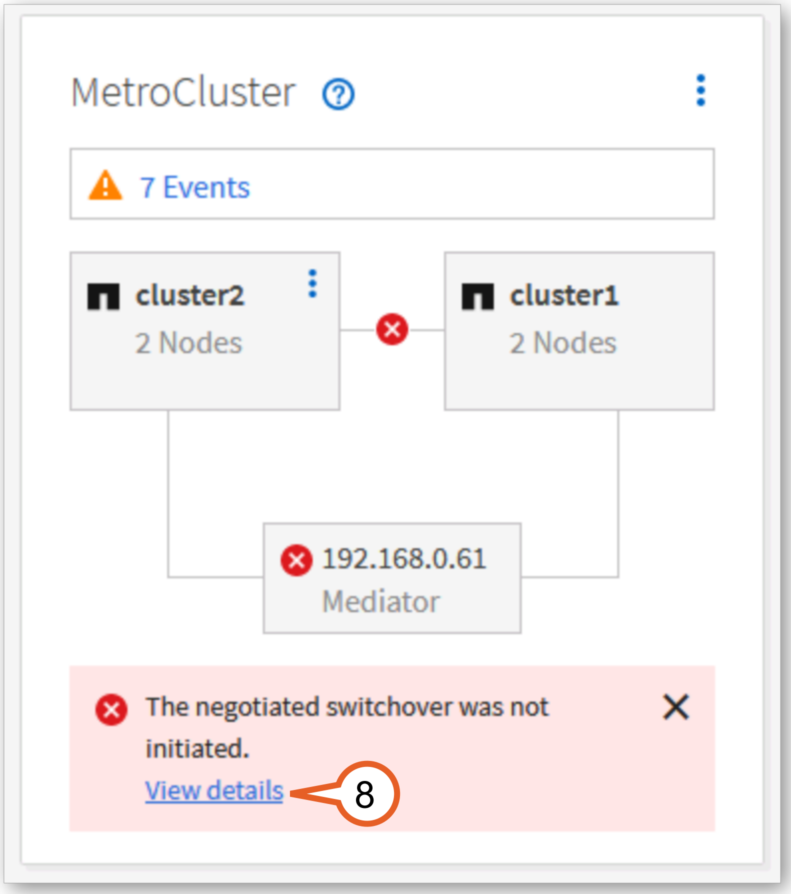

Figure 2-28:

9. A dialog details that the partner cluster (cluster1) cannot be reached. This is because it was shut down. Click **Initiate Unplanned Switchover**.

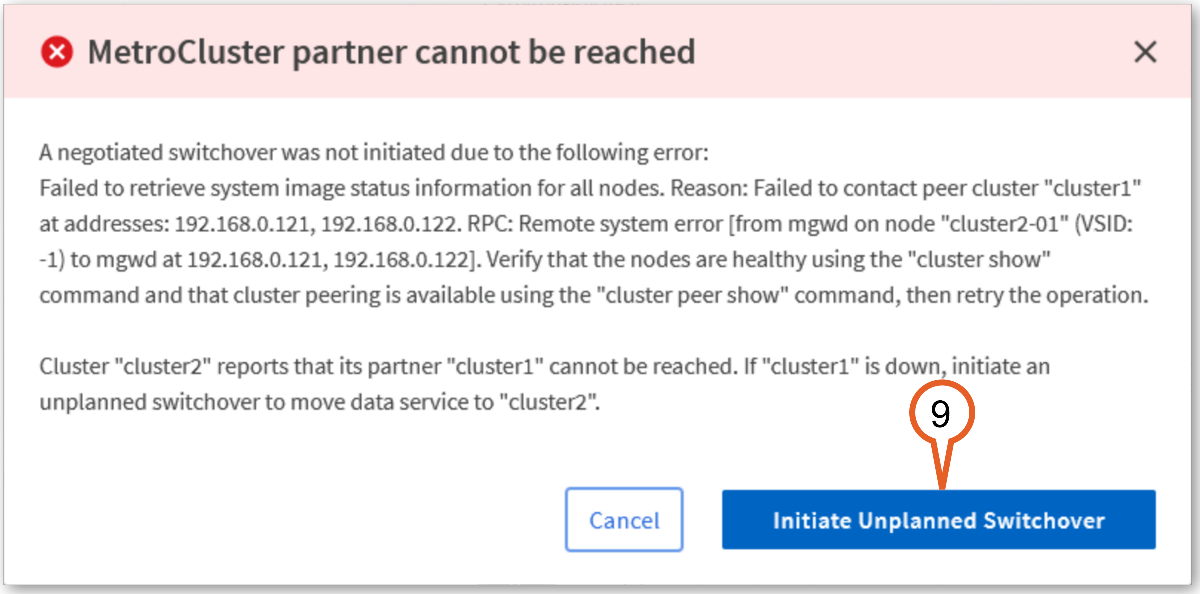

Figure 2-29:

The MetroCluster status panel changes to show it is performing an unplanned switchover.

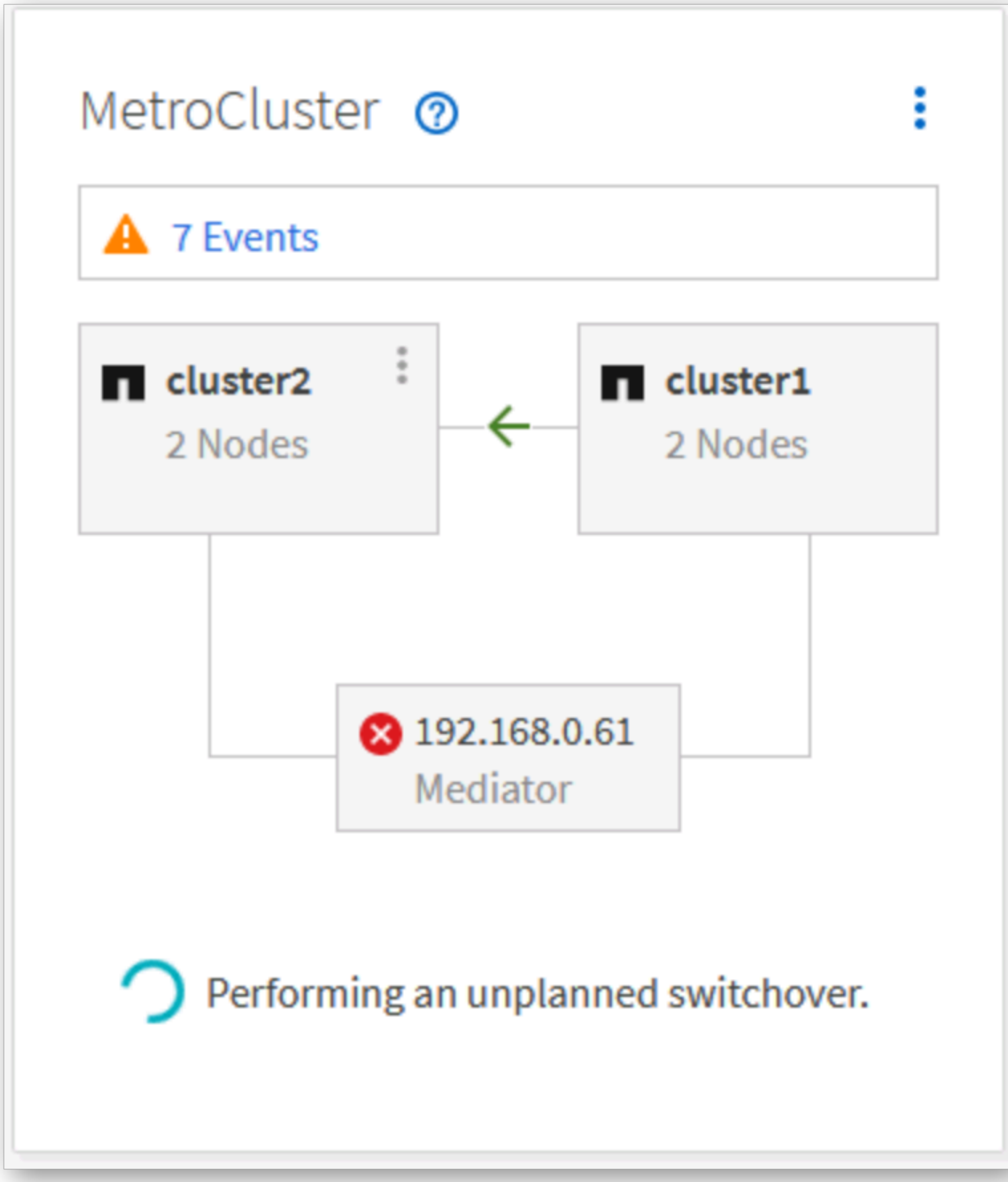
MetroCluster
7 Events
cluster2
2 Nodes
cluster1
2 Nodes
192.168.0.61
Mediator
Performing an unplanned switchover.

Figure 2-30:

10. When the switchover completes the icon changes from red to orange. Hover the mouse over the icon to see more information.

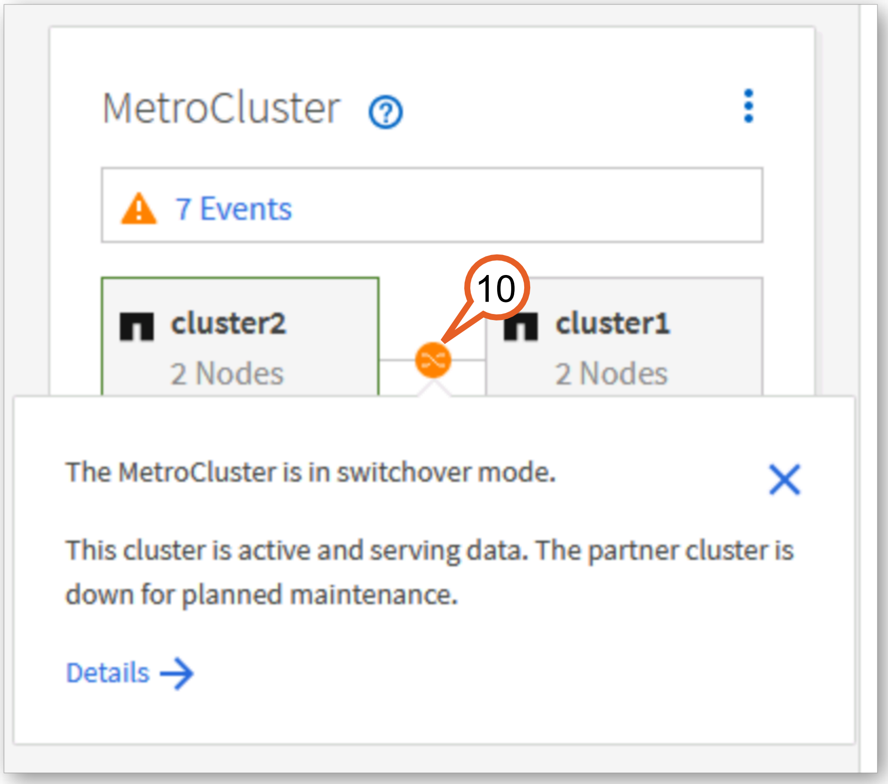

Figure 2-31:

11. Return to the vCenter tab to view the status. After several minutes “vm1” is restarted on “esx2”.

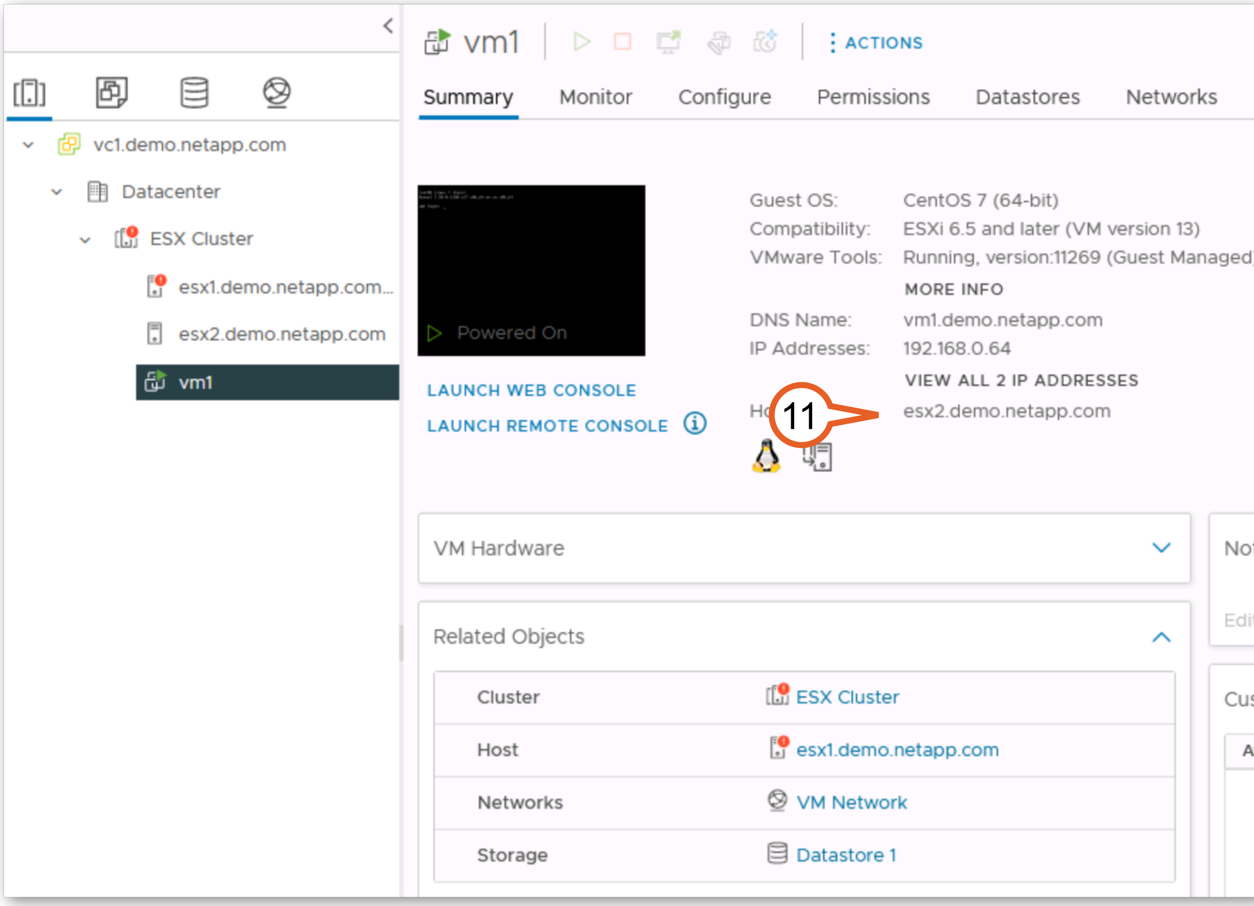

Figure 2-32:

Important: If the VM shows it is powered off or disconnected, do not attempt to power it on. Once the surviving hyper visor sees the storage online it will complete the host affinity failure action. This can take several minutes after the switchover is complete. It may be necessary to refresh the view.

12. Return to the tab for “cluster2 System Manager”. Click **Storage** to expand the menu.
13. Click **Storage VMs**.

14. Verify that both storage VMs are in the running state. When the switchover occurred, the “svm1_mc” storage VM was brought online on cluster2 and now hosts the data store that “vm1” is located in allowing “esx2” to bring the VM online.

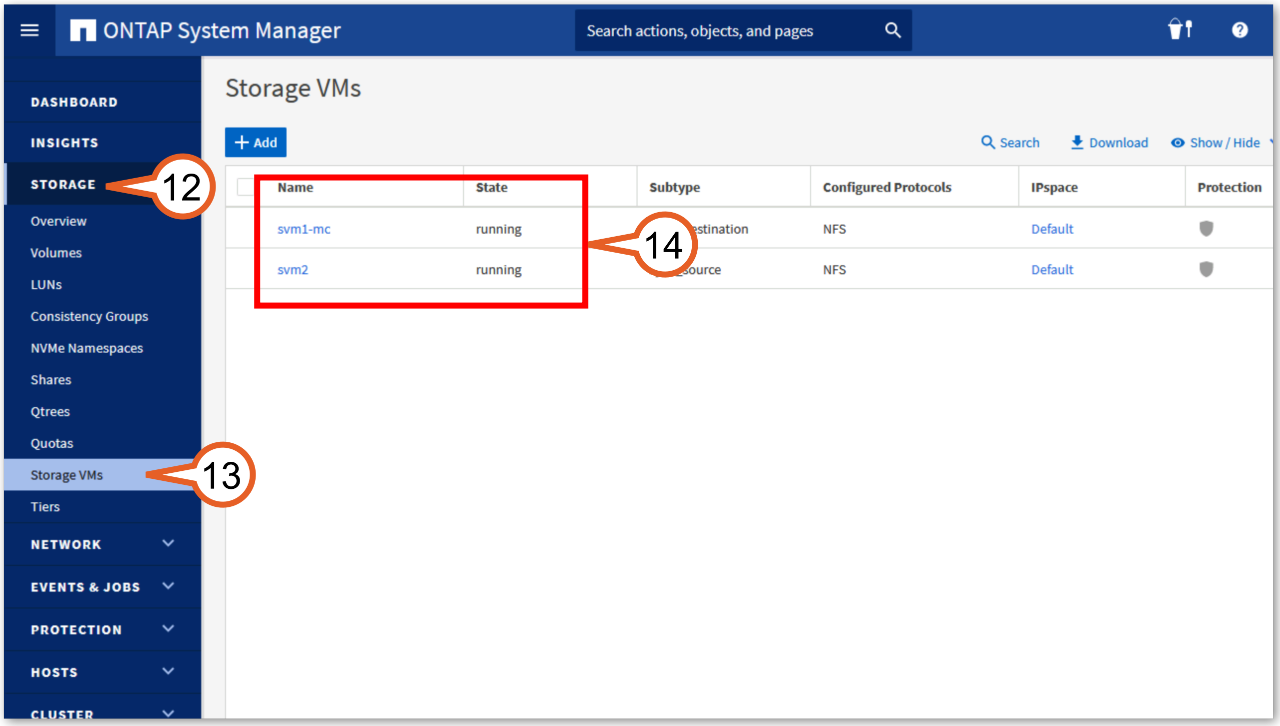

Figure 2-33:

In this activity you generated the equivalent of a site failure, verified the failure from multiple perspectives (vSphere and Storage), and monitored switchover of operations from Site A to Site B. Additionally, you saw how the vSphere HA cluster also moved the VM to the ESX2 server at Site B. In the next activity you will bring the failed site back online and recover from the failure.

2.3 MetroCluster Switchback

Following an unplanned or planned switchover event, you must perform certain steps to restore the MetroCluster to full health. This activity walks you through the following tasks:

Tasks
Power on the cluster at the failed-over site.
Prepare for and perform switchback.
Migrate the VM back to its normal location.

Note: One of the distinct differences between the IP configuration of MetroCluster and the Fibre Channel (FC) version is that in the FC version aggregate healing takes place before the controllers are brought back online. But because in the IP configuration of MetroCluster access to the disks is through the controllers, the controllers must be brought online before healing can take place. MetroCluster IP also performs auto healing when the cluster is back online.

1. Return to the “Power controls” browser tab. The display for “Site A” may or may not show all resources for “Site A” with the state of “powered off”.
2. If the resources do not all show the “powered off” state, click the **Refresh** button to update the status.
3. Click the green **Power On** button for the Full Site under “Site A”.

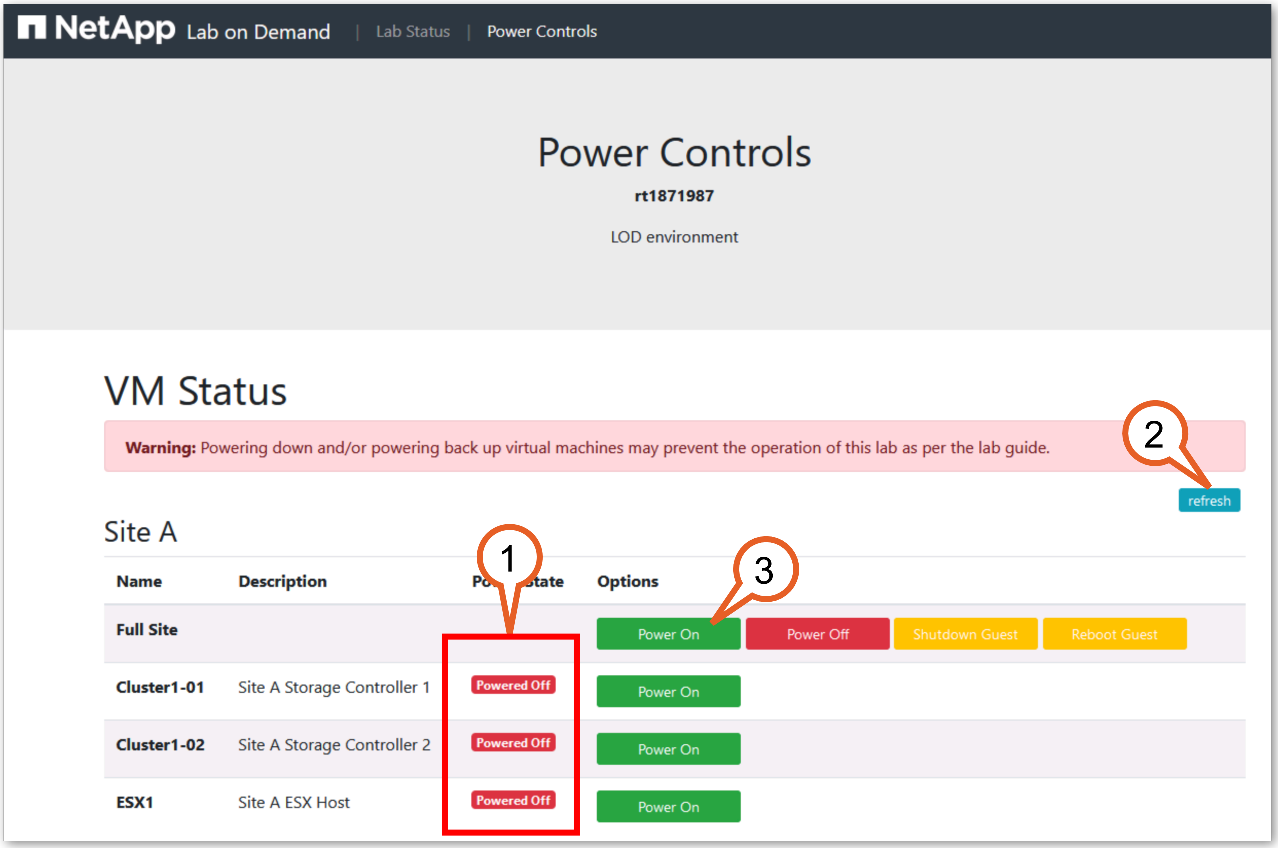

Figure 2-34:

4. Right-click **PuTTY** on the taskbar.
5. Select **PuTTY** to open a new session.

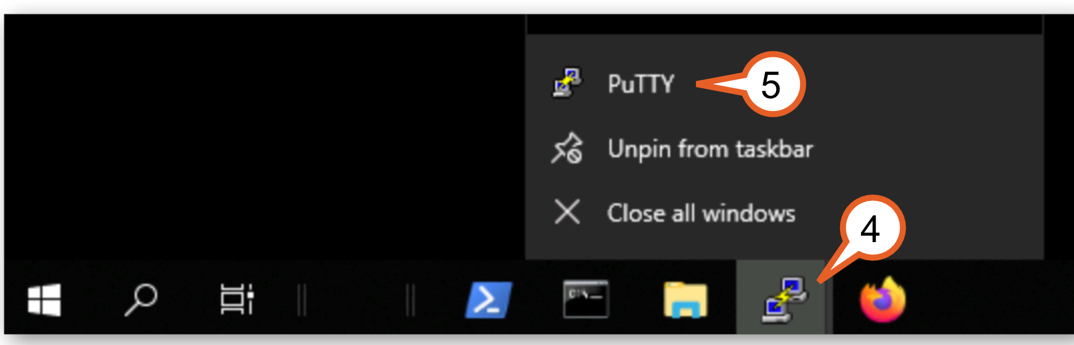

Figure 2-35:

6. Double-click **cluster1-01**.

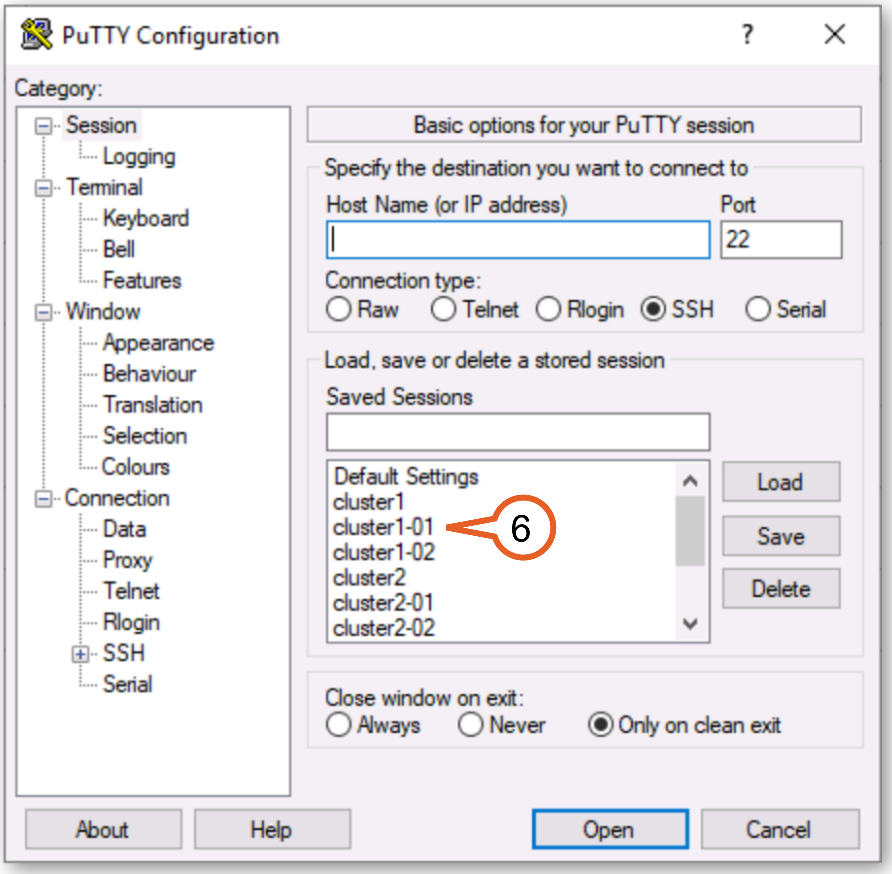

Figure 2-36:

7. You should see messages for node cluster1-01 booting ONTAP. Do not type anything in the window. It will take several minutes for the boot process to complete.

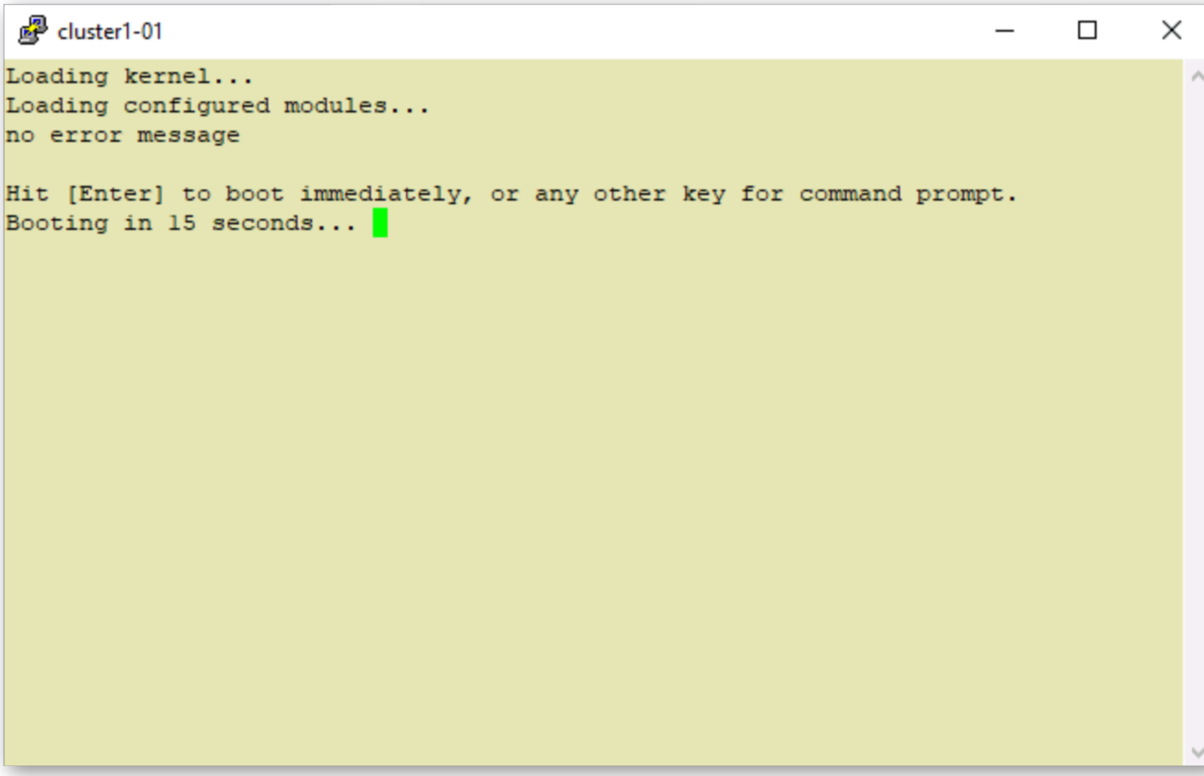

Figure 2-37:

When you see the login prompt, the boot process is complete.

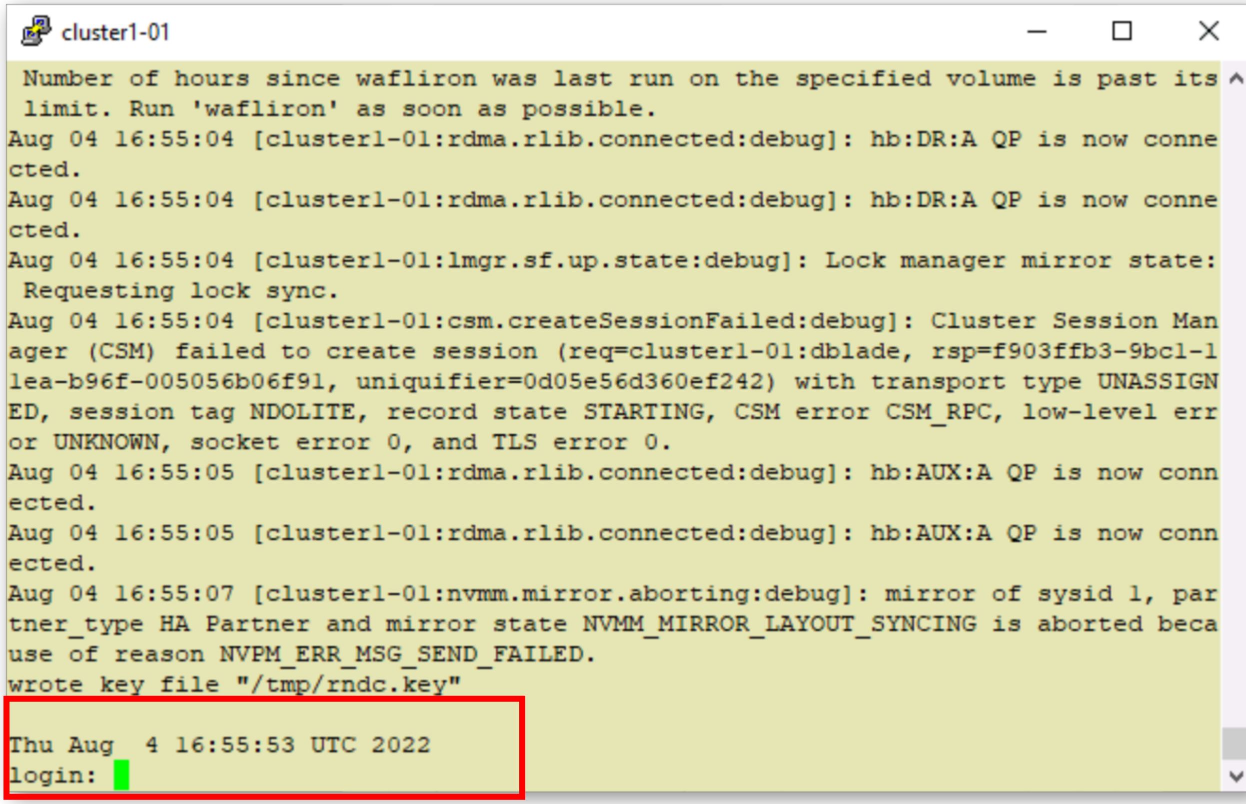
cluster1-01
Number of hours since wafliron was last run on the specified volume is past its
limit. Run 'wafliron' as soon as possible.
Aug 04 16:55:04 [cluster1-01:rdma.rlib.connected:debug]: hb:DR:A QP is now conne
cted.
Aug 04 16:55:04 [cluster1-01:rdma.rlib.connected:debug]: hb:DR:A QP is now conne
cted.
Aug 04 16:55:04 [cluster1-01:lmgr.sf.up.state:debug]: Lock manager mirror state:
Requesting lock sync.
Aug 04 16:55:04 [cluster1-01:csm.createSessionFailed:debug]: Cluster Session Man
ager (CSM) failed to create session (req=cluster1-01:dblade, rsp=f903ffb3-9bc1-1
1ea-b96f-005056b06f91, uniquifier=0d05e56d360ef242) with transport type UNASSIGN
ED, session tag NDOLITE, record state STARTING, CSM error CSM_RPC, low-level err
or UNKNOWN, socket error 0, and TLS error 0.
Aug 04 16:55:05 [cluster1-01:rdma.rlib.connected:debug]: hb:AUX:A QP is now conn
ected.
Aug 04 16:55:05 [cluster1-01:rdma.rlib.connected:debug]: hb:AUX:A QP is now conn
ected.
Aug 04 16:55:07 [cluster1-01:nvmm.mirror.aborting:debug]: mirror of sysid 1, par
tner_type HA Partner and mirror state NVMM_MIRROR_LAYOUT_SYNCING is aborted beca
use of reason NVPM_ERR_MSG_SEND_FAILED.
wrote key file "/tmp/rndc.key"
Thu Aug 4 16:55:53 UTC 2022
login:

Figure 2-38:

8. Return to the browser window for cluster2 System Manager. On the dashboard, view the MetroCluster status panel. Cluster1 should be online with a green icon showing it is ready for switchback.

Important: If the green icon does not appear, refresh your browser window and wait. It can take several minutes for the healing to complete and the status to update after both nodes have completed the boot process.

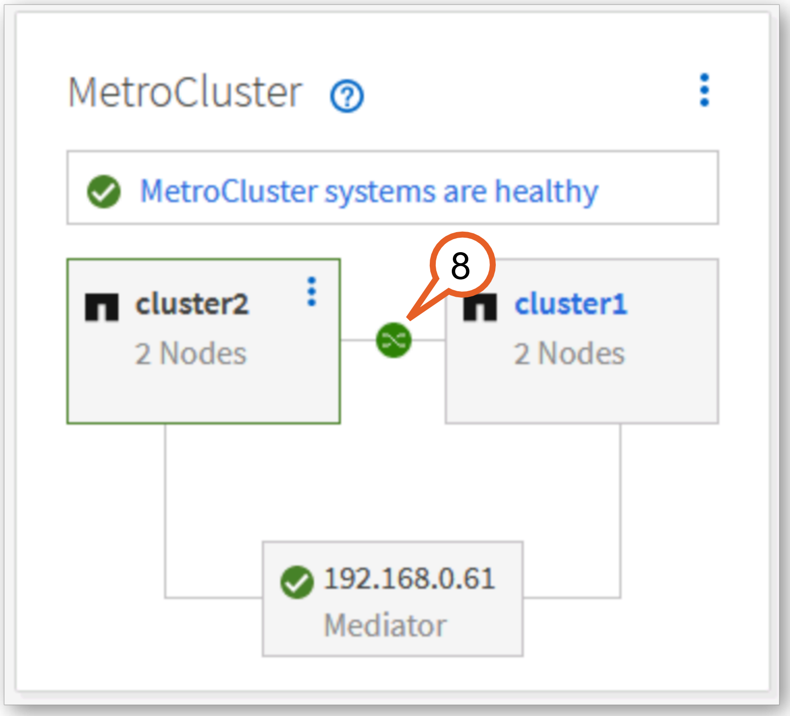

Figure 2-39:

9. Hover the mouse over the icon to see more information.

10. Click **Details**.

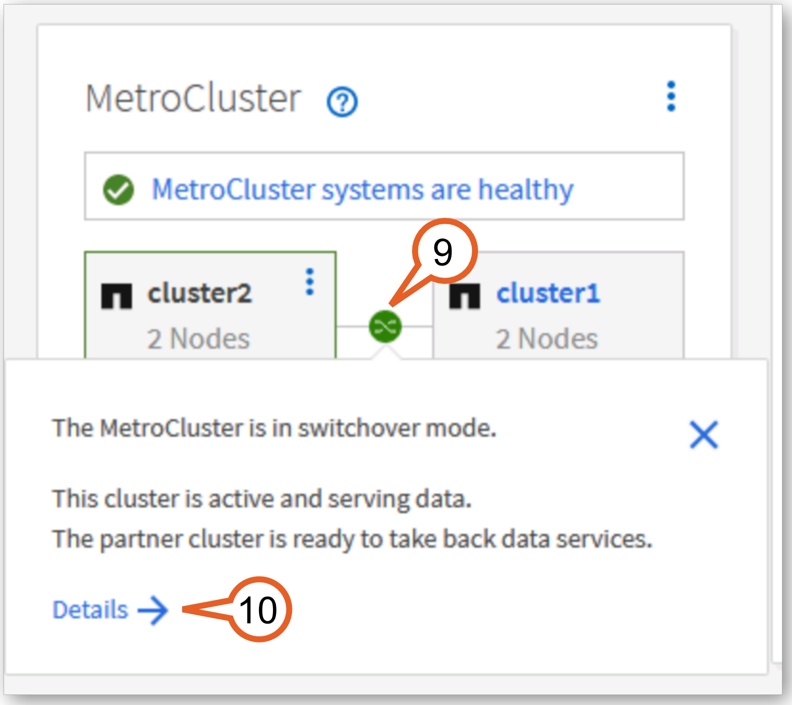

Figure 2-40:

11. **Important:** It can take several minutes for the auto healing to complete and the status to update after both nodes have completed the boot process.
12. View the latest events at the bottom of the page. It should show that auto healing has completed. (scroll down if necessary)

13. Click **Close**.

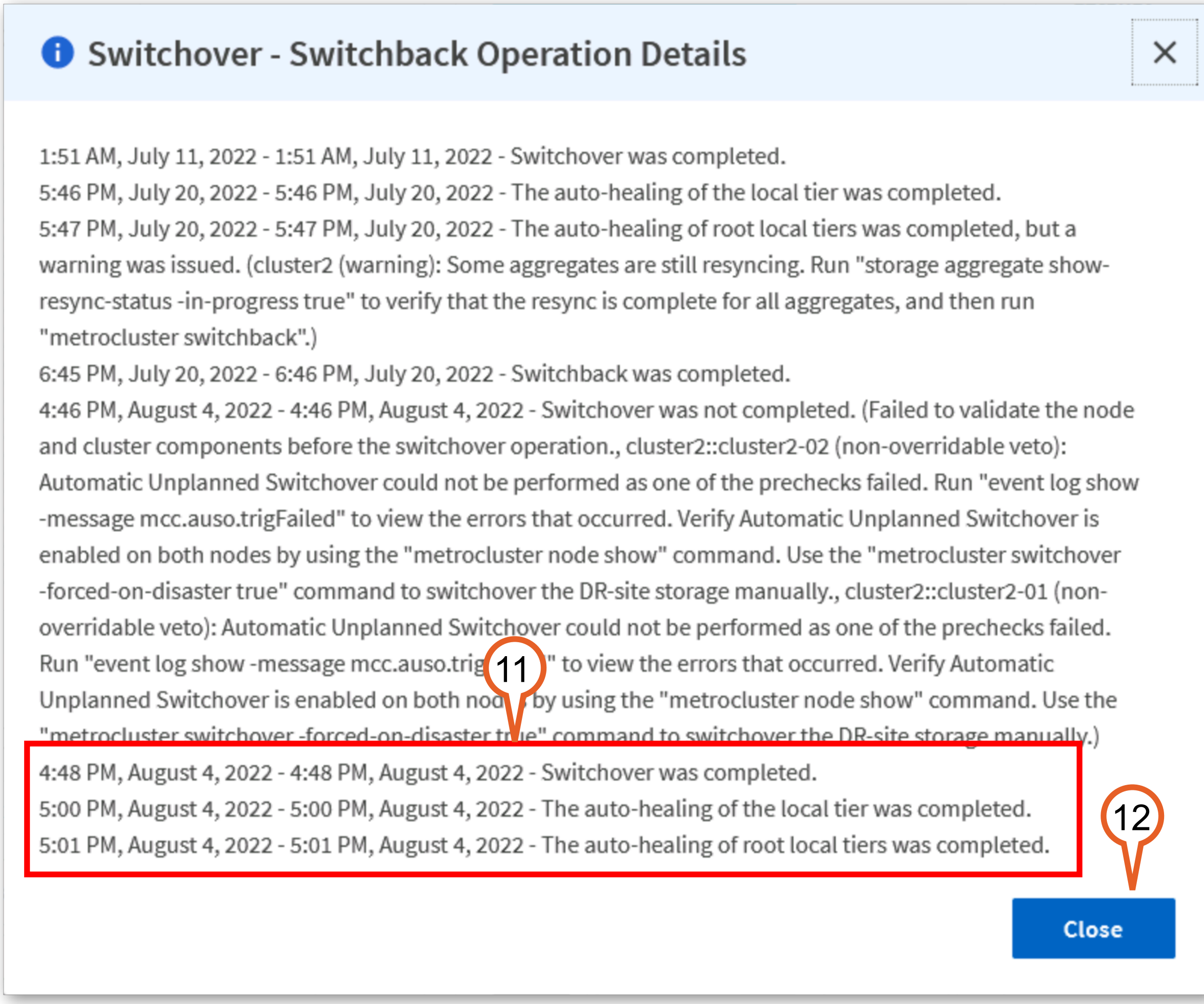

Figure 2-41:

14. Click the **menu** for "cluster2" in the MetroCluster panel.

15. Select **Switch back remote data services to the remote site**.

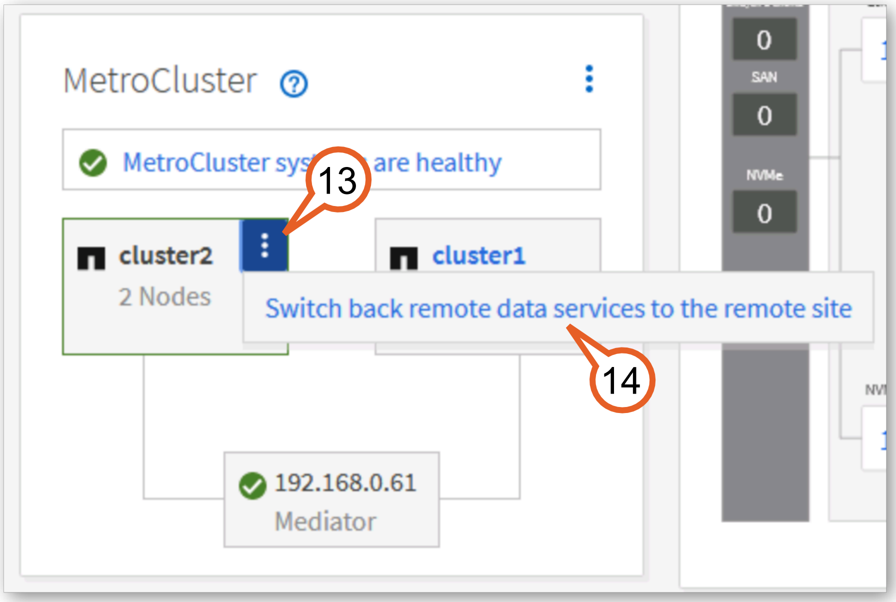

Figure 2-42:

16. In the dialog click **OK**.

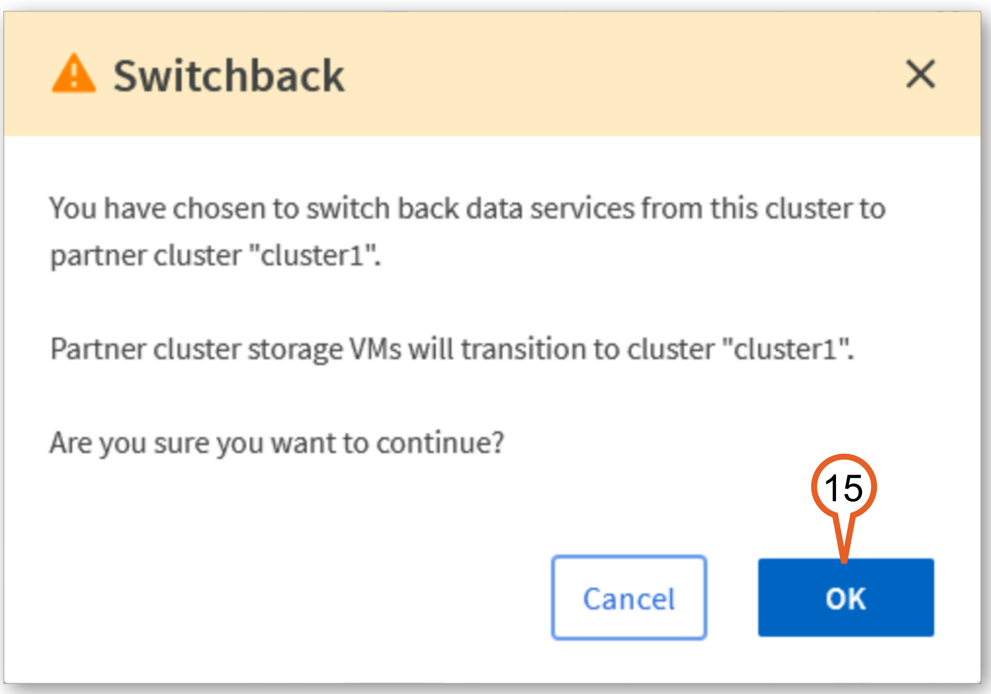

Figure 2-43:

17. Hover the mouse over the icon between cluster1 and cluster2. It indicates that services are being switched back.
18. Click the **X** to close the dialog.

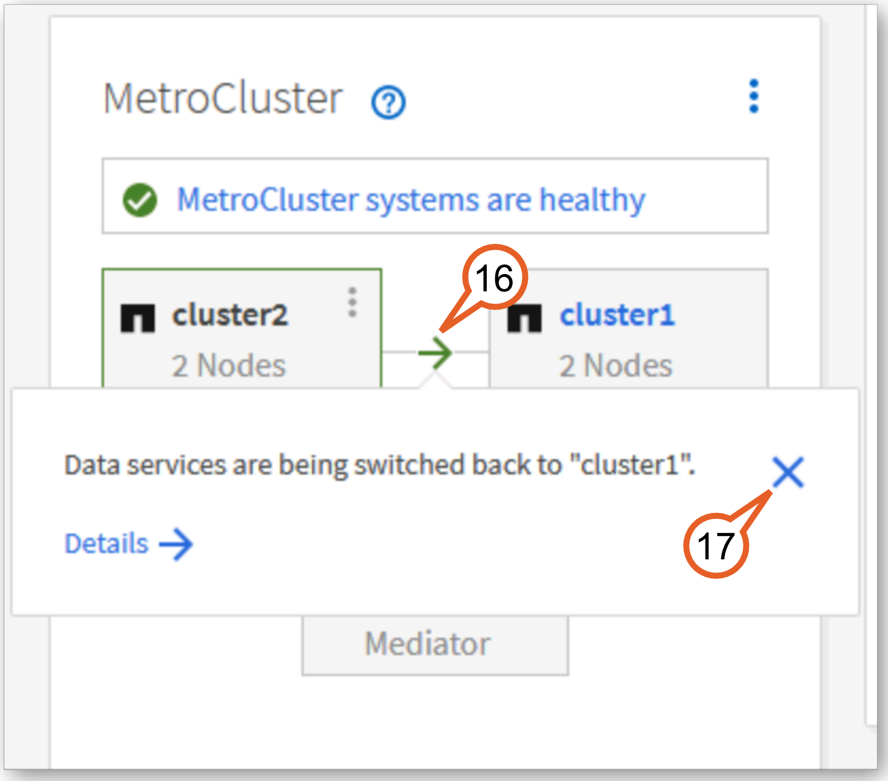

Figure 2-44:

19. Notice that the MetroCluster panel indicates it is “Performing a switchback”.

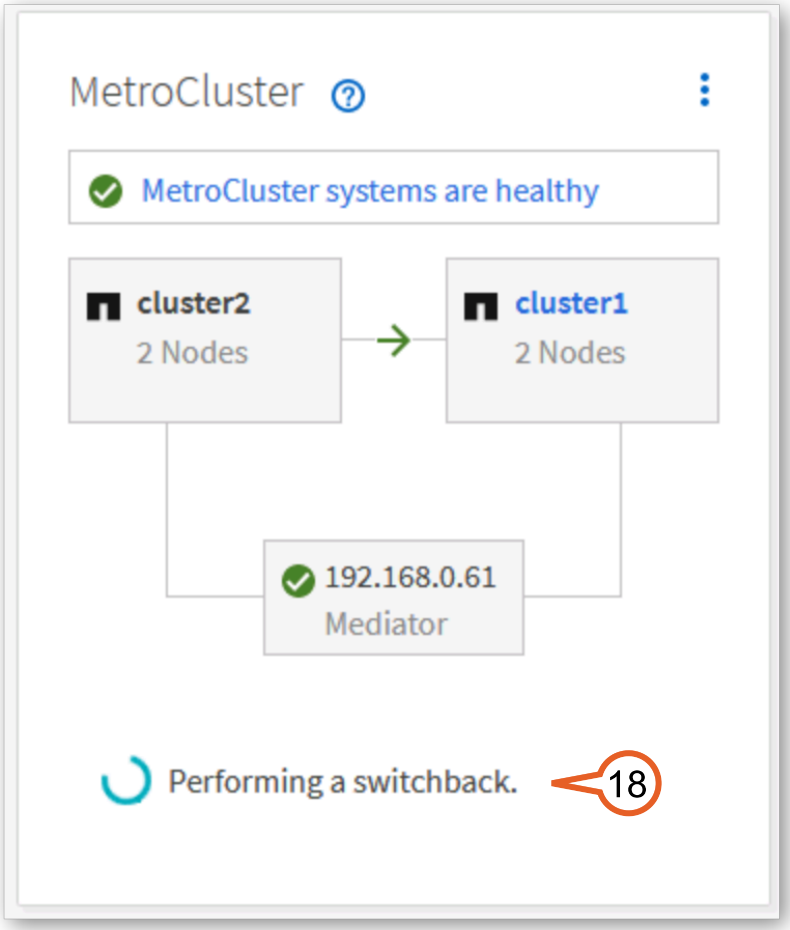

Figure 2-45:

20. Within a few minutes the switchback completes and the MetroCluster panel on the Dashboard returns to a normal status with a green check mark.

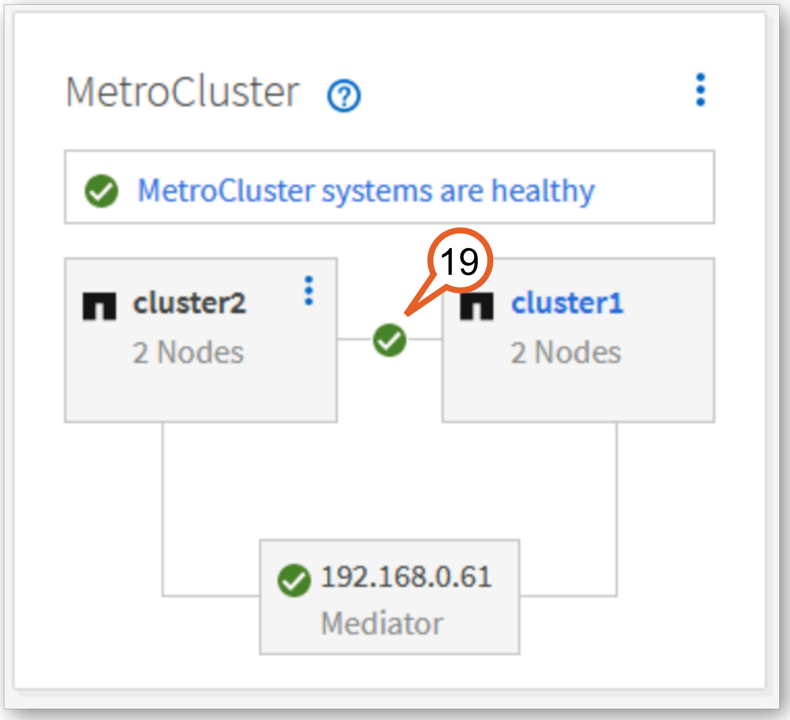

Figure 2-46:

21. Return to the browser tab for vSphere Client
22. Click **refresh** if necessary.

23. Note that “esx1” and “esx2” both appear without errors.

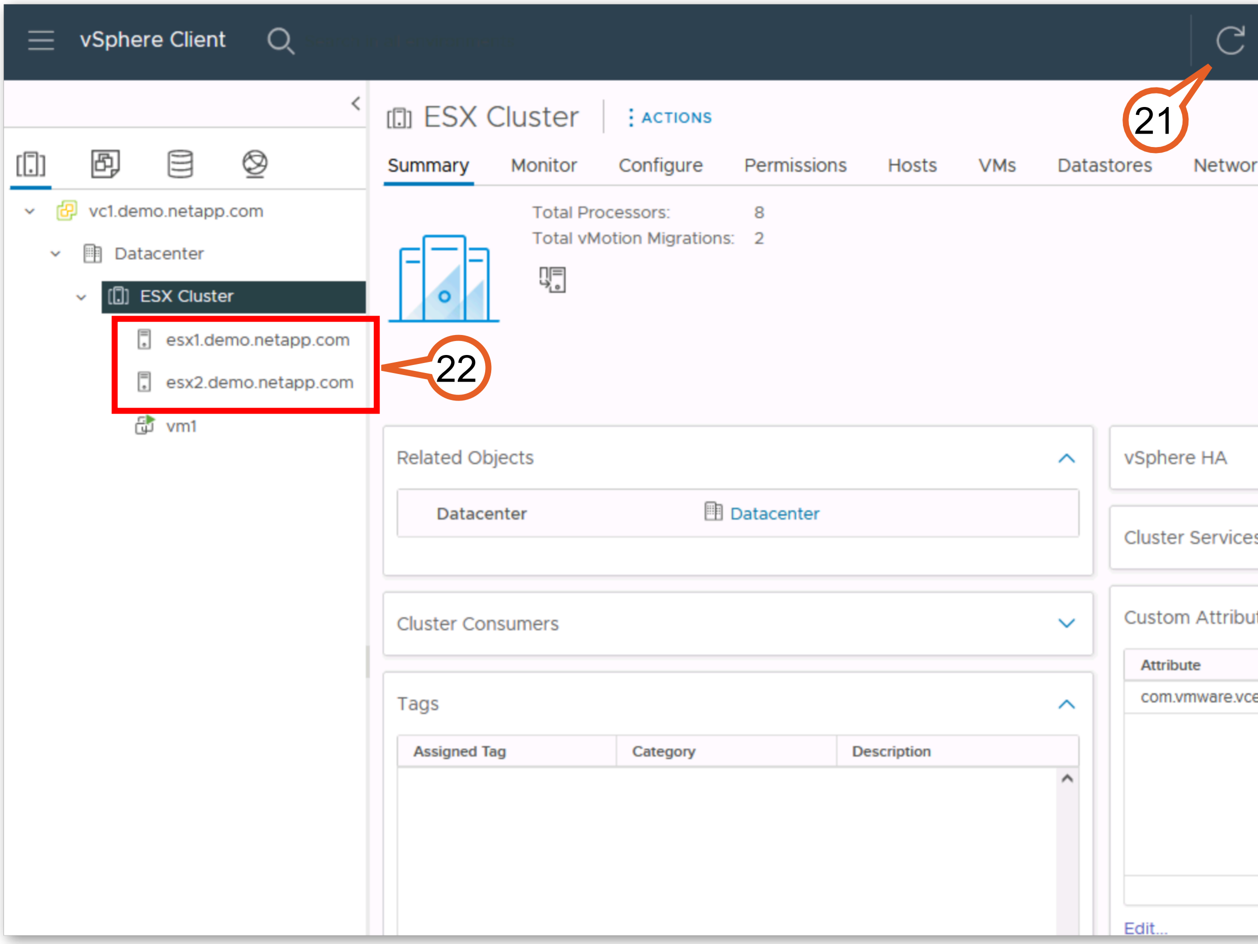

Figure 2-47:

24. Right-click **vm1**.

25. Select **Migrate**.

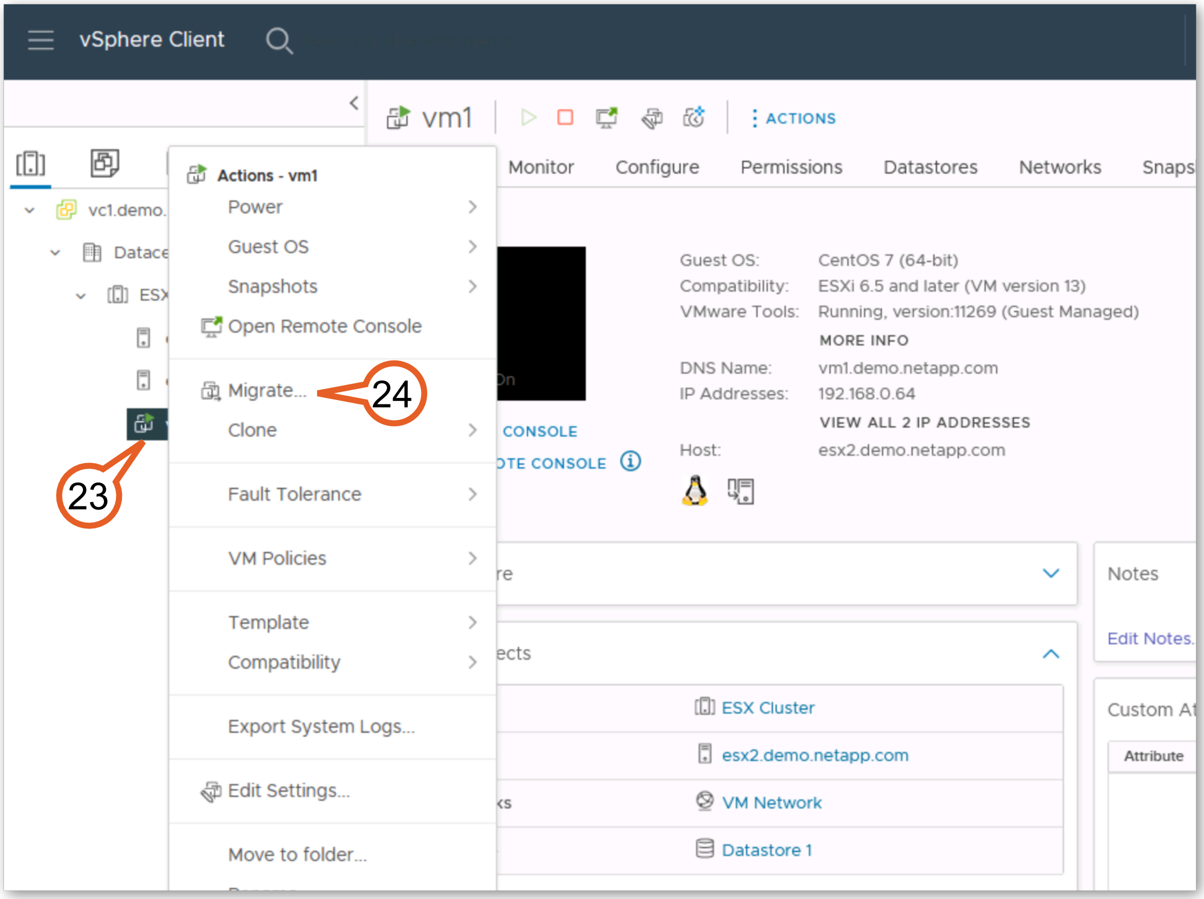

Figure 2-48:

26. Select **Change compute resource only**.

27. Click **Next**.

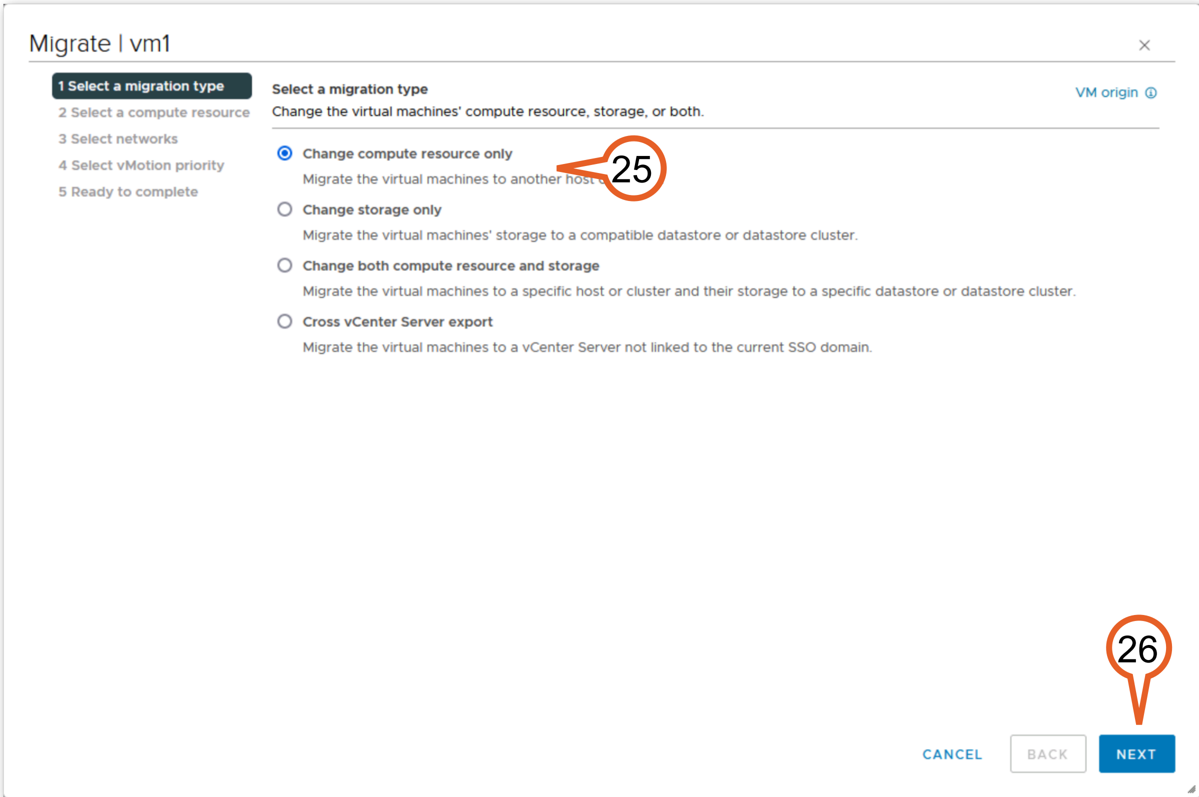

Figure 2-49:

28. Select **esx1**.

29. Click **Next**.

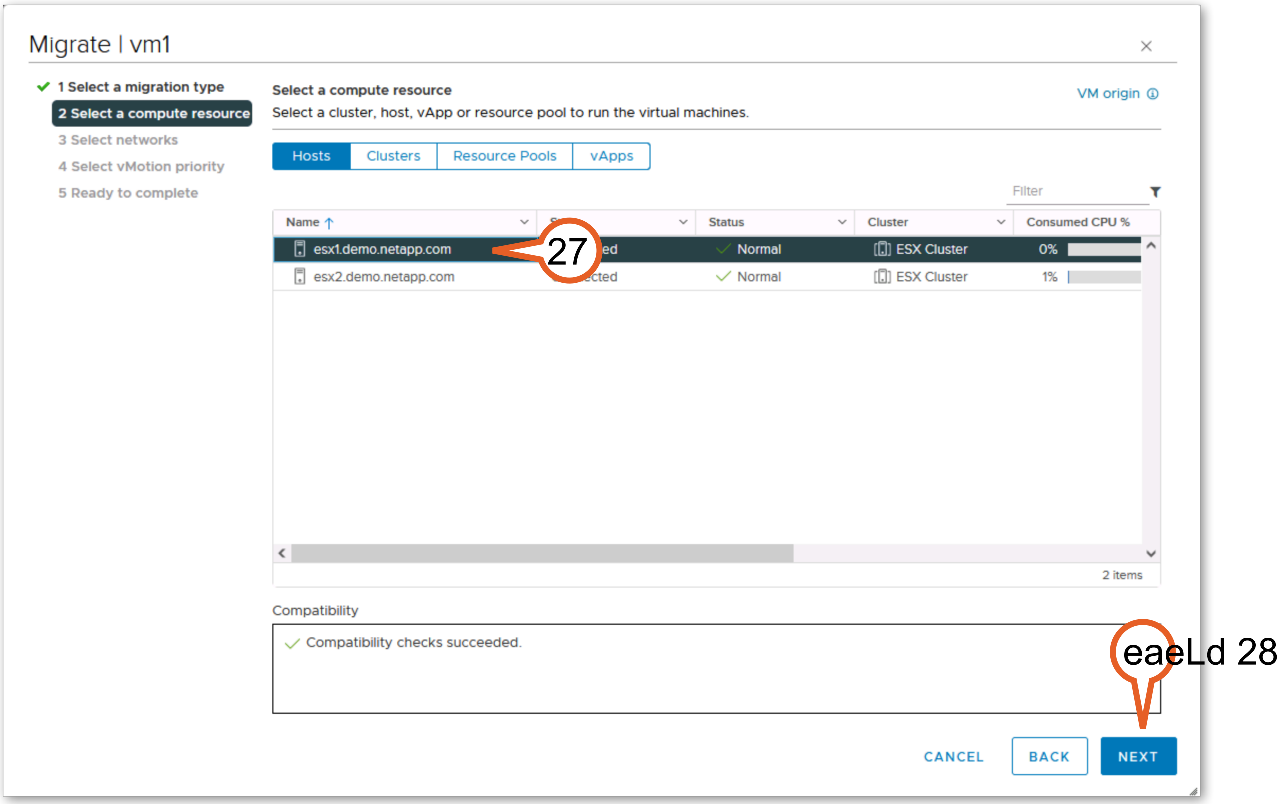

Figure 2-50:

30. Click **Next**.

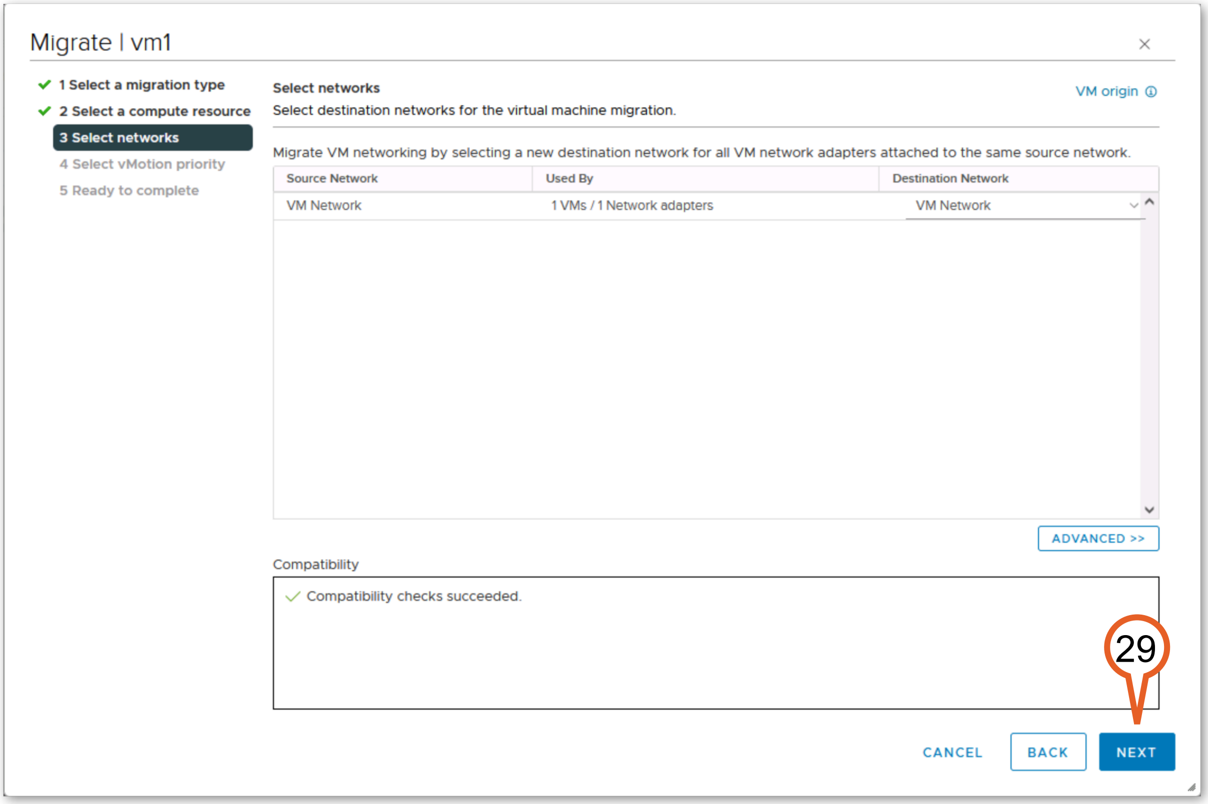

Figure 2-51:

31. Click **Next**.

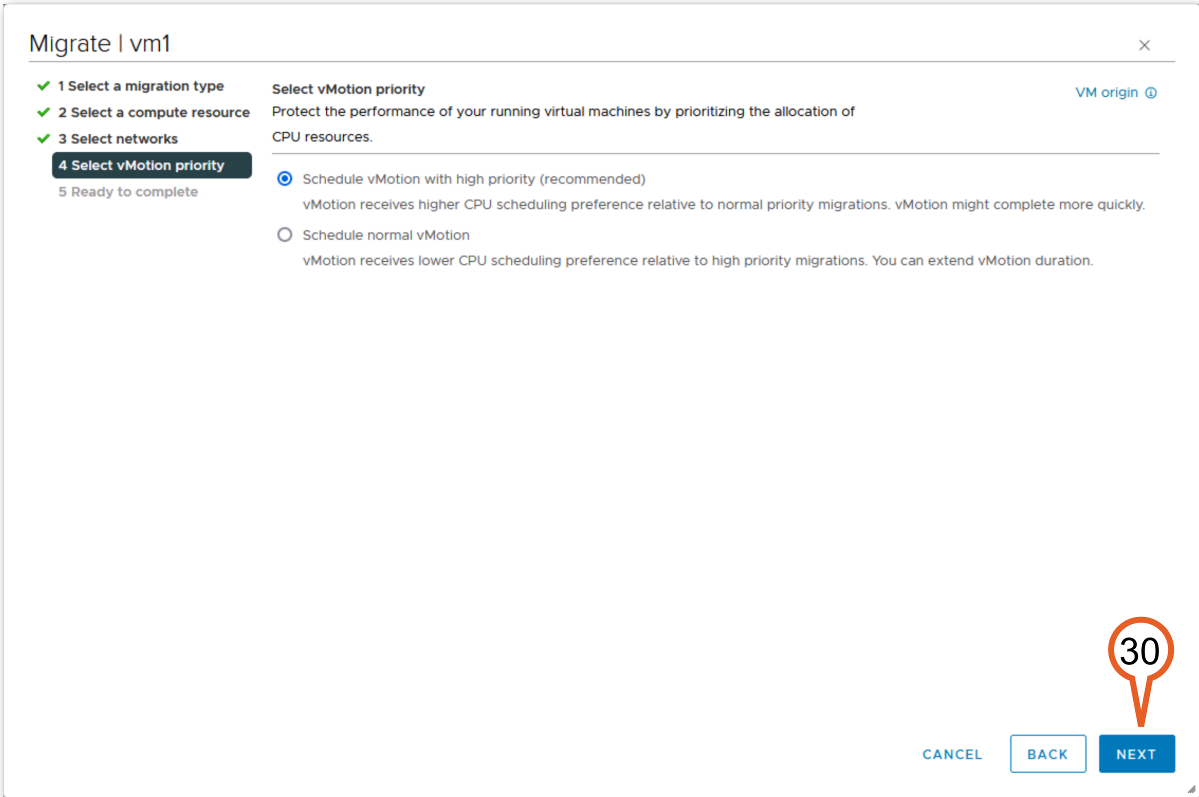

Figure 2-52:

32. Click **Finish** to begin the migration.

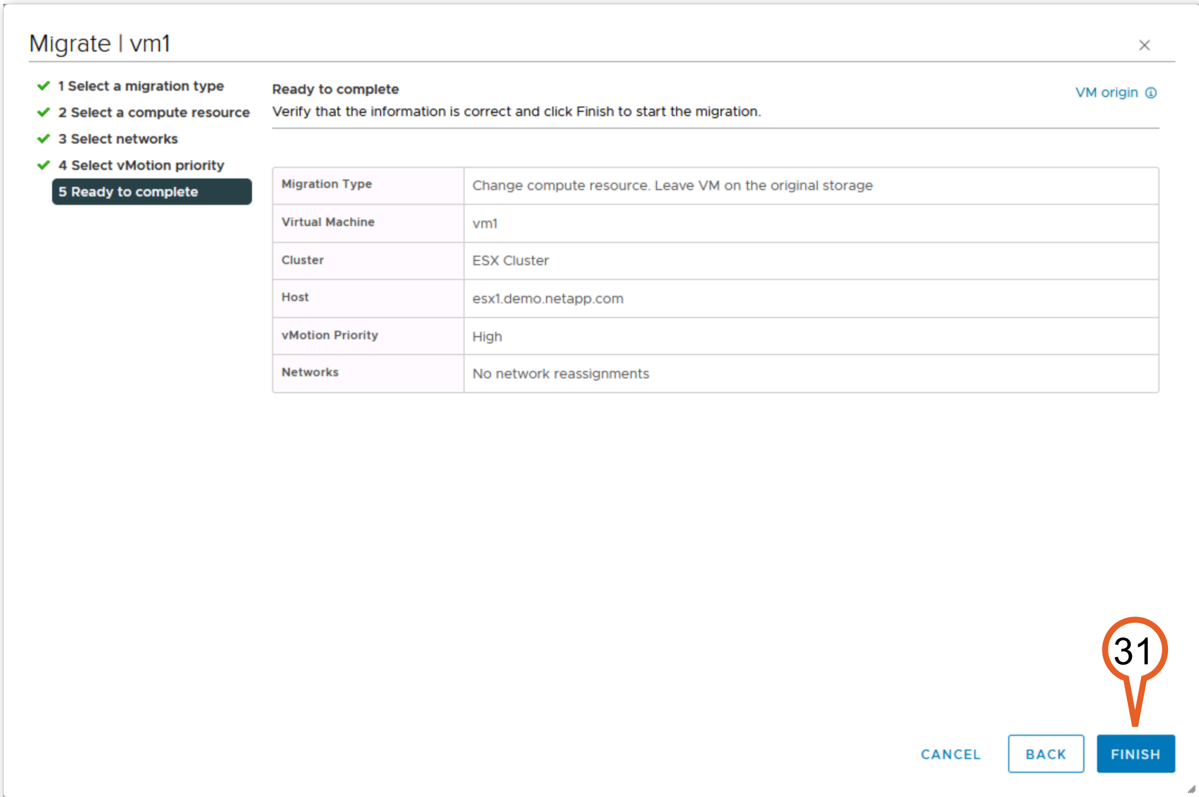

Figure 2-53:

33. Select **vm1**.

34. Note that after a moment or two it is once again running on “esx1.”

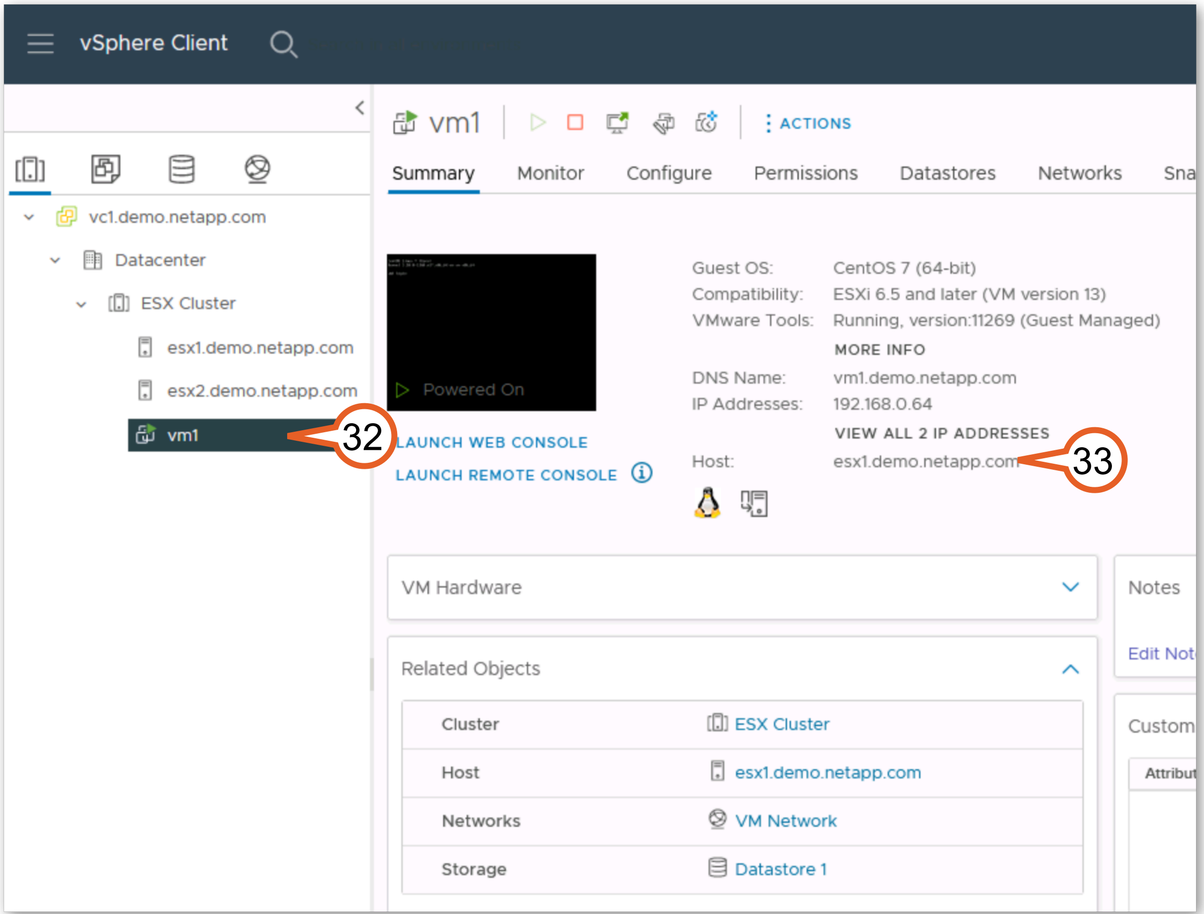

Figure 2-54:

Switching business operations over to an alternate location in case of site failure is vital, but at some point it is necessary to return that environment to its original condition as non-disruptively as possible. The steps you just completed in this activity showed you how easy it is to accomplish that. In the next activity, you will use the switchover on a normally operating cluster to test disaster recovery.

2.4 Planned Switchover: High Availability for Non-Disruptive Operations

A planned or negotiated switchover can be used in Non-Disruptive Operations (NDO). A negotiated switchover (NSO) cleanly shuts down processes on the partner site, and then switches over storage services from the partner site. You can use negotiated switchover to perform maintenance on a MetroCluster site, or test the switchover functionality. If you want to test the MetroCluster functionality or perform planned maintenance, you can perform a negotiated switchover in which one cluster is cleanly switched over to the partner cluster. Later, you switch back the services to the original site.

Rather than a forced switchover when one of the clusters is not reachable, this process involves performing the negotiated switchover prior to beginning the controlled power off event. In this activity you migrate the VM to compute at Site B, and have the NetApp MetroCluster serve storage from Site B in a controlled manner.

Tip: If the planned outage is for storage services only, it is possible to do the negotiated switchover of MetroCluster and not migrate the virtual machines. If the site to site bandwidth allows, the virtualization infrastructure can access storage from the remote site.

Tasks
Transfer of operations from Site A to Site B (cluster1 to cluster2).
(Optional) Transfer of operations from Site B back to Site A (cluster2 to cluster1) upon completion of maintenance.

2.4.1 Planned Switchover

In this activity, you act as an IT administrator that is tasked with testing the disaster recovery between sites. You perform the following tasks:

Tasks
Migrate the VM from the site to be switched over.
Power down the ESX server on the site to be switched over.
Perform a MetroCluster switchover.

1. First, vm1 needs to be migrated from site A to site B. Open or return to the VMware vSphere Web Client and right-click **vm1**.

2. Select **Migrate**.

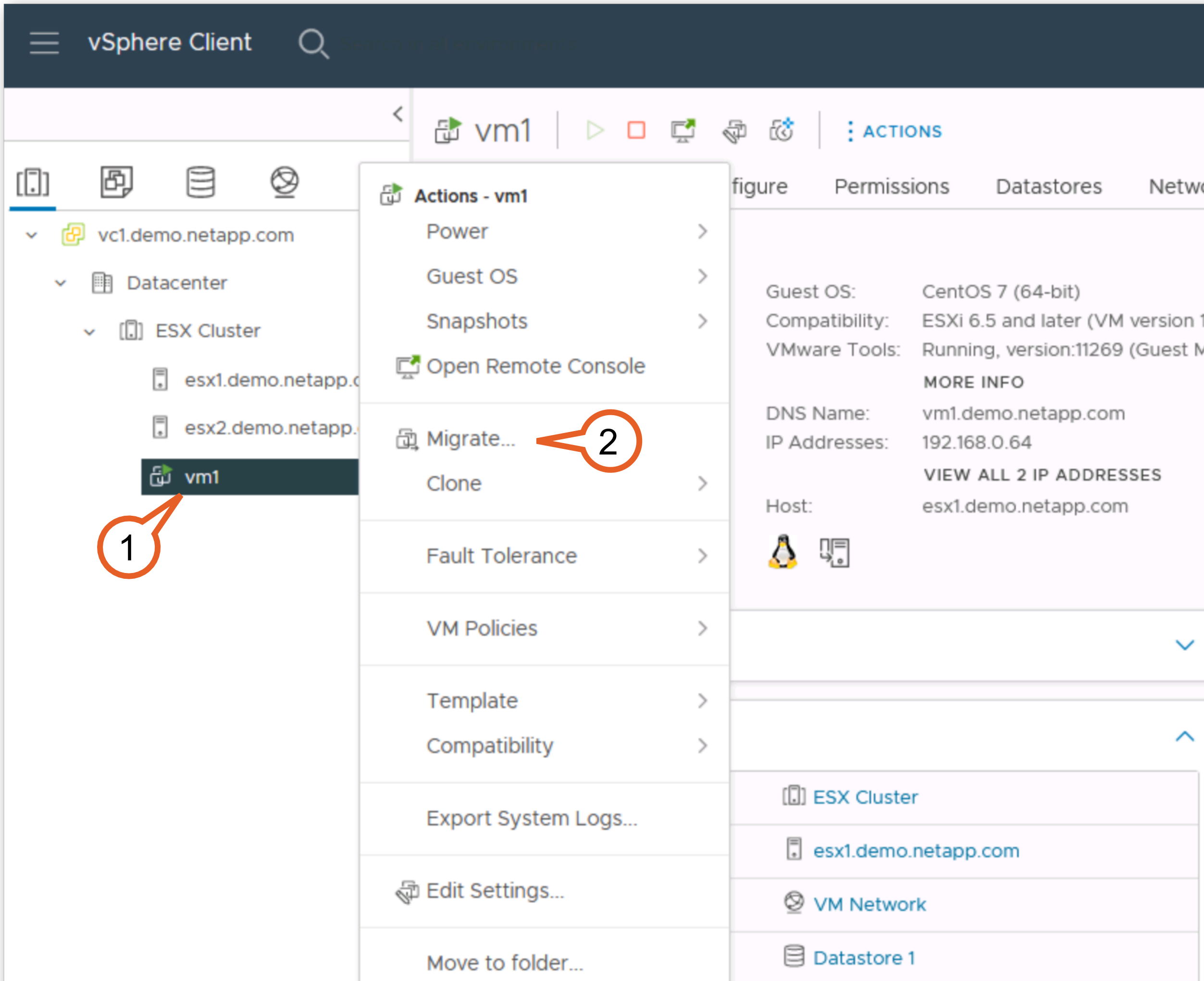

Figure 2-55:

3. Click **Next**.

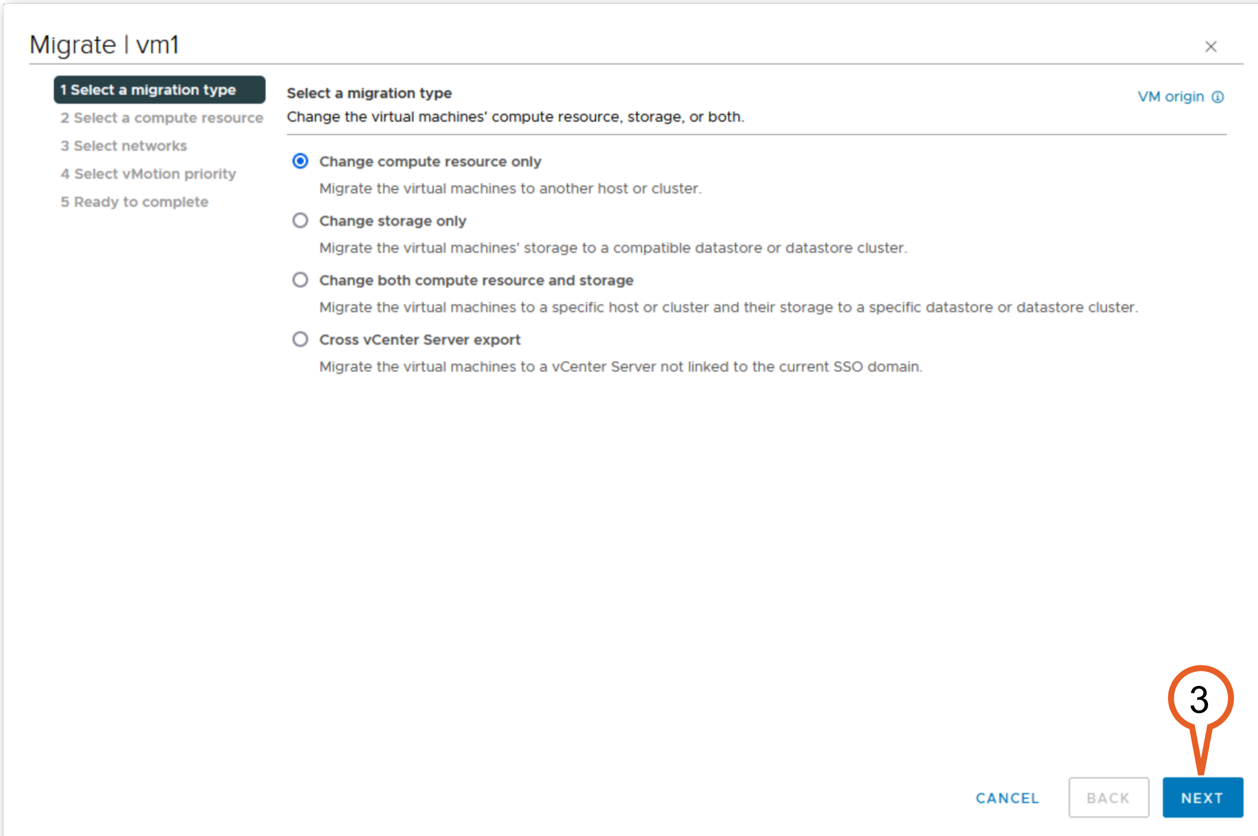

Figure 2-56:

4. Select **esx2**.

5. Click **Next**.

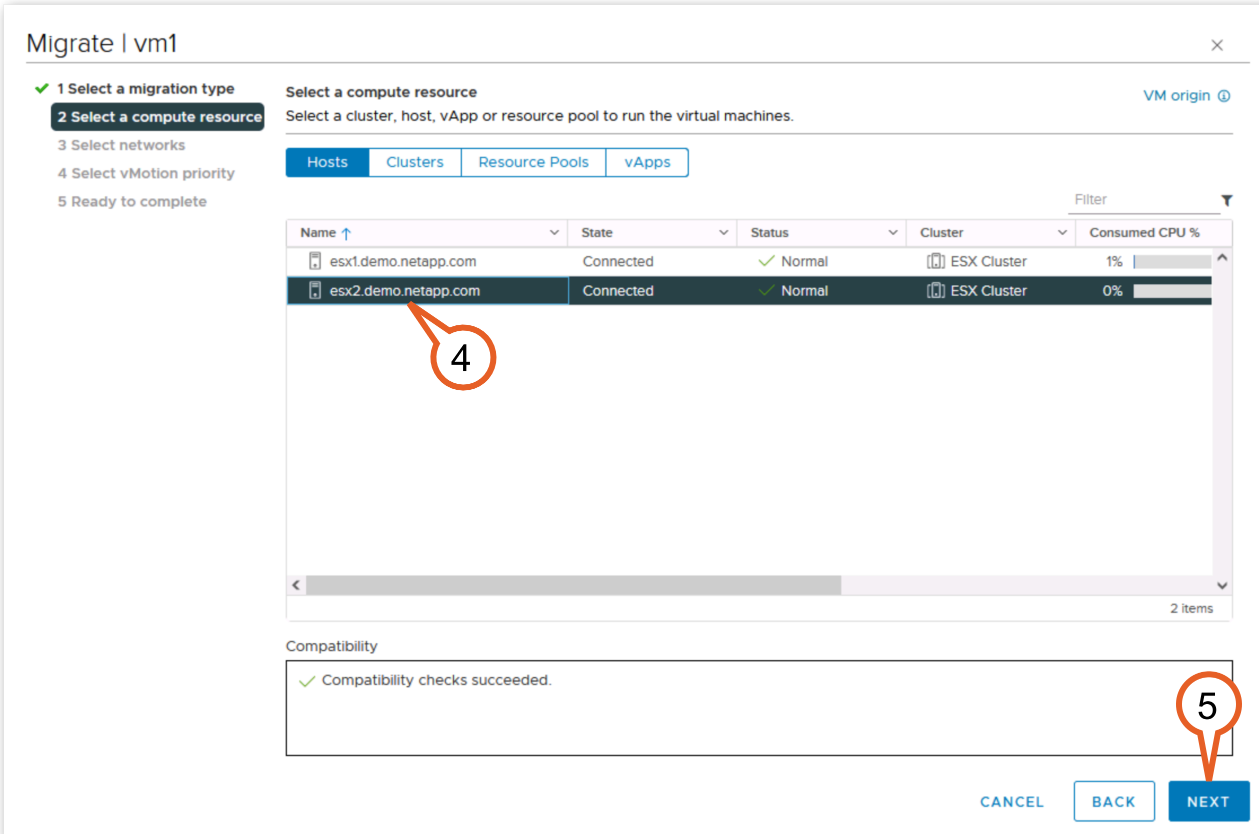

Figure 2-57:

6. Click **Next**.

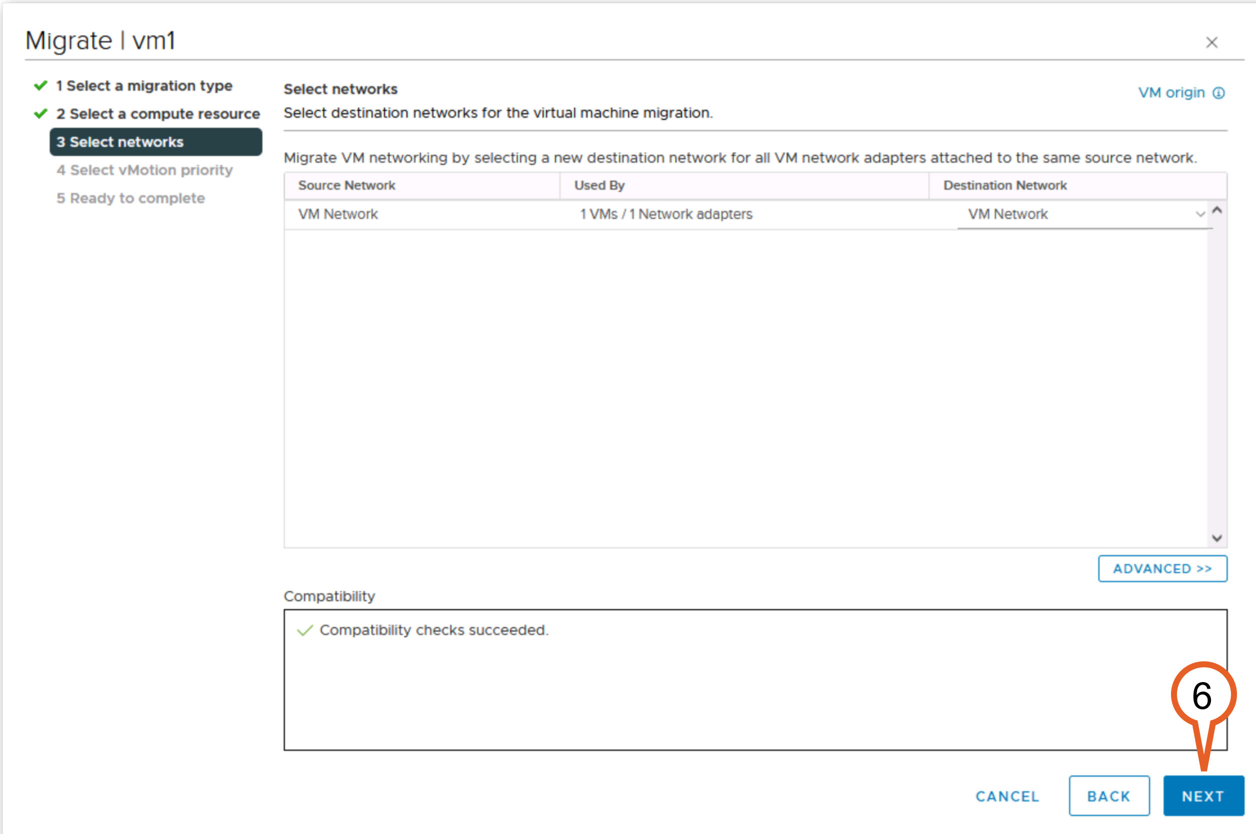

Figure 2-58:

7. Click **Next**.

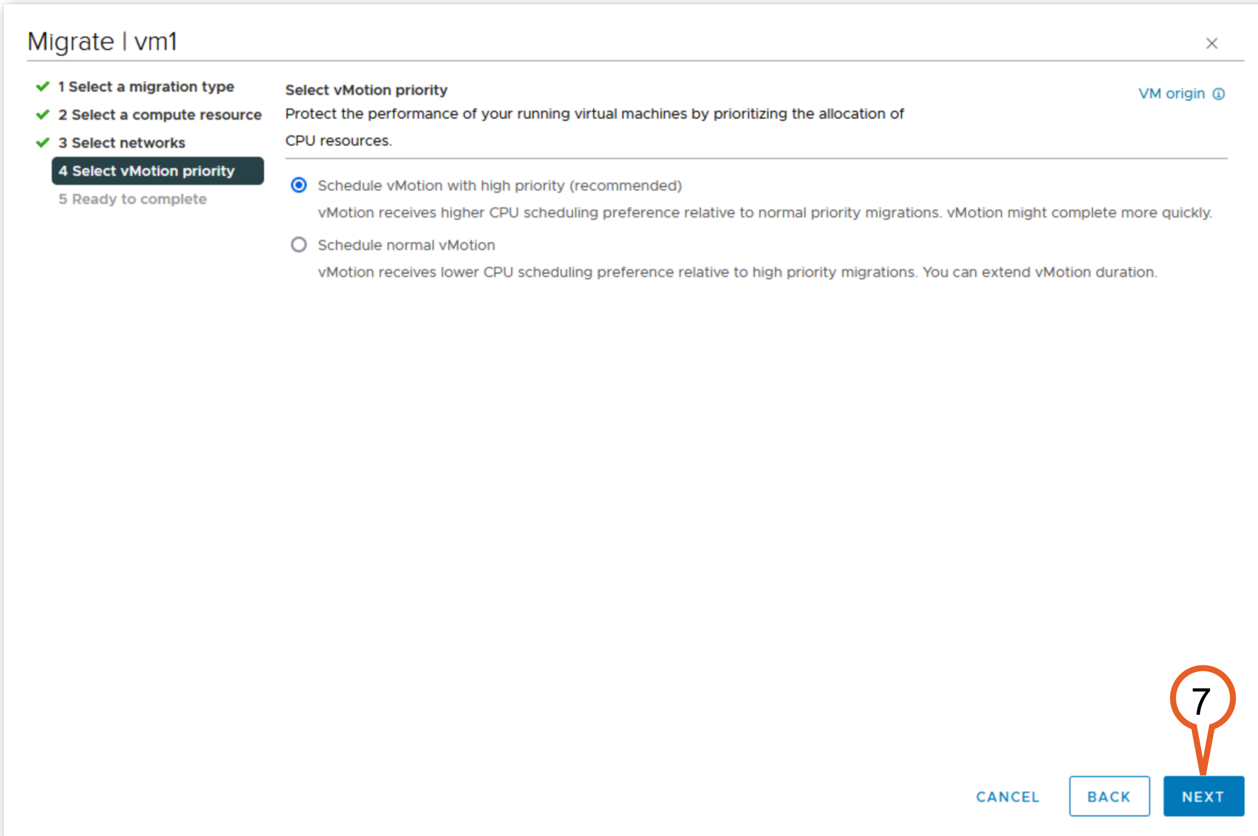

Figure 2-59:

8. Click **Finish**.

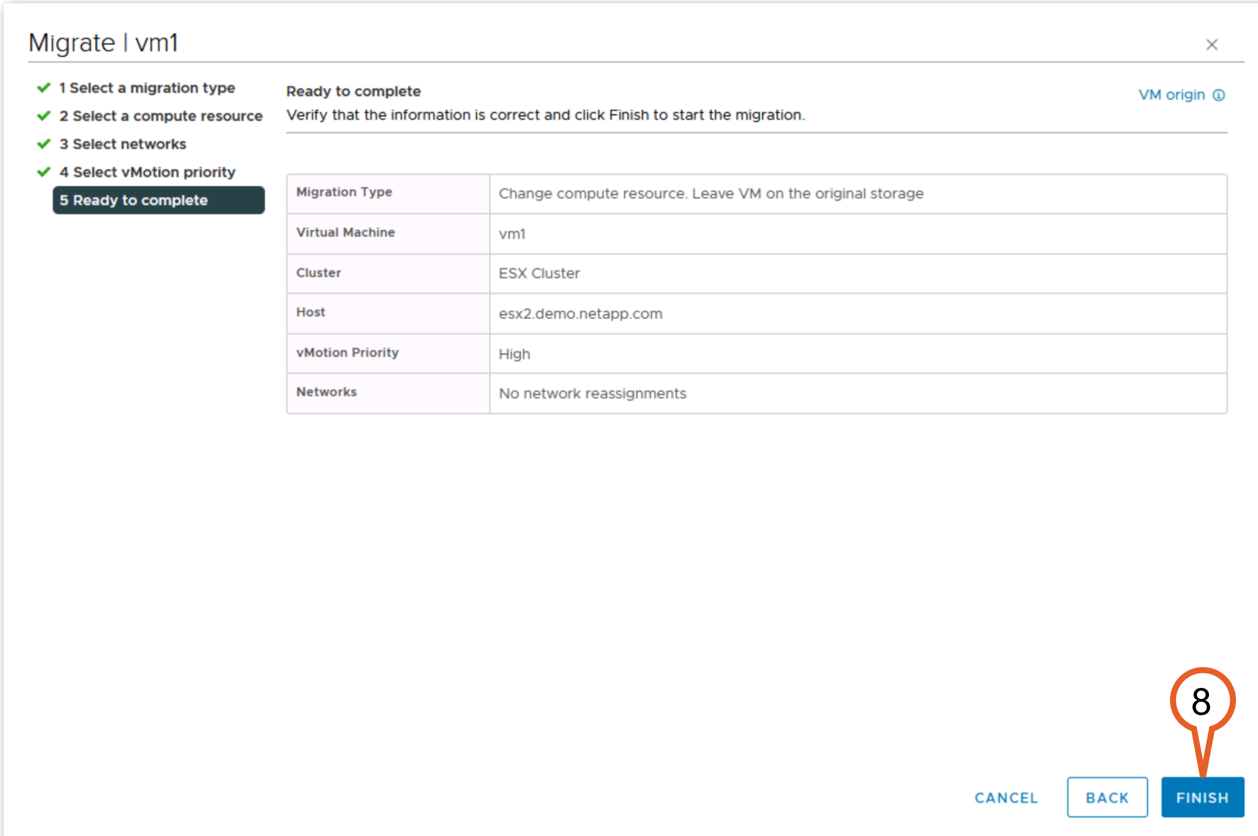

Figure 2-60:

9. After the VM task finishes migrating you should see “vm1” running on "esx2".

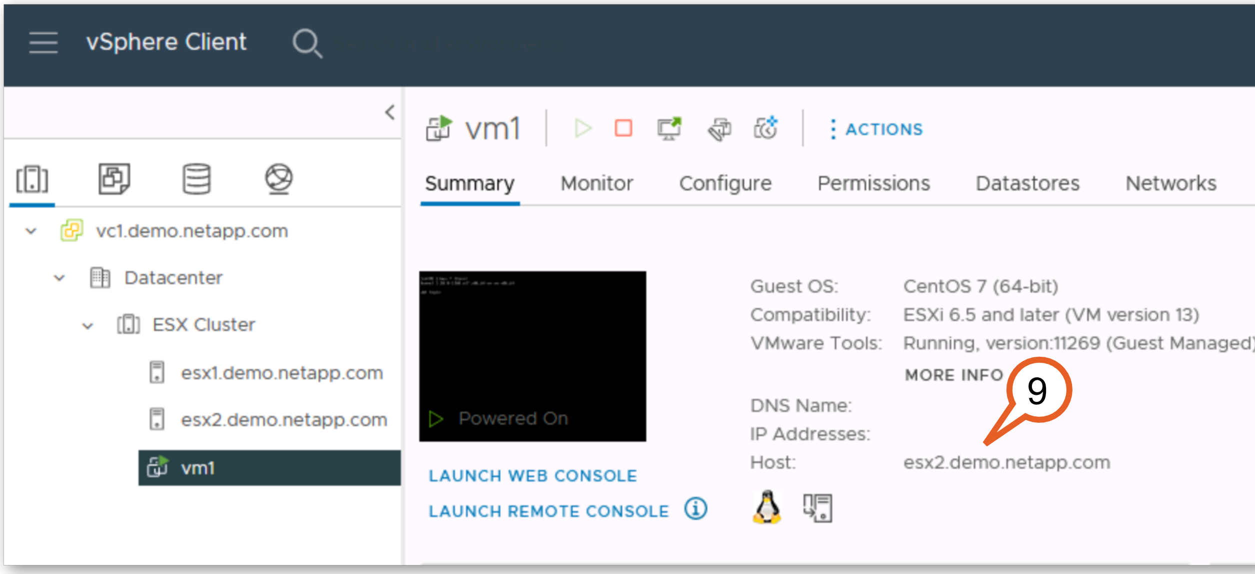

Figure 2-61:

10. Right-click **ESX1.**

11. From the menu select **Power** > **Shut Down**.

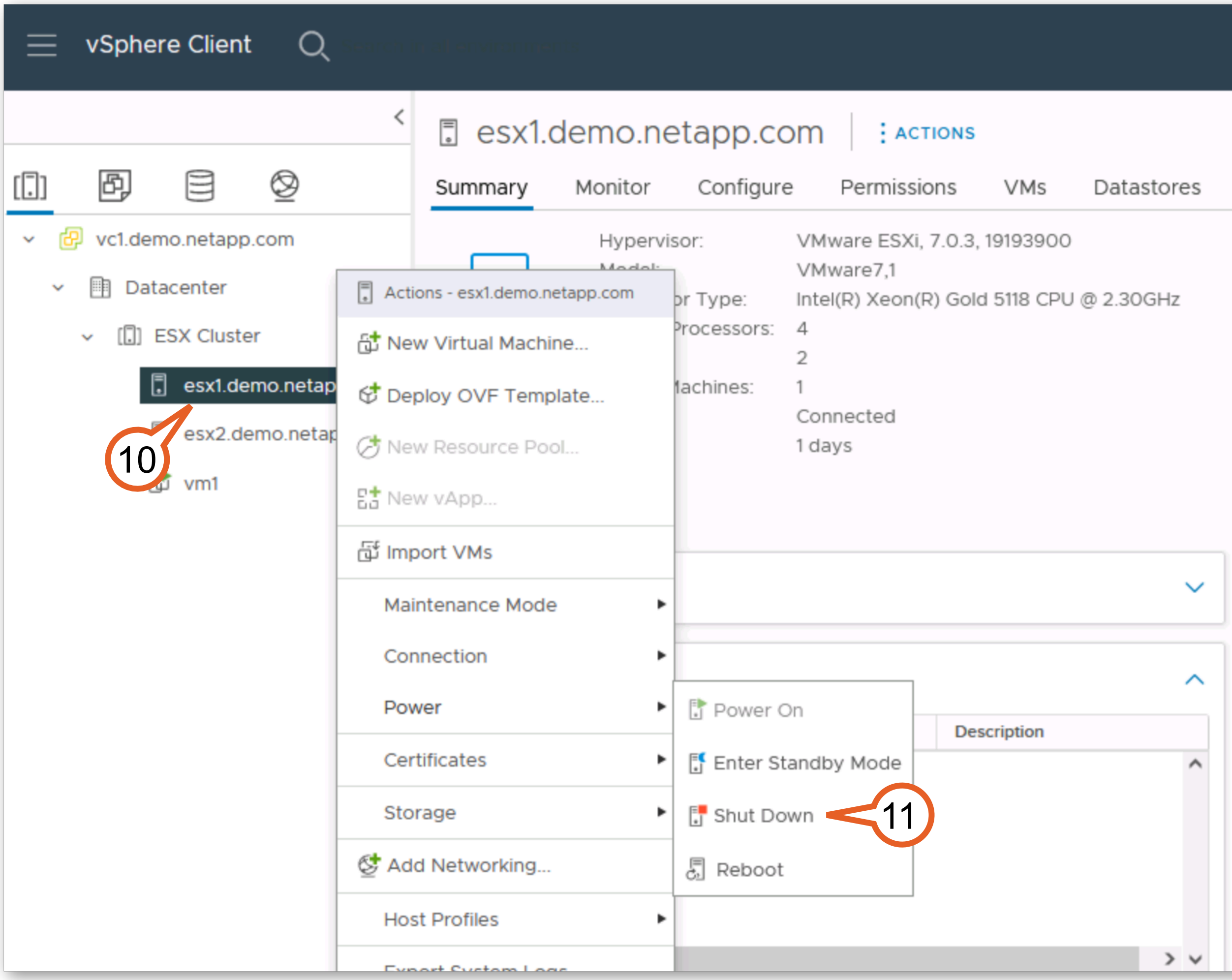

Figure 2-62:

12. Enter **`Planned Shutdown`** as the reason.

13. Click **OK**.

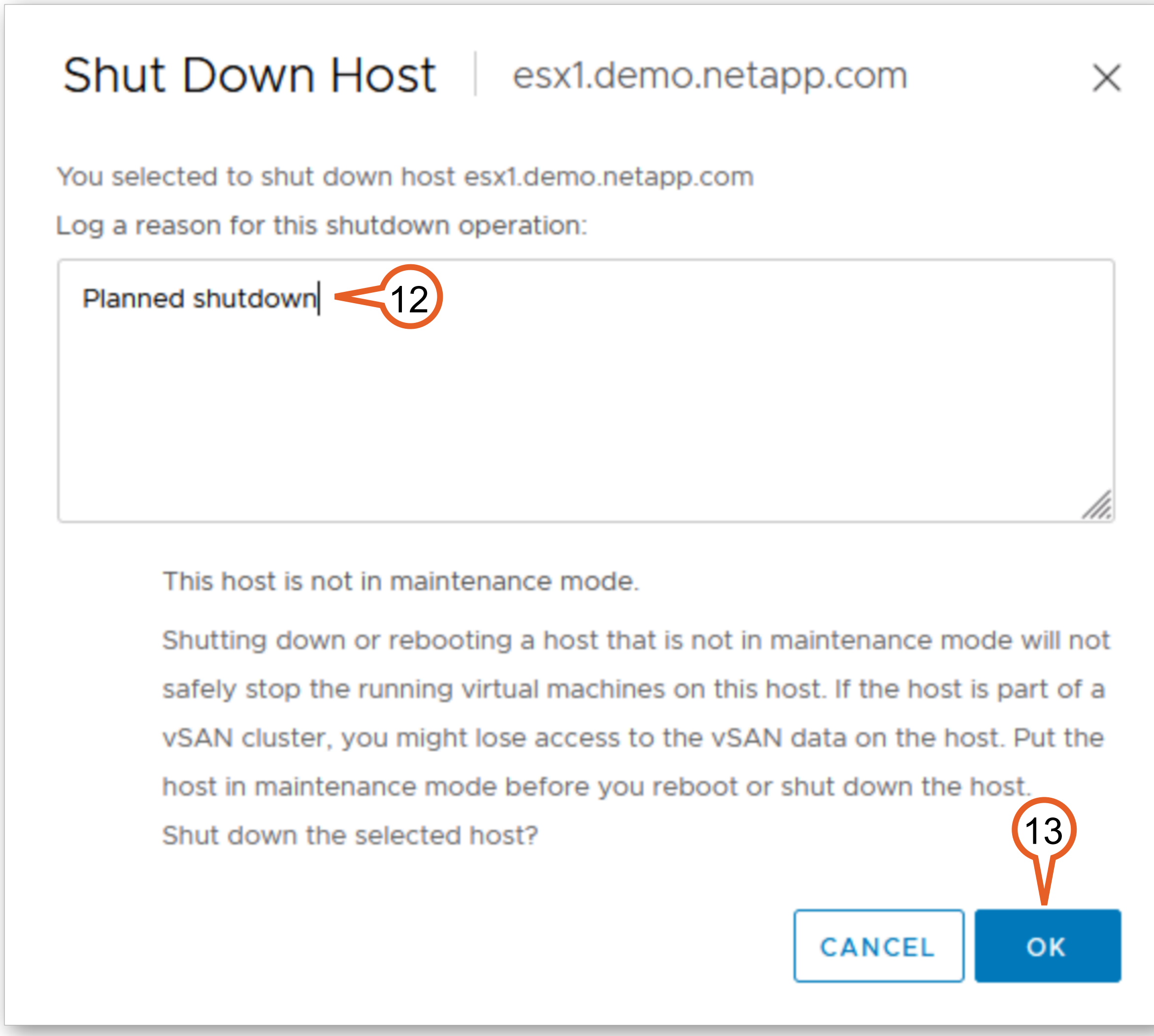

Figure 2-63:

Now that the VM is migrated from Site A to Site B and the ESX host is shut down, the next step is to switch over the MetroCluster.

14. Open or return to the browser tab for “cluster2 System Manager”, and log in if necessary.
15. In the MetroCluster panel on the Dashboard, click the **menu** for “cluster2”.

16. Select **Switch over remote data services to the local site**.

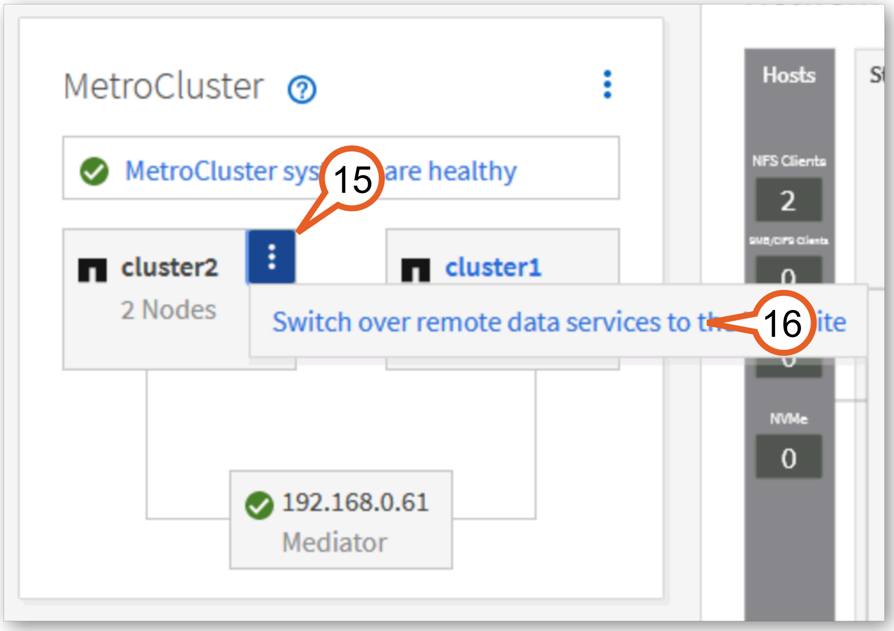

Figure 2-64:

17. Click **OK**.

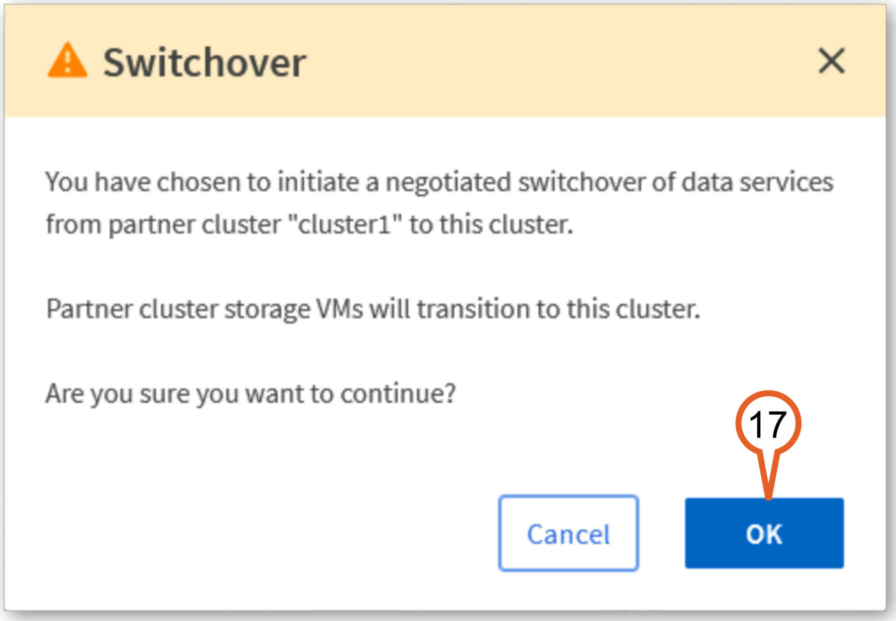

Figure 2-65:

The MetroCluster panel shows that ONTAP is performing a negotiated switchover.

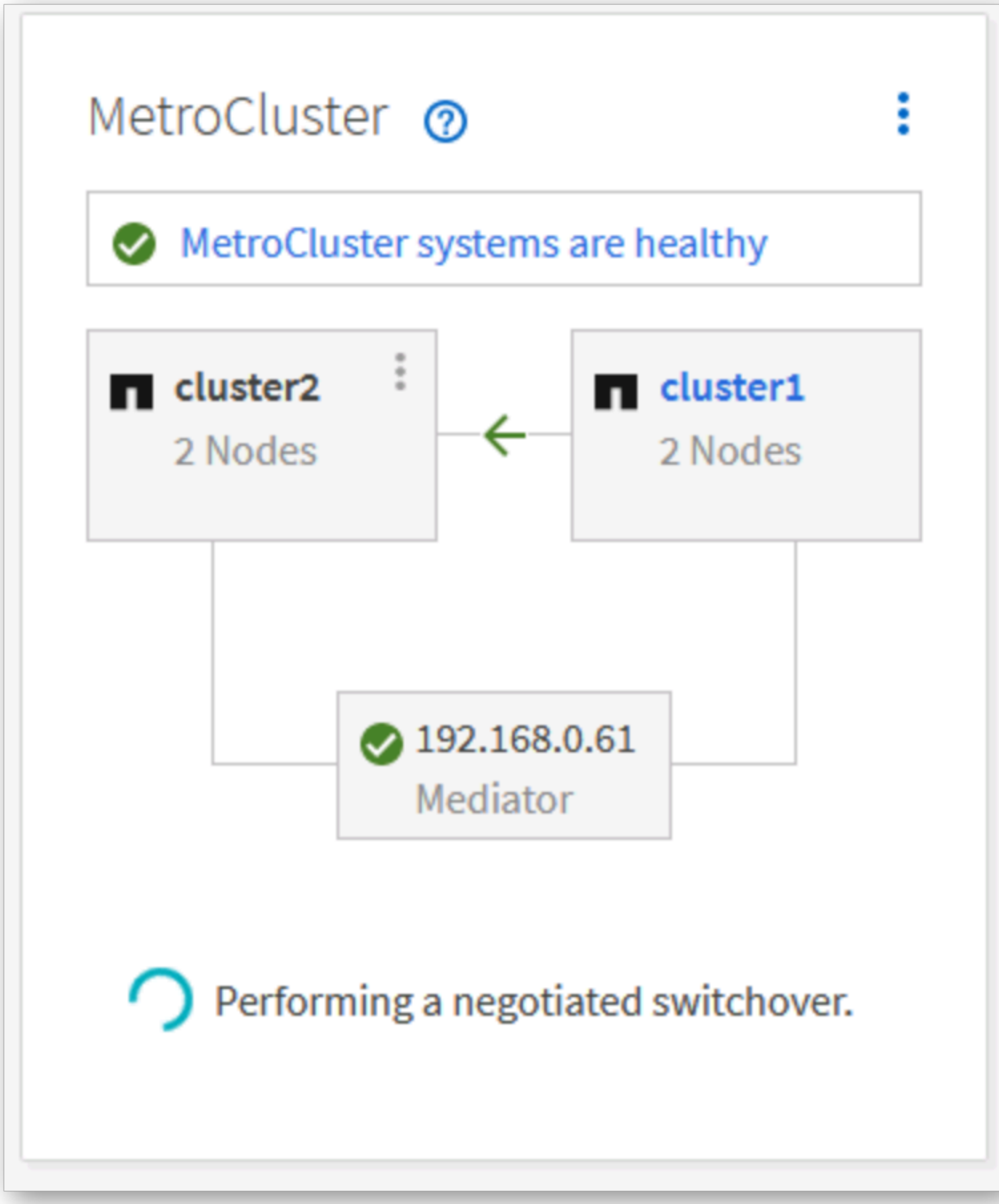

Figure 2-66:

In about a minute, the switchover is complete and the icon between the two clusters changes to a green switchover icon.

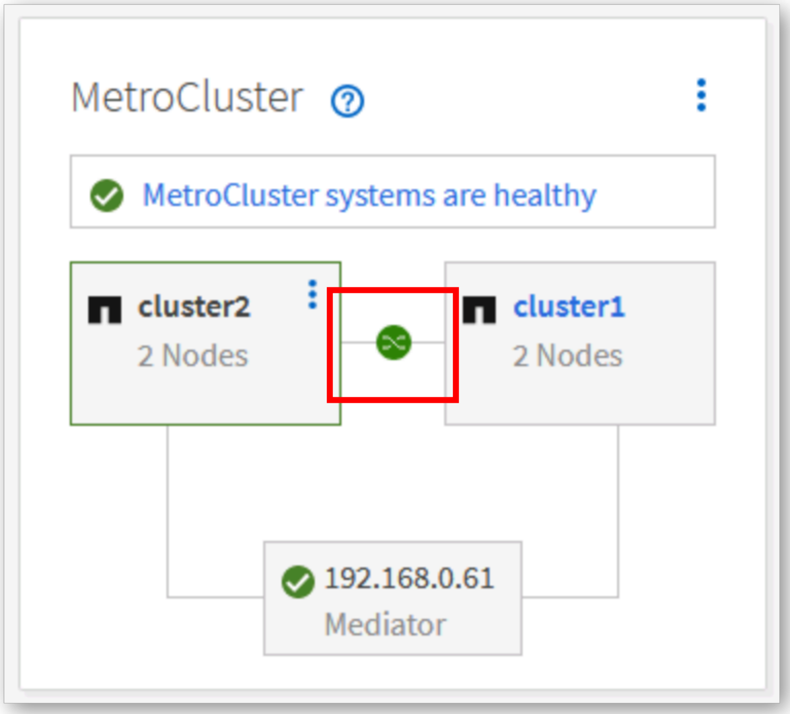

Figure 2-67:

18. On the left-hand navigation menu click **Storage** > **Storage VMs**.

19. Verify that “svm1-mc” is running.

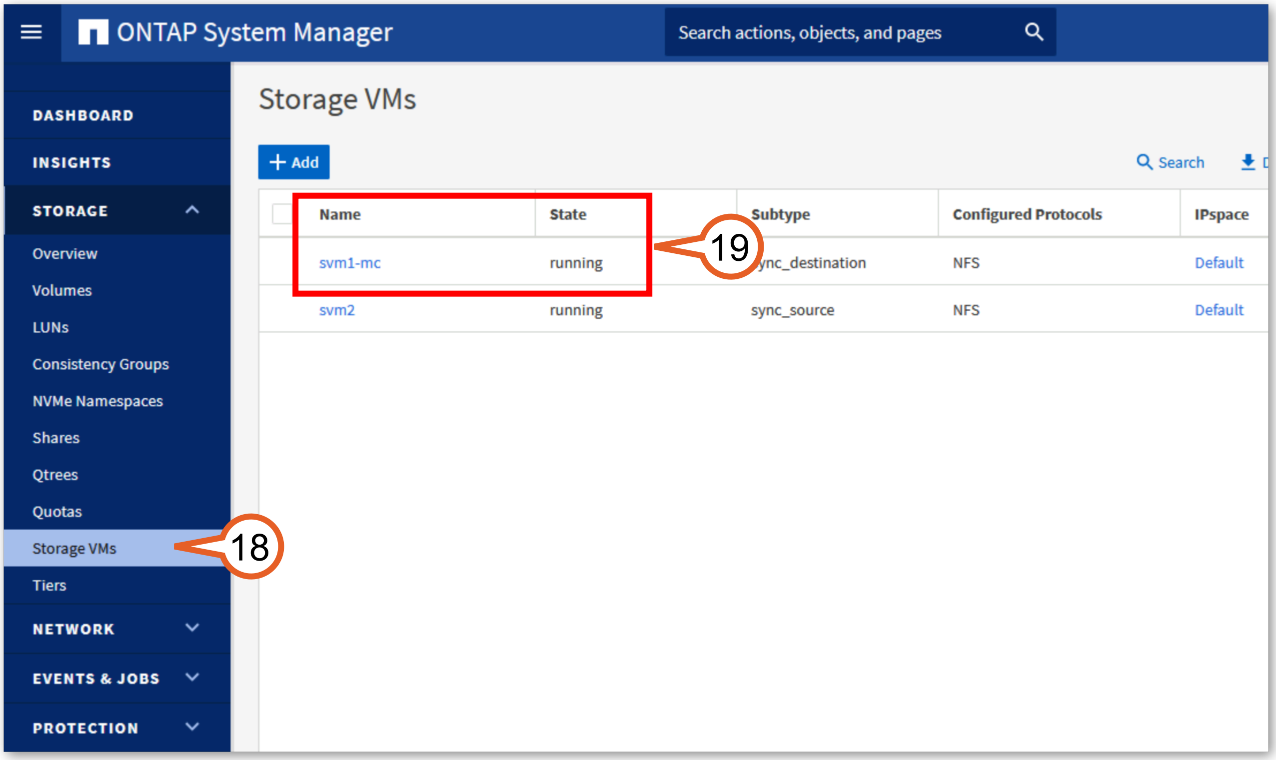

Figure 2-68:

This activity showed you how to complete a planned or negotiated switchover, including migration of a VM to the other location. This process does not impact business operations, meaning the virtual machines and storage are online and working during and after the switchover. This process is used to test disaster recovery and can also be used for non-disruptive operations where a site must be taken offline to perform maintenance. In the next activity, you return both sites to normal operation by switching back the services and migrating the virtual machine.

Tip: ONTAP updates use the local high-availability of the cluster to perform automatic non-disruptive upgrades (ANDU). Because the MetroCluster configuration uses redundancy, most maintenance tasks can be performed when the system is up and running, such as switch firmware updates.

2.4.2 Optional: Switchback from Planned Maintenance

In this activity you perform the following tasks:

Tasks
Perform a MetroCluster switchback from cluster2 to cluster1.
Power up the “esx1” server on Site A.
Migrate the “vm1” virtual machine back to its original location.

1. Return to “cluster2 System Manager” browser tab and log in if necessary.
2. On the Dashboard, in the MetroCluster status pane, click the **menu** for “cluster2”.

3. Select **Switch back remote data services to the remote site**.

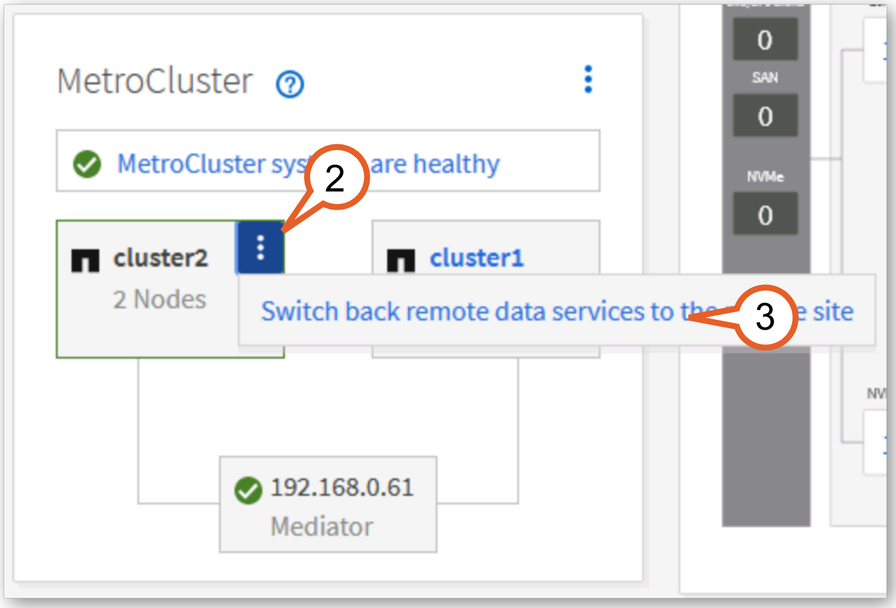

Figure 2-69:

4. Click **OK**.

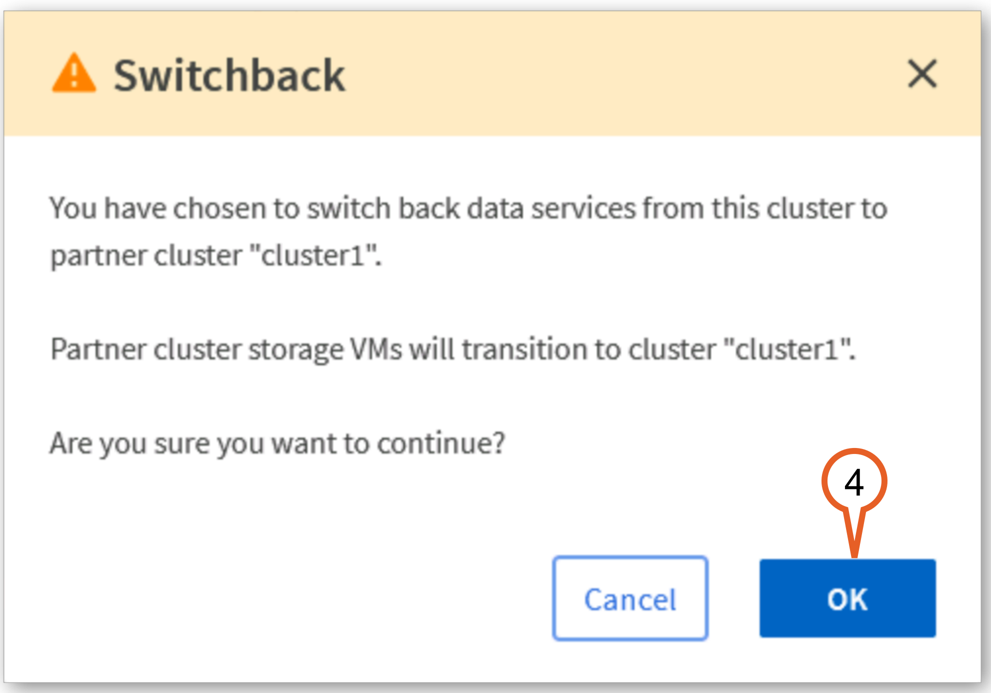

Figure 2-70:

5. The icon between the clusters changes to a green arrow pointing to cluster1. Monitor the progress of the switchback operation that takes a minute or two.

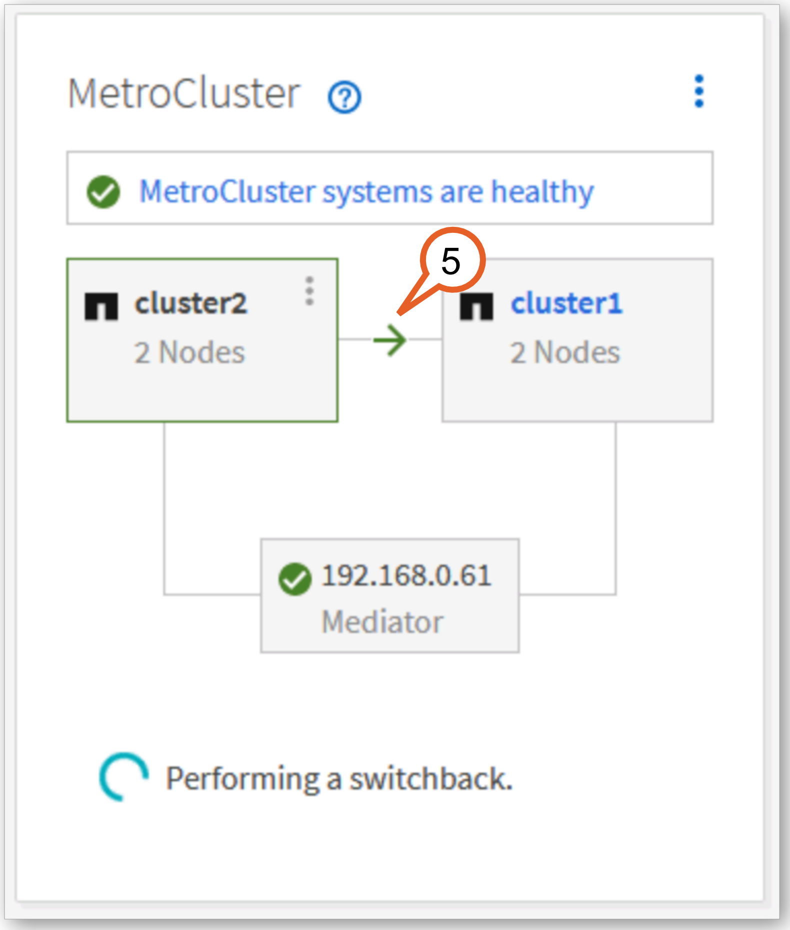

Figure 2-71:

When the operation is complete, the icon changes to a green check and the systems are all shown in a normal state.

6. Return to the “Power Controls” browser tab or click the **Power Controls** browser bookmark.

Note: You may have to click the **Refresh** button to see the current state.

7. Click the green **Power On** button for “esx1” under “Site A”.
8. The Status should change to “Powered On”.

9. You may have to click the **Refresh** button.

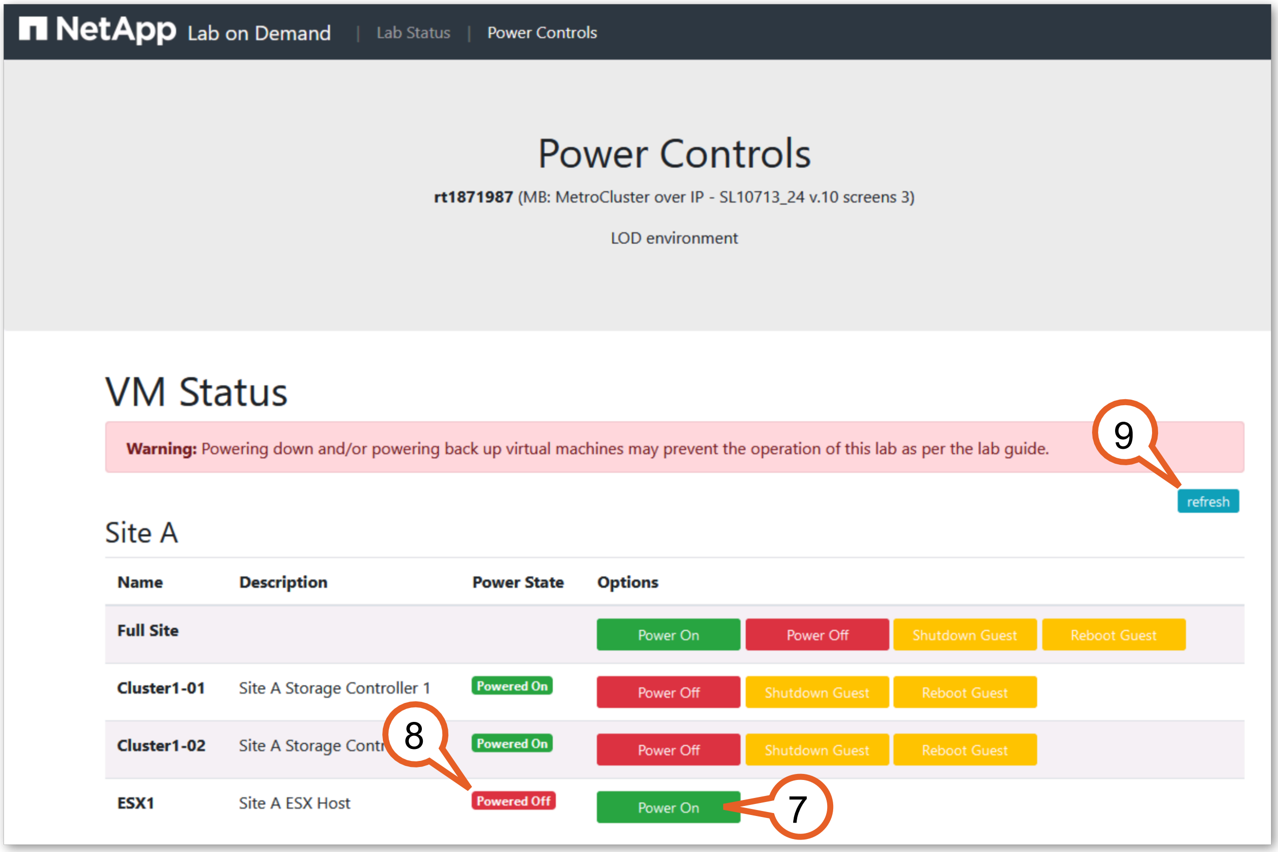

Figure 2-72:

Note: It can take a minute or two for changes with the Power Controls to take effect.

10. Open or return to the VMware vSphere Web Client and verify that the “esx1” host is operational. This could take several minutes.

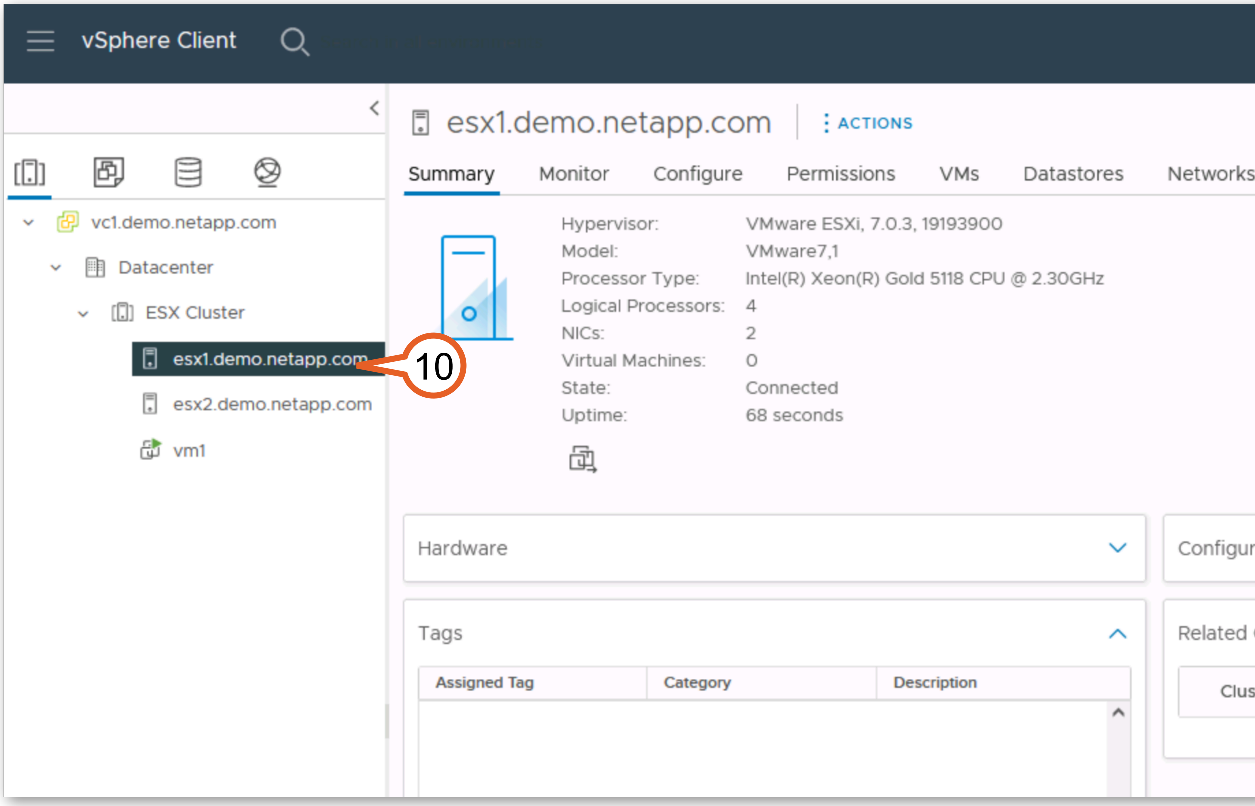

Figure 2-73:

11. Right-click **vm1**.

12. Select **Migrate**.

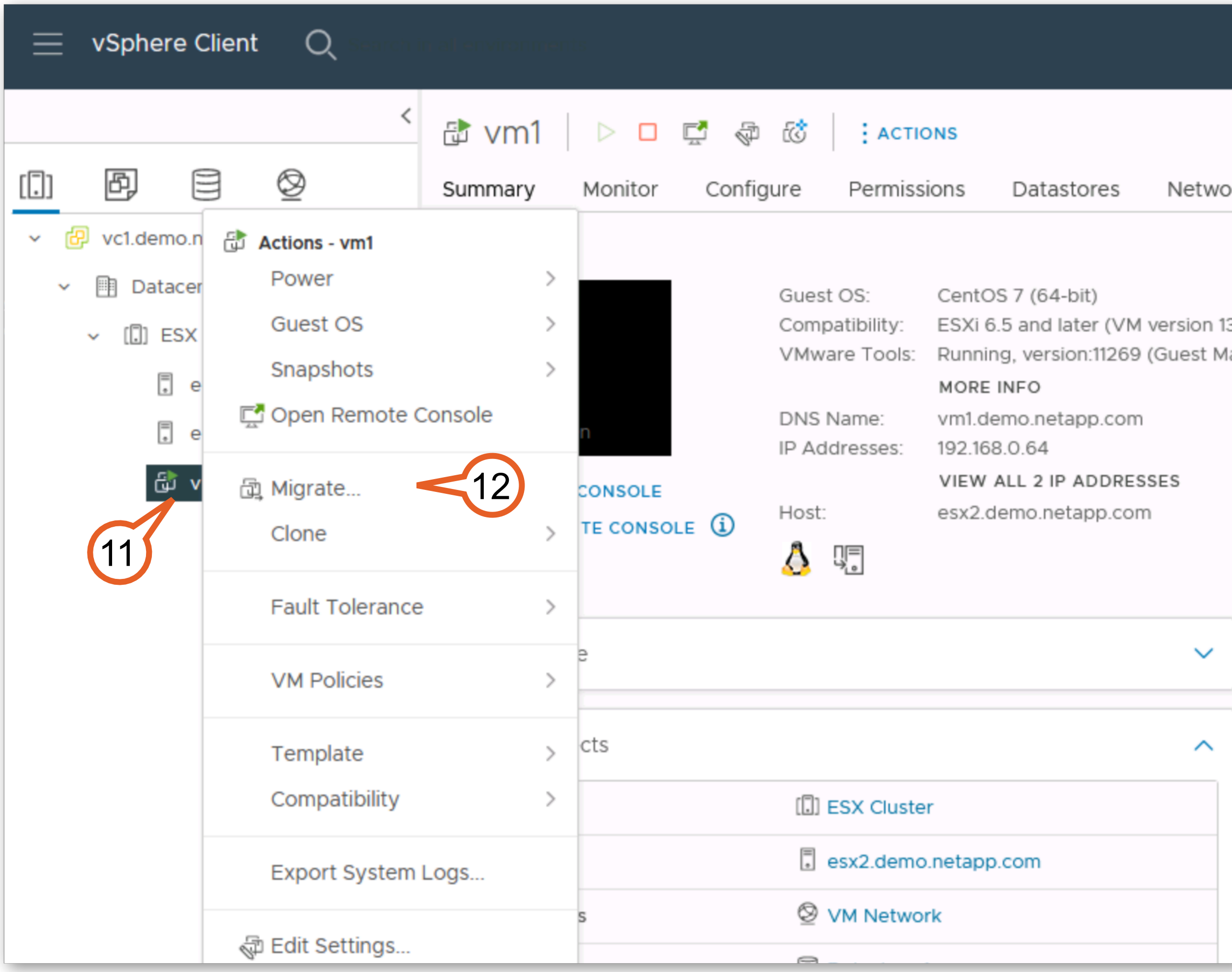

Figure 2-74:

13. Select **Change compute resource only**.

14. Click **Next**.

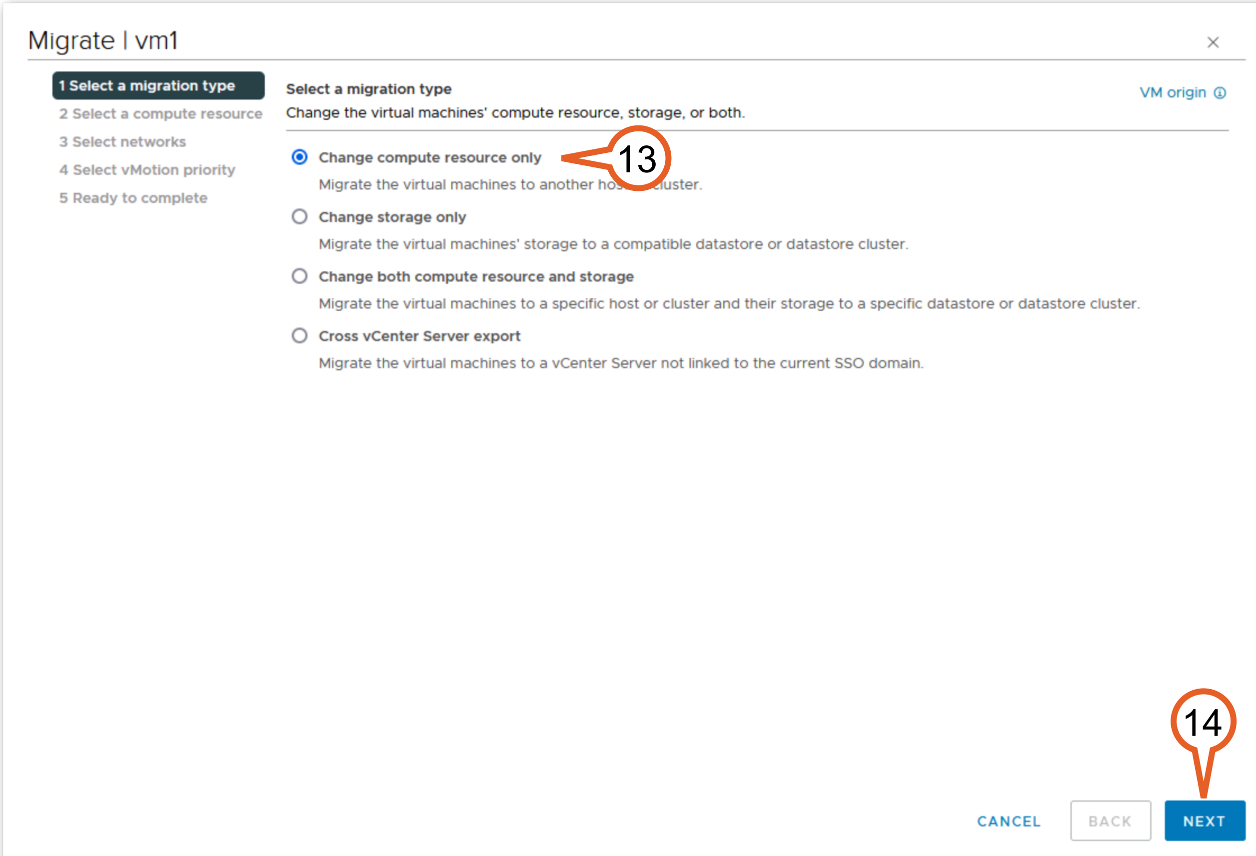

Figure 2-75:

15. Select **esx1**.

16. Click **Next**.

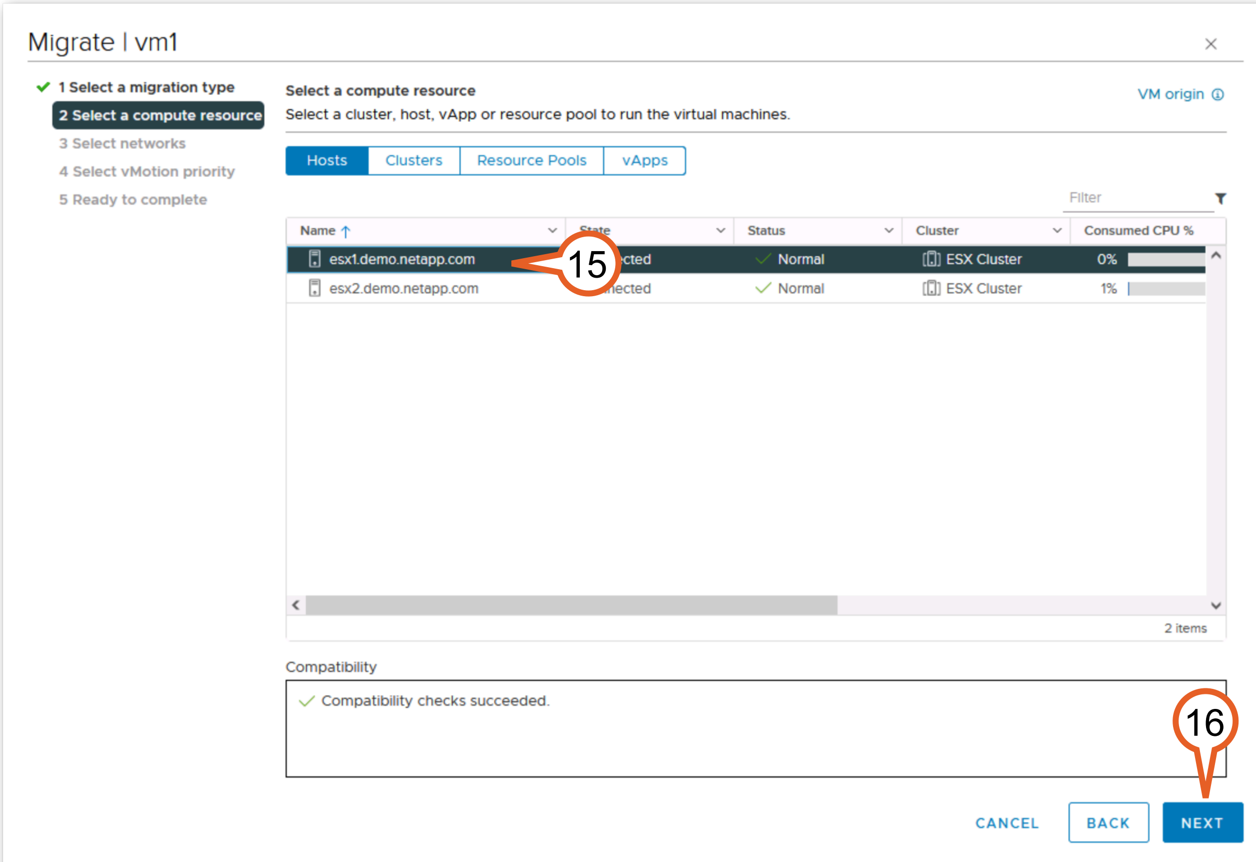

Figure 2-76:

17. Leave the default values and click **Next**.

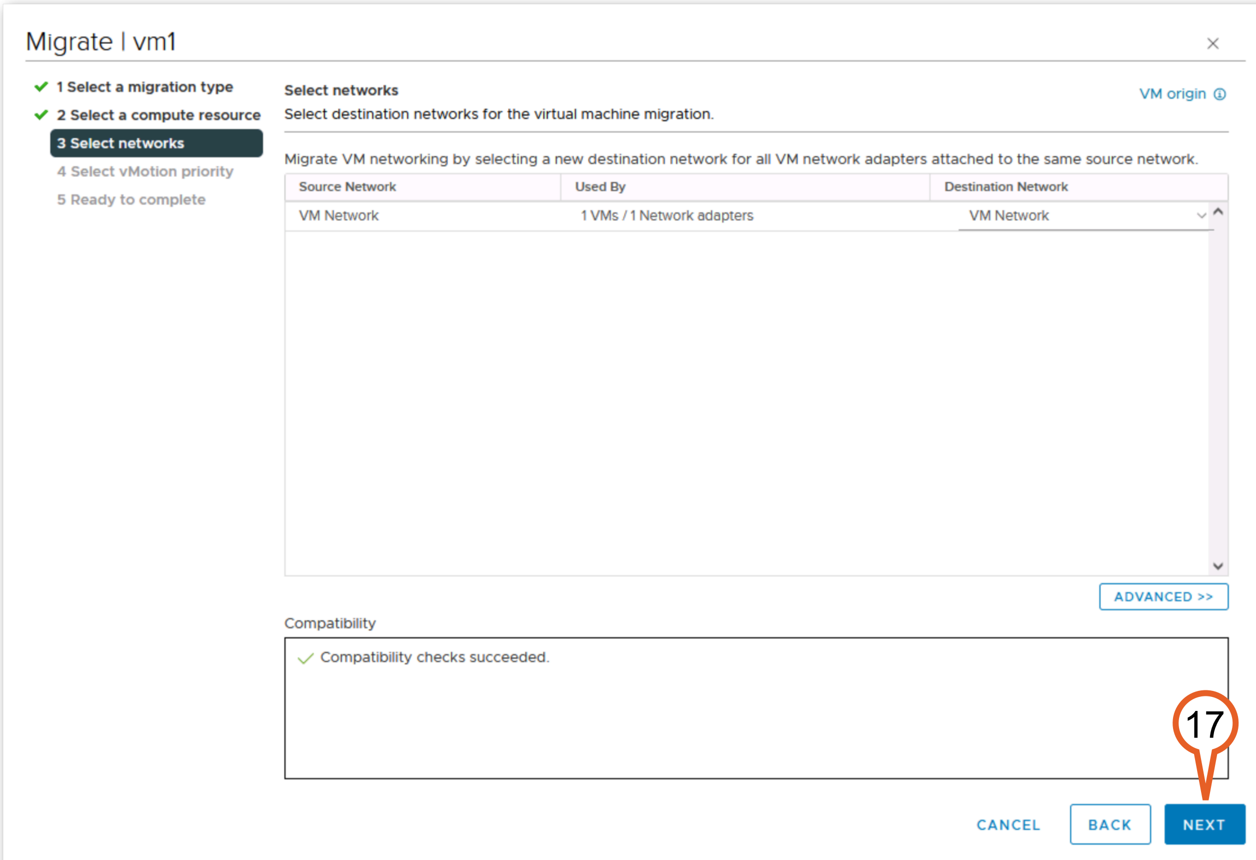

Figure 2-77:

18. Click **Next**.

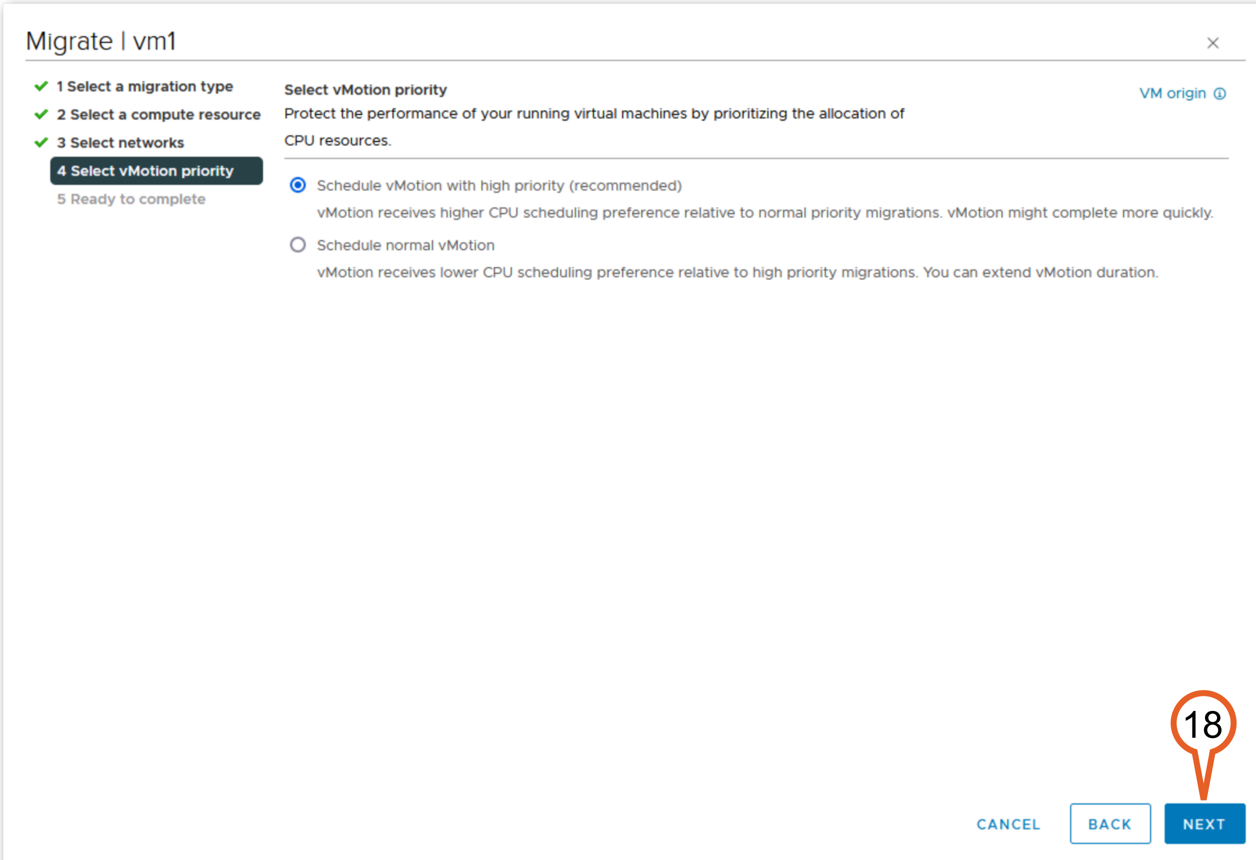

Figure 2-78:

19. Click **Finish**.

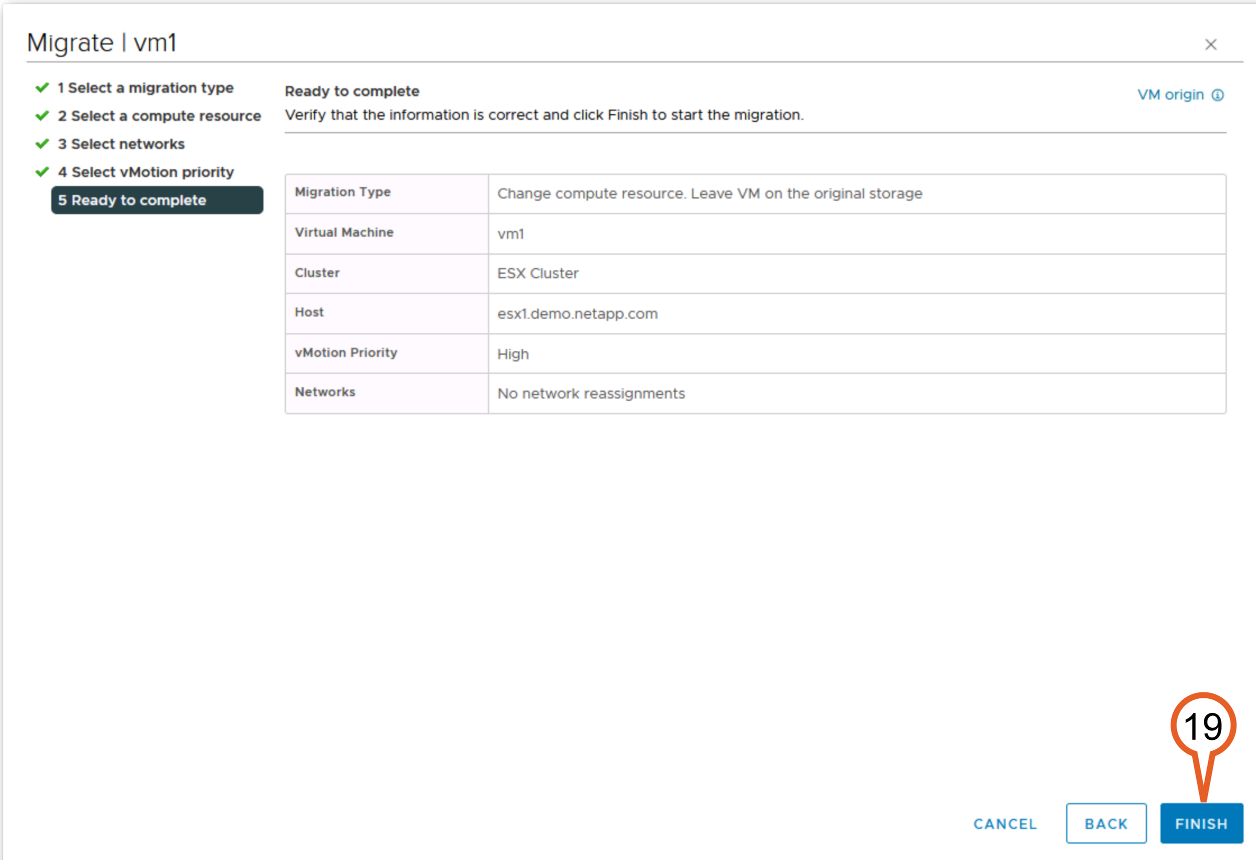

Figure 2-79:

20. Once finished, you will see that “vm1” is again running on “esx1”.

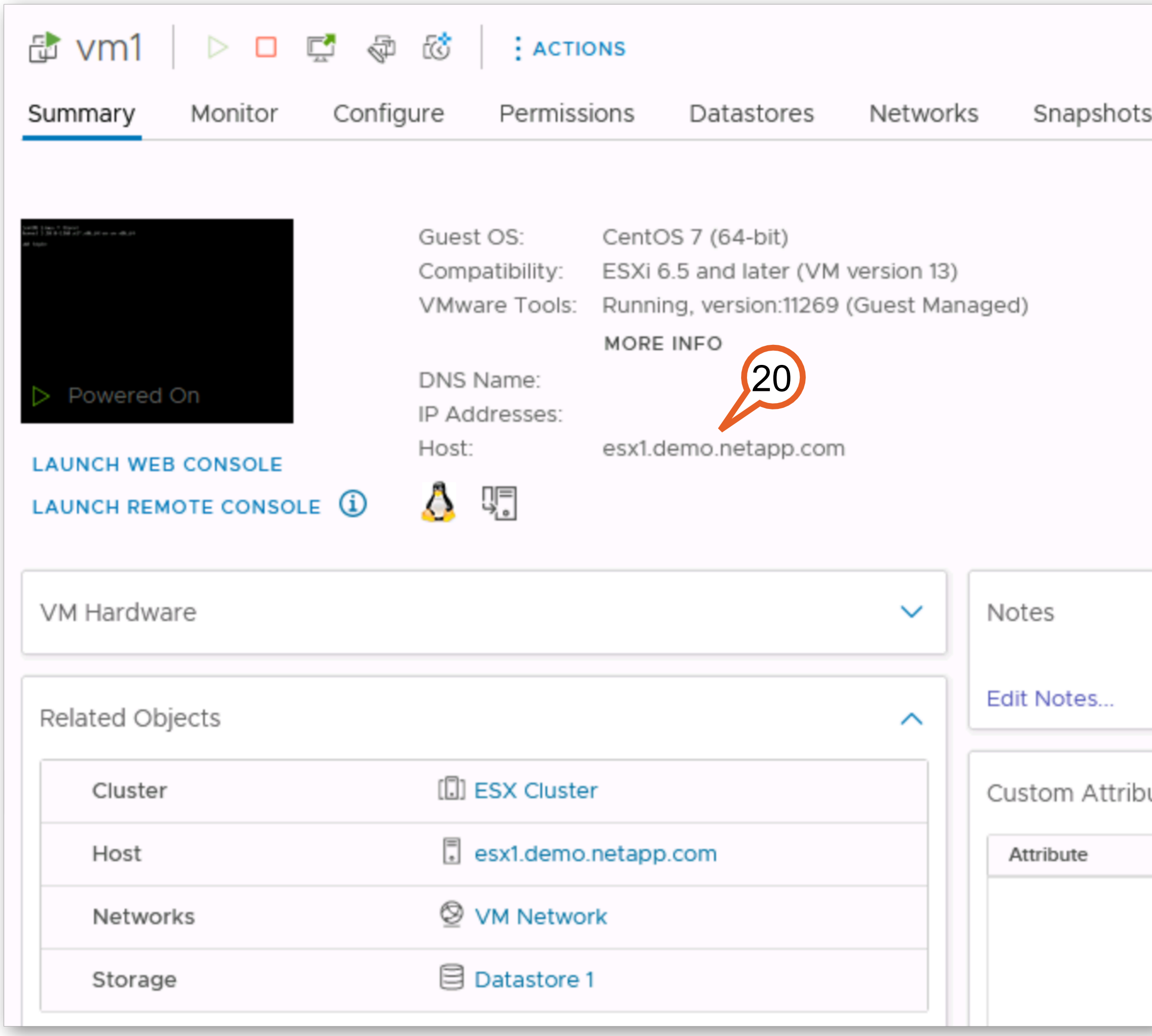

Figure 2-80:

This activity showed you how to complete a planned or negotiated switchover, including migration of a virtual machine to the other site in order to preserve non-disruptive operation.

3 Lab Summary

In this lab, you demonstrated how MetroCluster IP can combine with VMware vSphere HA clusters to provide automatic failover to an alternate location resulting in little, if any, disruption to business operations.

NetApp MetroCluster extends availability of mission critical data to reduce the risks associated with power or natural disasters at a data center. It complements the compute/server high availability provided by VMware HA Clusters by providing synchronously mirrored copies of data and the storage configuration.

When a site failure occurs, MetroCluster helps automate a switchover of storage services to the remote site. Using ONTAP Mediator, it is possible to have a swtichover occur automatically without interrupting access to storage.

When a site level failure is resolved, MetroCluster avoids additional downtime by performing resynchronization (healing) while business operations continue at the secondary site. As a result, switchback is quick, and disruption is minimized.

Downtime due to site maintenance and upgrades is avoided by performing a planned switchover of business operations to the secondary site. A planned swtichover is also an integral part of site operations allowing storage administrators to validate disaster recovery plans.

4 Additional Information

4.1 References

References for MetroCluster IP:

- ONTAP 9 MetroCluster Documentation
- Manage the ONTAP mediator service
- VMware vSphere 5.x, 6.x and 7.x support with NetApp MetroCluster (2031038)

4.2 Lab Environment

All of the servers and storage controllers presented in this lab are virtual devices, and the networks that interconnect them are exclusive to your lab session. The virtual storage controllers (vsims) offer nearly all the same functionality as do physical storage controllers, but at a reduced performance profile.

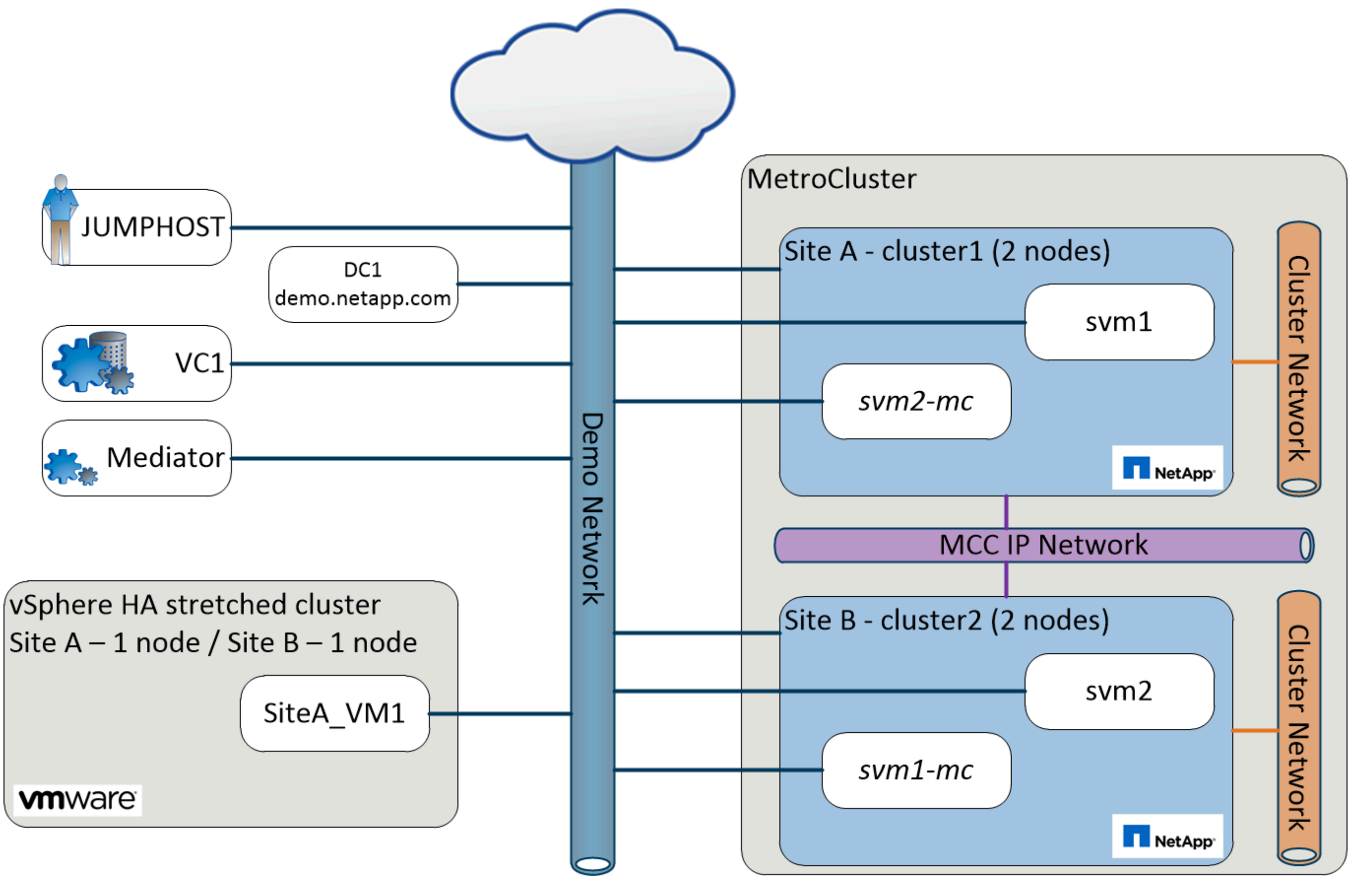

Figure 4-1:

Table 1: Table of Systems

Host Name	Operating System	Role/Function	IP Address
cluster1	ONTAP 9.11.1	cluster 1 - Site A	192.168.0.101
cluster1-01	ONTAP 9.11.1	cluster 1 node 1	192.168.0.111

Host Name	Operating System	Role/Function	IP Address
cluster1-02	ONTAP 9.11.1	cluster 1 node 2	192.168.0.112
cluster2	ONTAP 9.11.1	cluster 2 - Site B	192.168.0.102
cluster2-01	ONTAP 9.11.1	cluster 2 node 1	192.168.0.121
cluster2-02	ONTAP 9.11.1	cluster 2 node 2	192.168.0.122
jumphost	Windows Server 2019	primary desktop entry point for lab	192.168.0.5
dc1	Windows Server 2019	Active Directory / DNS	192.168.0.253
esx1	VMware vSphere 7 ESX host	Site A host 1	192.168.0.51
esx2	VMware vSphere 7 ESX host	Site B host 1	192.168.0.52
vc1	vCenter Appliance	VMware vSphere 7 vCenter	192.168.0.31
mcc-med	CentOS 7.6	linux client running NetApp Mediator	192.168.0.61
vm1	CentOS 7.7	nested VM	192.168.0.64
centos1	CentOS 7.7	linux host	192.168.0.62

Table 2: User IDs and Passwords

Host Name	User ID	Password	Comments
jumphost	DEMO\Administrator	Netapp1!	Domain Administrator
cluster1	admin	Netapp1!	Same for individual cluster nodes
cluster2	admin	Netapp1!	Same for individual cluster nodes
svm...	vsadmin	Netapp1!	svm administrator
vc1	administrator@demo.netapp.com	Netapp1!	Admin for vCenter
mcc-med	root	Netapp1!	Linux Administrator
mcc-med	mediatoradmin	Netapp1!	Mediator user
centos1	root	Netapp1!	Linux Administrator
vm1	root	Netapp1!	Linux Administrator

4.3 Lab Limitations

This lab has the following limitations:

- All of the servers and storage controllers presented in this lab are virtual devices. Consequently, any operations involving movement of large quantities of data will not exhibit performance representative of real systems.
- Due to the differences between the IP-based and the Fibre Channel-based configurations of NetApp MetroCluster, there are differences between the Lab-on-Demand offerings for each version. Refer to the introduction of this lab guide for details.

- Due to the configuration of this lab environment, the virtual machines containing the ONTAP clusters cannot be powered off. To simulate a “site failure” they will be shut down instead.

4.4 Version History

Version	Date	Document Version History
1.0.0	Jun 2018	Initial Release
1.1.0	Jun 2020	Upgraded ONTAP 9.7, vSphere 6.7
1.1.1	July 2020	Correct typo in Lab Environment for VC admin
1.1.2	November 2020	Replaced external Power Controls with Web based
1.2	August 2022	ONTAP 9.11.1

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  - Notice for ONTAP 9.2
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